FirstPass STUDIO

USER GUIDE

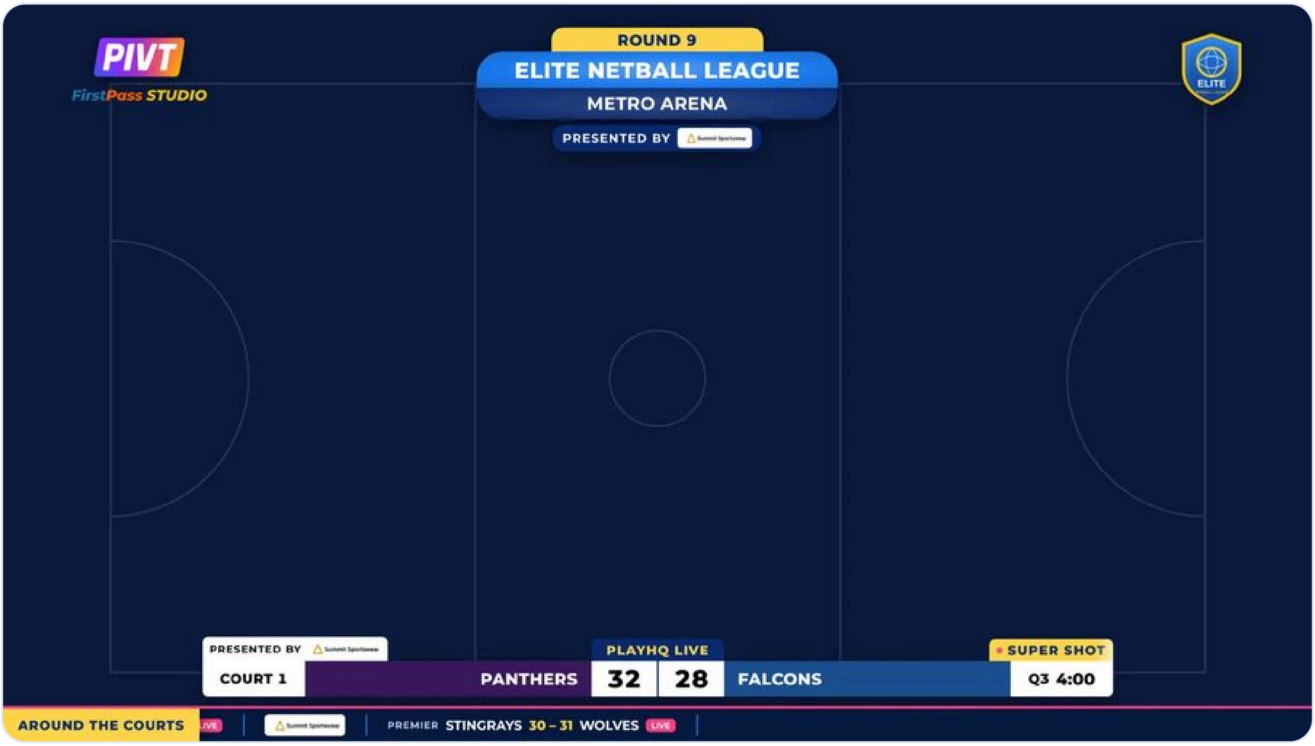
Version 1 · October 2026 · by PIVT

ABOUT THIS GUIDE

Graphics for the stream, from the first whistle to the last.

FirstPass Studio draws broadcast graphics for a netball livestream, fed straight from PlayHQ and sent to your switcher. This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

Screenshots use the fictional Elite Netball League.


PIVT
FirstPass STUDIO
ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY
ELITE
PRESENTED BY
COURT 1
PLAYHQ LIVE
PANTHERS
32
28
FALCONS
SUPER SHOT
Q3 4:00
AROUND THE COURTS
LIVE
PREMIER STINGRAYS 30 – 31 WOLVES
LIVE

CHAPTER 1 OF 11

What Studio is

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

IN THIS CHAPTER

1. The words used in this guide
2. The pages you work in
3. The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

The words used in this guide

Studio

The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air

Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page

One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source

A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene

Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)

A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence

A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition

Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

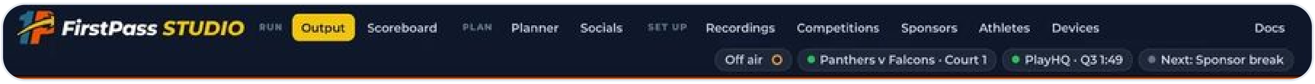
The data folder

Where Studio saves everything, outside the app, so updates never touch it.

• The pages you work in

Every Studio window has the same bar at the top, with the FirstPass Studio mark and the tabs in three groups by what you are doing. **Run** is the night itself: **Output**, where the show is built and sent out, and **Scoreboard control**, which runs the board. **Plan** is before the night: the **Production Planner** plans the court and the commentators, and **Socials** makes the night's pictures for social media. **Set up** is done once: **Recordings**, **Competitions setup**, **Sponsors**, **Athletes** and **Devices**. **Docs** sits on its own at the right: it is this guide, with the bar on top. In the app, ⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices and Docs.

The right of the bar carries the status chips, on every page, so you can see the night from wherever you are: **ON AIR** in orange like a tally light, with a dot per output that is solid in the output's colour while it sends and hollow while it doesn't (or **Off air**); the game on the board, its dot orange before the clock has started, green from the first clock start or score until an hour after the game ends, then grey; PlayHQ, whose dot is the link alone (green when PlayHQ answers, amber when a reading is overdue or reads **your score held** while PlayHQ is behind a hand score, red when it keeps failing, purple and **PlayHQ not live** when PlayHQ answers but the game isn't being scored live, grey when scores are by hand) and whose words give the quarter and clock, or **game completed** once PlayHQ closes the game (**full time** when the last quarter is run down by hand); the sequence playing with its step and the time left on it, or the triggered sequence counting down to start; **Run order ▶** with what Play next plays and ⏭ with what follows it (also what Skip leaves at the head), shown whatever is playing and gone when the run order is empty; **ATEM** while a switcher is set up (green when it answers, red when it doesn't), and **REC** while a game records (grey while Record games waits for one). The chips are status, not links; the tabs take you to a page.



The bar on every page: the tabs grouped as Run, Plan and Set up, and the chips for the night: off air, the game on the board, PlayHQ with the quarter and clock, and what the run order plays next.

The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.

CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

IN THIS CHAPTER

1 Install
2 On air and off air
3 The 1P menu
4 Before you go on air

Install

1. Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).
2. The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.
3. Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).
4. Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

On air and off air

The pages zoom like Safari's, for a small screen: **Zoom In** (⌘+), **Zoom Out** (⌘−) and **Actual Size** (⌘⌥0) in the View menu, which shows the level, or a pinch on the trackpad. Every Studio window zooms together and the level is kept on that Mac; while it isn't 100% a chip at the right of the bar shows it, and a click on the chip puts it back. The Docs tab, this guide, has its contents at the left on a wide window.

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The bar's chips show the game on the board, whether PlayHQ is answering (amber when overdue, red after a minute of errors, purple when the game isn't being scored live) and which outputs are on air, on every page; the Output page's status board shows the score and the rest of the night. If anything is wrong, the page you keep open says so.

The 1P menu

The 1P icon in the menu bar is Studio's home: it opens Studio, its pages and this guide, backs the settings up and restores them, and has **Open at login**, **Appearance** (the Studio pages follow the Mac's light or dark look unless pinned here; the graphics never change) and **Check for updates**. Chapter 10 lists every item. Updates download in the background and install when the app quits, never mid-stream.

LINKS

Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/` , so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

Before you go on air

1. **Open the app.** The Output page opens and Studio goes on air. Keep a window open all night.
2. **Right competition.** Competitions setup shows the one in use at the top. Check its PlayHQ link and the scoring window, and that the Production Planner has tonight's court.
3. **Right game.** On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.
4. **Scores arrive.** Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.
5. **Scene and output.** On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.
6. **Sound.** Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.
7. **Stream Deck.** Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.
8. **Sponsors.** Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

More in the Reference: **The Mac app**.

CHAPTER 3 OF 11

Game night, the automated way

This is how most nights run: one operator, PlayHQ scoring the game, a base scene that stays on, automatic layers that fire by themselves, and sequences that handle the run-up, the breaks and full time.

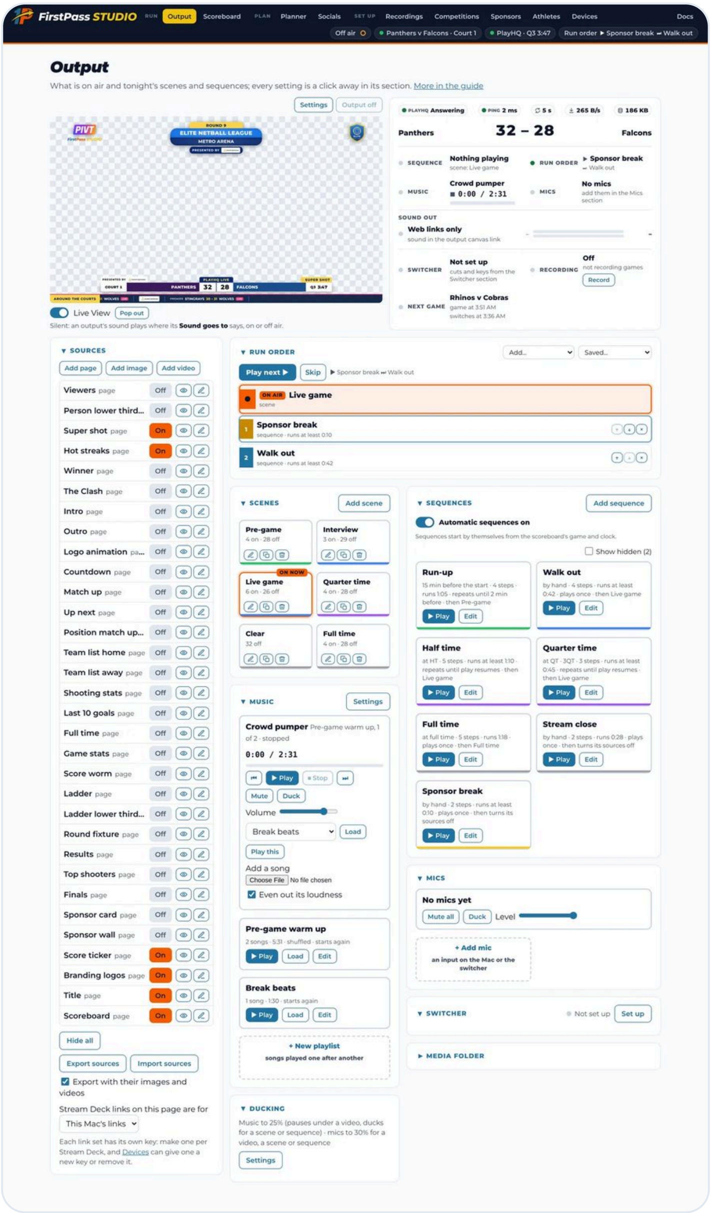
IN THIS CHAPTER

1 How it fits together
2 Scoreboard control
3 The Output page
4 The base scene
5 Automatic layers
6 Scenes
7 Sequences
8 Music, playlists and mics
9 Video output
10 A night, start to finish
11 The Production Planner
12 Socials
13 Recordings

How it fits together

Four things do the work, and each one is set up once:

- **Follow PlayHQ.** The scoreboard reads the live score, quarter and clock from the same feed the PlayHQ game page uses, every 5 seconds by default (as often as every second) while the game is on, from 2 minutes before it starts until 2 minutes after PlayHQ finishes it, while someone live-scores the game on PlayHQ. Either side of the game Studio is kinder to PlayHQ: every 30 seconds once electronic scoring opens (so a scorer who starts early is caught), every 5 minutes otherwise. Automatic game selection moves the board to each next game on your court as its electronic scoring starts.
- **The base scene.** A saved scene with the sources that stay on through the game: the scoreboard, the title banner, the branding logos and the score ticker.
- **Automatic layers.** Sources that stay on but only show when something happens: the super shot celebration, the hot streaks and game moments, the super shot tab on the scoreboard, the naming-rights sponsor's tab. They watch the board and play by themselves.
- **Sequences.** Steps played in order for the parts of the night that aren't live play: the run-up, the walk out, quarter and half time, full time and the close. A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.


FirstPass STUDIO
RUN
Output
Scoreboard
PLAN
Planner
Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 3:47
Run order ▶ Sponsor break ⏭ Walk out
Settings
Output off
PIVT
ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY
COURT 1
PANTHERS
PLAYHQ LIVE
32
28
FALCONS
SUPER SHOT
Q3 3:47
AROUND THE COURTS
WOLVES
STINGRAYS 30 - 31 WOLVES
Live View
Pop out

Output

What is on air and tonight's scenes and sequences; every setting is a click away in its section. More in the guide

Silent: an output's sound plays where its **Sound goes to** says, on or off air.

PLAYHQ Answering · PING 2 ms · 5 s · 265 B/s · 186 KB

Panthers 32 – 28 **Falcons**

SEQUENCE: **Nothing playing** scene: Live game

RUN ORDER: ▶ **Sponsor break** ⏭ Walk out

MUSIC: **Crowd pumper** ■ 0:00 / 2:31

MICS: **No mics** add them in the Mics section

SOUND OUT: **Web links only** sound in the output canvas link

SWITCHER: **Not set up** cuts and keys from the Switcher section

RECORDING: **Off** not recording games

NEXT GAME: **Rhinos v Cobras** game at 3:51 AM switches at 3:36 AM

▼ SOURCES

Add page
Add image
Add video

Viewers	page	Off
Person lower third...		Off
Super shot	page	On
Hot streaks	page	On
Winner	page	Off
The Clash	page	Off
Intro	page	Off
Outro	page	Off
Logo animation	pa...	Off
Countdown	page	Off
Match up	page	Off
Up next	page	Off
Position match up...		Off
Team list home	page	Off
Team list away	page	Off
Shooting stats	page	Off
Last 10 goals	page	Off
Full time	page	Off
Game stats	page	Off
Score worm	page	Off
Ladder	page	Off
Ladder lower third...		Off
Round fixture	page	Off
Results	page	Off
Top shooters	page	Off
Finals	page	Off
Sponsor card	page	Off
Sponsor wall	page	Off
Score ticker	page	On
Branding logos	page	On
Title	page	On
Scoreboard	page	On

Hide all
Export sources
Import sources

☑ Export with their images and videos

Stream Deck links on this page are for: This Mac's links

Each link set has its own key: make one per Stream Deck, and Devices can give one a new key or remove it.

▼ RUN ORDER

Add...
Saved...
Play next ▶
Skip

▶ Sponsor break ⏭ Walk out

ON AIR **Live game** scene

1 **Sponsor break** sequence · runs at least 0:10

2 **Walk out** sequence · runs at least 0:42

▼ SCENES

Pre-game 4 on · 28 off

Interview 3 on · 29 off

ON NOW **Live game** 6 on · 26 off

Quarter time 4 on · 28 off

Clear 32 off

Full time 4 on · 28 off

▼ MUSIC

Crowd pumper Pre-game warm up, 1 of 2 · stopped

0:00 / 2:31

▶ Play
■ Stop
Mute
Duck
Volume
Break beats
Load
Play this

Add a song: Choose File No file chosen

☑ Even out its loudness

Pre-game warm up 2 songs · 5:31 · shuffled · starts again

Break beats 1 song · 1:30 · starts again

▶ Play
Load
Edit

+ New playlist
songs played one after another

▼ DUCKING

Music to 25% (pauses under a video, ducks for a scene or sequence) · mics to 30% for a video, a scene or sequence

▼ SEQUENCES

Add sequence

Automatic sequences on

Sequences start by themselves from the scoreboard's game and clock.

☐ Show hidden (2)

Run-up 15 min before the start · 4 steps · runs 1:05 · repeats until 2 min before · then Pre-game

Walk out by hand · 4 steps · runs at least 0:42 · plays once · then Live game

Half time at HT · 5 steps · runs at least 1:10 · repeats until play resumes · then Live game

Quarter time at QT · 3QT · 3 steps · runs at least 0:45 · repeats until play resumes · then Live game

Full time at full time · 5 steps · runs 1:18 · plays once · then Full time

Stream close by hand · 2 steps · runs 0:28 · plays once · then turns its sources off

Sponsor break by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

▼ MICS

No mics yet

Mute all
Duck
Level

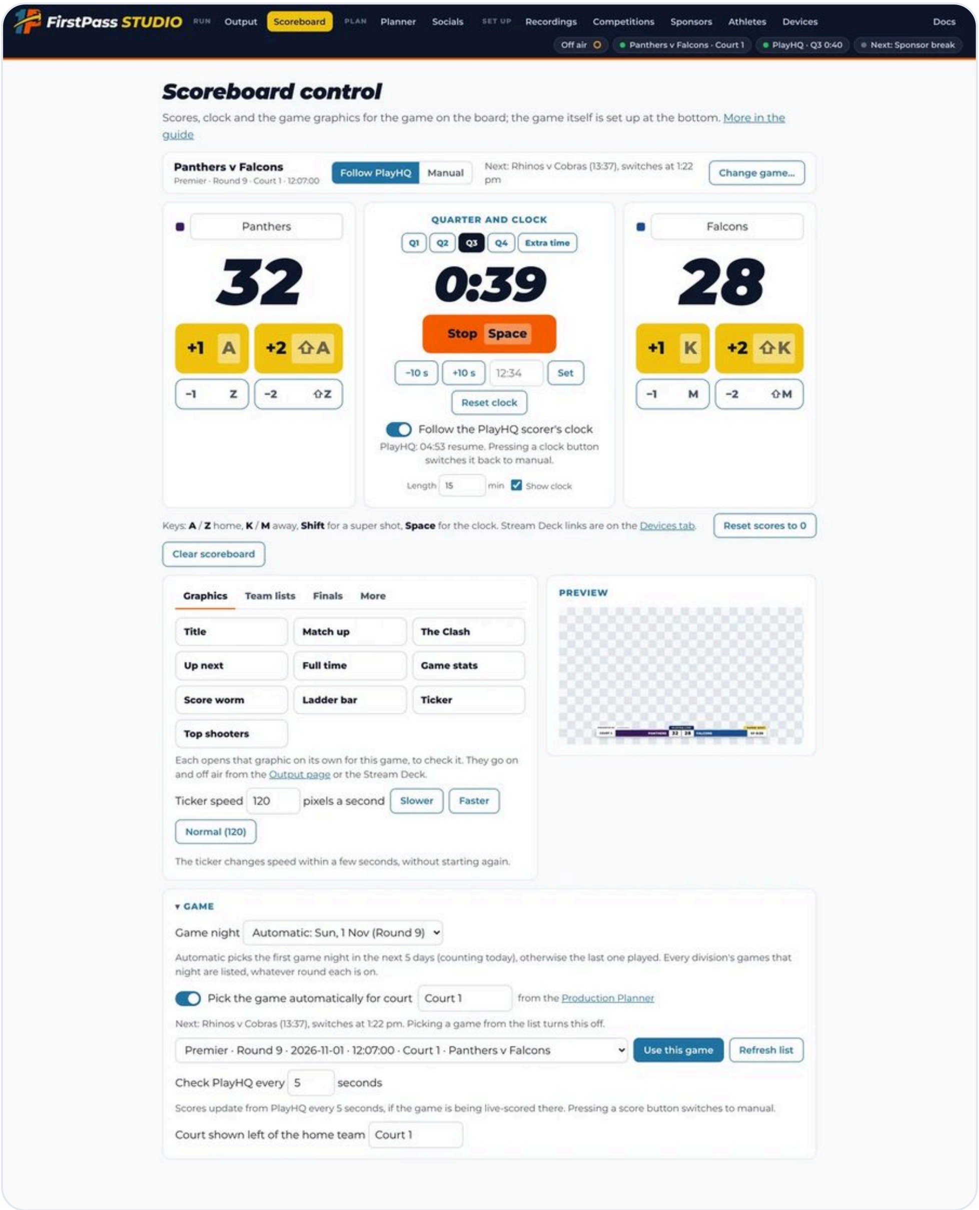
+ Add mic
an input on the Mac or the switcher

▼ SWITCHER: Not set up

▶ MEDIA FOLDER

The Output page is the whole show on one page: video output, music, sequences, scenes, the Scene View, and the sources with their settings.

- # Scoreboard control


FirstPass STUDIO
RUN
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PLAN
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Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 0:40
Next: Sponsor break
Follow PlayHQ
Manual
Change game…
QUARTER AND CLOCK
Q1
Q2
Q3
Q4
Extra time
+1 A
+2 ⇧A
−1 Z
−2 ⇧Z
Stop Space
−10 s
+10 s
Set
Reset clock
+1 K
+2 ⇧K
−1 M
−2 ⇧M
Reset scores to 0
Clear scoreboard
Graphics
Team lists
Finals
More
Title
Match up
The Clash
Up next
Full time
Game stats
Score worm
Ladder bar
Ticker
Top shooters
Slower
Faster
Normal (120)
PREVIEW
▾ GAME
Use this game
Refresh list

Scoreboard control

Scores, clock and the game graphics for the game on the board; the game itself is set up at the bottom. More in the guide

Panthers v Falcons
Premier · Round 9 · Court 1 · 12:07:00

Next: Rhinos v Cobras (13:37), switches at 1:22 pm

Panthers 32

0:39

Falcons 28

12:34

Follow the PlayHQ scorer's clock
PlayHQ: 04:53 resume. Pressing a clock button switches it back to manual.

Length 15 min ☑ Show clock

Keys: **A** / **Z** home, **K** / **M** away, **Shift** for a super shot, **Space** for the clock. Stream Deck links are on the Devices tab.

Each opens that graphic on its own for this game, to check it. They go on and off air from the Output page or the Stream Deck.

Ticker speed 120 pixels a second

The ticker changes speed within a few seconds, without starting again.

Game night Automatic: Sun, 1 Nov (Round 9)

Automatic picks the first game night in the next 5 days (counting today), otherwise the last one played. Every division's games that night are listed, whatever round each is on.

Pick the game automatically for court Court 1 from the Production Planner

Next: Rhinos v Cobras (13:37), switches at 1:22 pm. Picking a game from the list turns this off.

Premier · Round 9 · 2026-11-01 · 12:07:00 · Court 1 · Panthers v Falcons

Check PlayHQ every 5 seconds

Scores update from PlayHQ every 5 seconds, if the game is being live-scored there. Pressing a score button switches to manual.

Court shown left of the home team Court 1

Scoreboard control: the game in one bar, the two scores and the clock as the biggest things on the page with their keys printed on them, then the graphics, team lists and finals as tabs with a preview of the board.

The page is laid out for the person scoring. The bar at the top names the game on the board with its division, round, court and time, holds the **Follow PlayHQ** / **Manual** switch, and says which game the board will switch to next; **Change game…** opens the **Game** section at the bottom, where the game night, court, automatic selection and PlayHQ settings live (it folds to its title once the night is set). Under the bar are the two scores with their +1 and +2 pads, the quarter buttons and the clock with **Start** / **Stop**, and the keys printed on each button. Beneath them, tabs for the game graphics (each opens on its own, to check it), the team lists, the finals and the rest, with a preview of the board beside them.

ALWAYS A GAME ON

When Studio starts, it puts a game on the board with Follow PlayHQ on: the game on now on your court, otherwise the first game of tonight on that court. Games are picked by game night, not by round: the list holds every division's games that date, whatever round each is up to, so one division on Round 10 and another on Finals Round 1 sit side by side, each game showing its own round. Tonight is found by date (the first game night in the next 5 days, counting today), so a Tuesday night is found from the Friday before. You can pick another night from the **Game night** list, which names each night's rounds; its games on your court are then followed by their time (the one due now, otherwise the next to come). Set it back to Automatic afterwards.

AUTOMATIC GAME SELECTION

With **Pick the game automatically for court** ticked, the board shows the first game of the night on that court, then switches to each next game when its electronic scoring starts (**Electronic scoring starts N minutes before the game** on the setup page, 10 by default, 15 in the sample league), with the quarter back to Q1 and a full clock. In finals it keeps the final on until PlayHQ marks it finished, in case of extra time. Picking a game from the list by hand turns automatic selection off; tick it again to go back. The court comes from the Production Planner: when a game night arrives, its planned court is set here with **from the Production Planner** beside it, and automatic selection goes back on. A night with no plan uses the **Fallback court to follow** from Competitions setup (**fallback from Competitions setup**). Changing the court here, or picking a game by hand, lasts for that night only (**changed by hand for tonight**), even if Studio restarts; the Planner flags it so the commentators can be moved, and the next night goes back to the plan.

Court shown left of the home team is the court's name on the scoreboard and the other graphics. Left blank, it is PlayHQ's name for the game's court, shown in grey in the box, so a show court PlayHQ names (the Jerome Gillbard Centre, say) appears by its name. Type a name to show instead; it lasts for that night, even if Studio restarts, and clearing the box goes back to PlayHQ's. Following a court still uses PlayHQ's name, so a name typed here never changes which games are picked.

FOLLOW PLAYHQ

With **Follow PlayHQ** on, the score and quarter update every 5 seconds from PlayHQ's live scoring, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock. The clock matches the scores: switching to Manual, or pressing any score button, turns clock-following off as well, and the control page says so. Press a clock button to run the clock by hand while scores still follow PlayHQ. To line the clock up with the venue, type the venue's time in **Match venue clock**.

NOTHING BEFORE THE WINDOW

Studio doesn't ask PlayHQ's live feed for a game until its electronic scoring window opens, whether the game was picked automatically or by hand. The control page says when it will start checking.

THE NIGHT'S OTHER GAMES

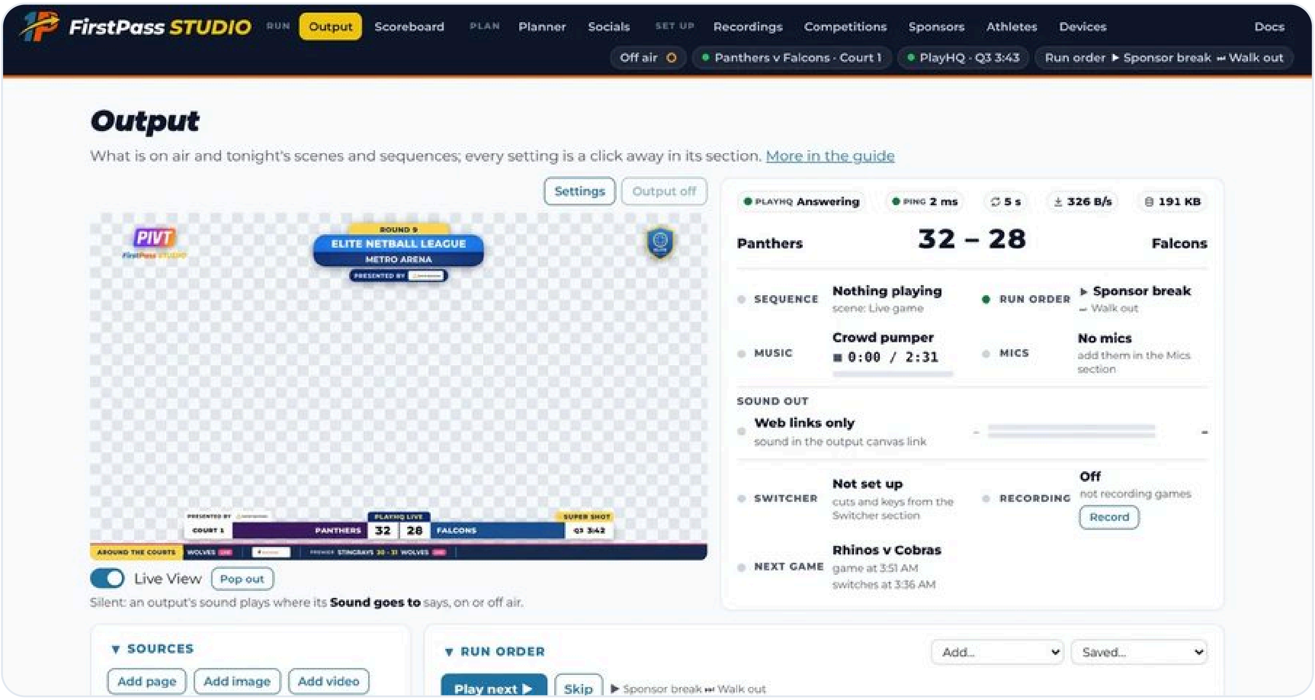
With **Also follow the night's other games, for their stats** on (it starts on), Studio follows every other game of the game night too, in the background, while it's on air: the same goals, quarters, runs and moments as the game on the scoreboard, from PlayHQ's live scoring, so each one gets its Game story, Score worm and Moments of the game on the Socials tab. Nothing of them goes on air: no super shot song, no moments, no warnings. Under the switch, Scoreboard control lists the games it's following and where each is up to. If you pick one of them for the scoreboard part way through, the scoreboard carries on from its goals; the game it leaves goes on being followed. Each game costs one more small PlayHQ request every 5 seconds while it's being played (every 30 seconds from when its scoring opens until 2 minutes before it starts), and a game PlayHQ doesn't answer for waits longer before it's asked again.

More in the Reference: **Scoreboard control in detail**.

The Output page

The Output page builds Studio's own scene: sources drawn as layers on one 1920 × 1080 frame, top layer first, and sends it to a display, a Blackmagic device or NDI for your switcher.

It is one page, laid out for the night. The picked output's picture sits at the top with its tabs directly above it, **HD** or **4K**, **Settings** and **Output on** in a row over them, and a status board beside it with a PlayHQ line above the score, always on one line (**PlayHQ** answering, overdue, failing, **Holding** while PlayHQ is behind your hand score, **Not live** in purple when the scorebench seems to be off line (hover for why), waiting while PlayHQ isn't scoring the game yet, or by hand; **Ping**, how long PlayHQ took to answer, green under half a second, amber to a second and a half, red slower or with no answer; ↻ and how often PlayHQ is checked, as set by **Check PlayHQ every** on Scoreboard control, grey while nothing is being checked; the download arrow, the **bandwidth** from PlayHQ a second over the last minute; the disc stack, the **data**, everything downloaded from PlayHQ since Studio went on air), then a line each for the sequence playing, the run order (▶ what Play next plays, ⏭ what follows it; or the sequence armed or starting by itself, counting down, in days when it's days away), the music (with why it's ducked: a scene or sequence, a stinger or by hand), the mics (**Heard**, **Silent** in amber with the mic's name when one has heard nothing for 10 seconds, **Not heard**, **Waiting**, **Muted** or **No mics**, with the reason under it and an amber line while they duck), then, across the board, **Sound out**: a row for each place each output's sound goes, as its **Sound goes to** says, under the output's name when there's more than one. Each row has a light (green while it's sending, amber while it starts or when the SDI's sound ran dry in the last minute, red with the reason when a destination fails, grey while the output is off), the device with the mics ticked for it under it, what it plays at (the SDI: PCM 16-bit, 48 kHz, stereo, 1.5 Mbps; one of the Mac's sound outputs: its sample rate and channels, as macOS doesn't say its bit depth), a meter for each side with a tick for the highest in the last 3 seconds (amber from −12 dB, red from −3 dB, as on the mics), and the louder side's level. The SDI's meter is the sound going into it; the others are what Studio plays to the device. **Web links only** means the output's sound is in the output canvas link; **No Audio**, in place of the meters, means the output's **Sound goes to** says **No sound**. Under it sit the switcher and the recording (with **Record** or **Stop** under it) side by side, and, across the board, the next game: its teams, when it starts and when the scoreboard switches to it. Under them the Sources run down the left, and to their right the run order, then two columns that line up all the way down: Scenes, Music and Ducking in one, the Sequences, Mics, the switcher and the media folder in the other, each section sitting right under the one above it. Every setting is a click away in its section: a source's settings open under its row, the pencil on a scene its name, colour and switcher, **Settings** on Music and Mics unfolds theirs, and **Set up** on the switcher its address and keys. A name that's too long for its space ends in …; hover over it to read it in full.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Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 3:43
Run order ▶ Sponsor break ⏭ Walk out

Output

What is on air and tonight's scenes and sequences; every setting is a click away in its section. More in the guide

Settings
Output off
PIVT
FirstPass STUDIO
ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY
COURT 1
PLAYHQ LIVE
PANTHERS
32
28
FALCONS
SUPER SHOT
Q3 3:42
AROUND THE COURTS
WOLVES
STINGRAYS
WOLVES
Live View
Pop out

Silent: an output's sound plays where its **Sound goes to** says, on or off air.

● PLAYHQ Answering ● PING 2 ms 5 s 326 B/s 191 KB

Panthers 32 – 28 Falcons

SEQUENCE: Nothing playing — scene: Live game

RUN ORDER: ▶ Sponsor break — Walk out

MUSIC: Crowd pumper — ■ 0:00 / 2:31

MICS: No mics — add them in the Mics section

SOUND OUT

Web links only — sound in the output canvas link

SWITCHER: Not set up — cuts and keys from the Switcher section

RECORDING: Off — not recording games

NEXT GAME: Rhinos v Cobras — game at 3:51 AM — switches at 3:36 AM

Record
▼ SOURCES
Add page
Add image
Add video
▼ RUN ORDER
Add...
Saved...
Play next ▶
Skip
▶ Sponsor break ⏭ Walk out

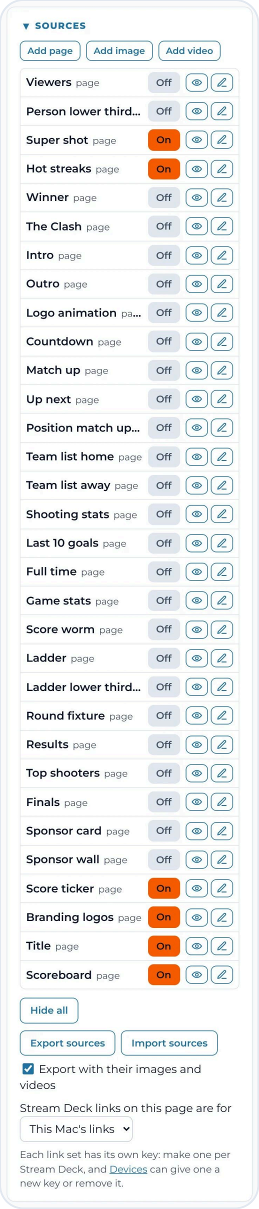
The top of the Output page in the third quarter: the picture with Settings and Output on over it, and the status board beside it with the PlayHQ line (answering, ping, how often it's checked, bandwidth, data) over the score, the run order, the music, the mics, Sound out (Web links only here, as no output is on), the switcher, the recording and the next game.

Every section folds away with the arrow beside its title (or a click on the title), so the ones you don't use, say Music or Switcher, take one line. Studio remembers which are folded on that Mac, even after it restarts, and buttons in a folded section's title row, like

Output on, still work. The page carries no explanations; this guide has them, and **More in the guide** under each page's title opens its part.

SOURCES

Add page adds any Studio graphic (or any web address), **Add image** a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV. A new source goes in hidden with its settings open under its row, ready to change. Each row has its On / Off, an eye to preview it on its own and a pencil for its settings. The list is top layer first: drag a source up or down to move it between layers, or use **Move up** and **Move down** in its settings. **Add all unused pages** puts every Studio graphic in the scene at once, hidden.



Viewers page	Off
Person lower third…	Off
Super shot page	On
Hot streaks page	On
Winner page	Off
The Clash page	Off
Intro page	Off
Outro page	Off
Logo animation pa…	Off
Countdown page	Off
Match up page	Off
Up next page	Off
Position match up…	Off
Team list home page	Off
Team list away page	Off
Shooting stats page	Off
Last 10 goals page	Off
Full time page	Off
Game stats page	Off
Score worm page	Off
Ladder page	Off
Ladder lower third…	Off
Round fixture page	Off
Results page	Off
Top shooters page	Off
Finals page	Off
Sponsor card page	Off
Sponsor wall page	Off
Score ticker page	On
Branding logos page	On
Title page	On
Scoreboard page	On

☑ Export with their images and videos

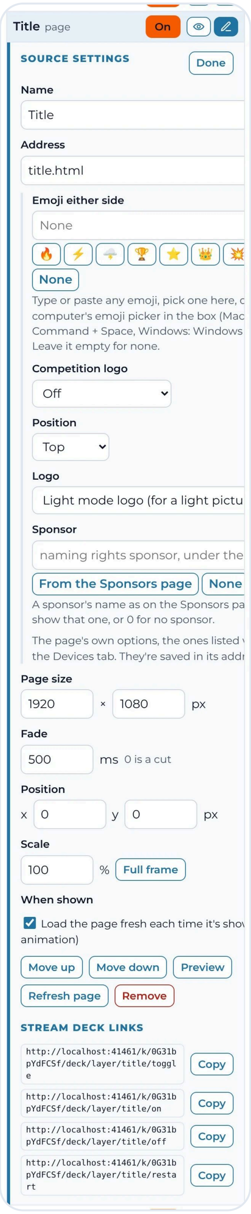
Stream Deck links on this page are for
This Mac's links

Each link set has its own key: make one per Stream Deck, and Devices can give one a new key or remove it.

The Sources list, top layer first. The orange On buttons are the base scene and the automatic layers; everything else waits for a scene, a sequence or a key.

SOURCE SETTINGS

Click a source, or its pencil, and its settings open under its row; **Done** closes them. A graphic's link options (position, heading, division, round, animate away and so on) show as fields under its address, so there is nothing to look up. **On / Off** fades it in or out over its own **Fade** time (0 is a cut). **Position and scale** place it on the frame. **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays; untick it to keep a page running so it shows instantly. **Preview** (and the eye in its row) opens the source on its own, half size, on a checkerboard.



SOURCE SETTINGS

Name: Title

Address: title.html

Emoji either side: None

Type or paste any emoji, pick one here, c computer's emoji picker in the box (Mac Command + Space, Windows: Windows Leave it empty for none.

Competition logo: Off

Position: Top

Logo: Light mode logo (for a light pictu

Sponsor: naming rights sponsor, under the

A sponsor's name as on the Sponsors pa show that one, or 0 for no sponsor.

The page's own options, the ones listed the Devices tab. They're saved in its addr

Page size: 1920 × 1080 px

Fade: 500 ms 0 is a cut

Position: x 0 y 0 px

Scale: 100 %

When shown

☑ Load the page fresh each time it's sho animation)

STREAM DECK LINKS

```
http://localhost:41461/k/0G31bpYdFCSf/deck/layer/title/toggle
http://localhost:41461/k/0G31bpYdFCSf/deck/layer/title/on
http://localhost:41461/k/0G31bpYdFCSf/deck/layer/title/off
http://localhost:41461/k/0G31bpYdFCSf/deck/layer/title/restart
```

Source settings for the title banner: its options are fields, and the Stream Deck links for this source are copied from here.

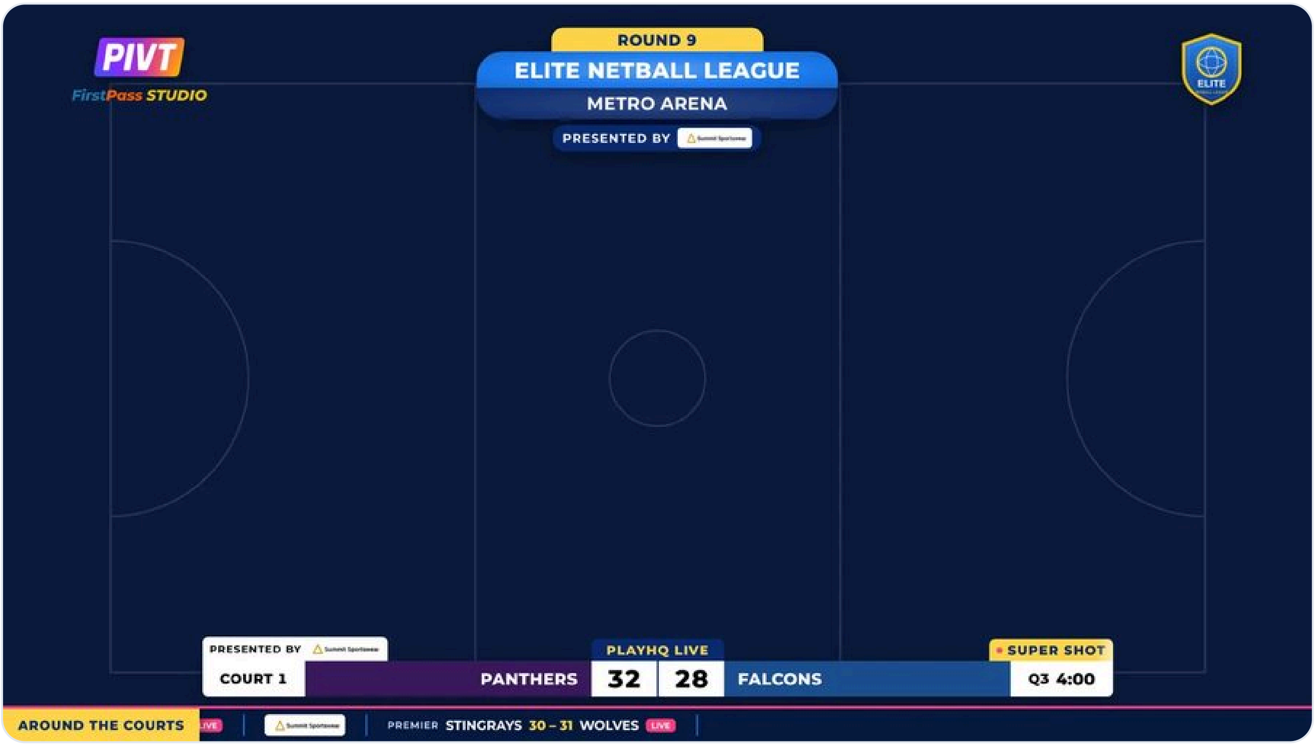
Videos play from the beginning each time they're shown, can loop, and fade out when they finish unless looping; a video that fades out switches its source off, so its key goes dark and the next On plays it again. The Scene View only watches: opened or switched to an output part way through a video, it joins where that output has got to, and never starts the video again on the output. A ProRes 4444 `.mov` is converted to a transparent

WebM when it is added, keeping its alpha. **Export sources** saves the whole scene with its media as one zip to keep or move to another Mac; **Import sources** puts one back.

More in the Reference: **The Output page in detail**.

The base scene

The base scene is what viewers see for most of the game. In the sample league it is the **Live game** scene: the scoreboard at the bottom, the title banner at the top with the PIVT and FirstPass Studio logos on the left and the competition logo on the right, and the score ticker along the bottom edge with the other courts' scores.


PIVT
FirstPass STUDIO
ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY
ELITE
PRESENTED BY
COURT 1
PLAYHQ LIVE
PANTHERS
32
28
FALCONS
SUPER SHOT
Q3 4:00
AROUND THE COURTS
LIVE
PREMIER STINGRAYS 30 – 31 WOLVES
LIVE

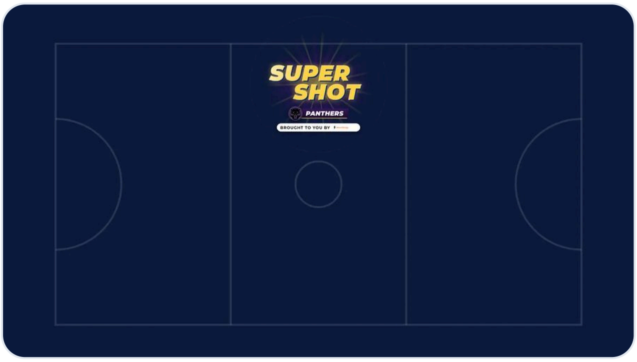
The Live game scene. The court behind it stands in for your vision; the switcher keys the graphics over the cameras.

- **Scoreboard.** Court in a pink box on the left, both teams in their club colours, the score at screen centre, quarter and clock on the right. "PlayHQ live" shows while scores follow PlayHQ; a "Presented by" tab carries the naming-rights sponsor.
- **Title.** Round, competition and venue, with the naming-rights sponsor's pill under it.
- **Branding logos.** PIVT and FirstPass Studio left of the title, the competition logo right of it, so other graphics can be turned on and off around them.
- **Score ticker.** Scrolling scores from tonight's other games, led by a Partner-and-up sponsor's tagline. Its speed is set on the control page or from the Stream Deck.

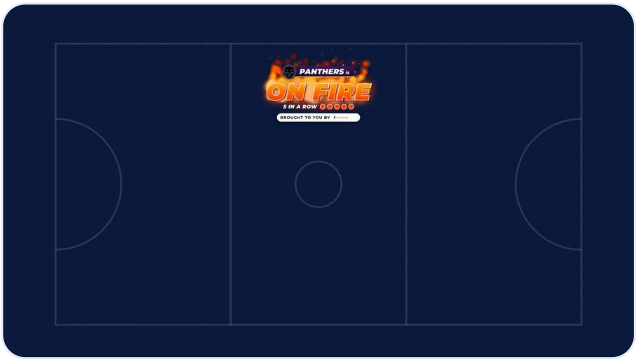
Build it once: turn those sources on, then press **Save what's on now** under Scenes and name it. From then on, one click (or one Stream Deck key) brings the show back to this look.

Automatic layers

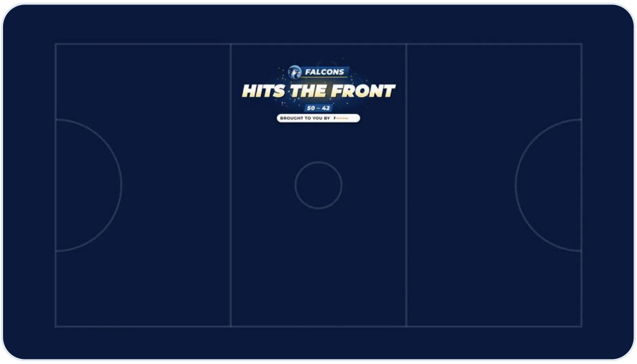
These sources are left on in the base scene and play by themselves from the scoreboard. They add no PlayHQ requests of their own; they ask Studio, which already has the score.


SUPER SHOT
PANTHERS
BROUGHT TO YOU BY

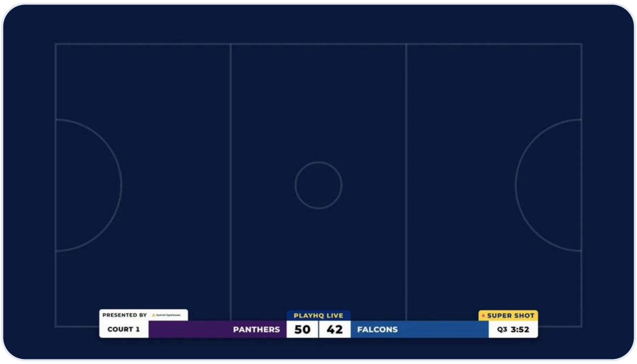
Super shot. A 3-second burst with the team's logo and name in their club colour, played for each successful super shot, from PlayHQ's super shot count or a +2 press. It plays small, centred just under the title.


PANTHERS is
ON FIRE
5 IN A ROW
BROUGHT TO YOU BY

Hot streaks. "Heating up" at 3 goals in a row, "On fire" at 5, "Goal for goal" when 10 in a row go one each way, from the goal order Studio works out from the score.


FALCONS
HITS THE FRONT
50 - 42
BROUGHT TO YOU BY

Game moments on the same page: hits the front, all square (level in the last 2 minutes), coming back, on the siren, back to back super shots. One goal plays one moment at most; the Commentator stats page lists them all.

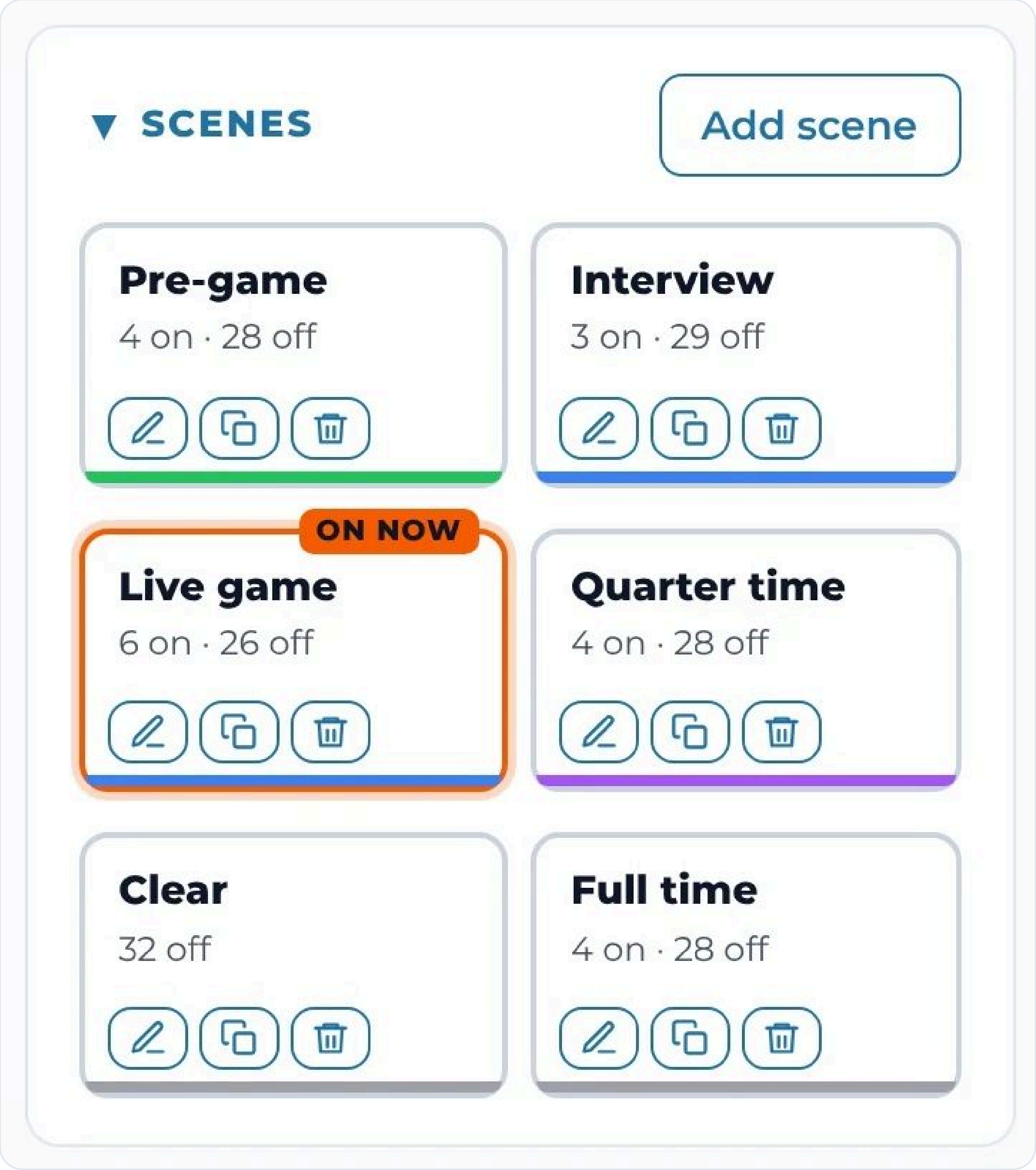

PRESENTED BY
COURT 1
PANTHERS
PLAYHQ LIVE
50
42
FALCONS
SUPER SHOT
Q3 3:52

Super shot tab. In a super shot division, a yellow tab rises above the clock for the last minutes of each quarter (set per competition) and sinks away at the siren.

Each celebration can carry a sponsor's "Brought to you by" tag (Chapter 7). A division without super shots plays no super shot animation and counts no super goals; that is set on the competition setup page, per division if needed.

Scenes

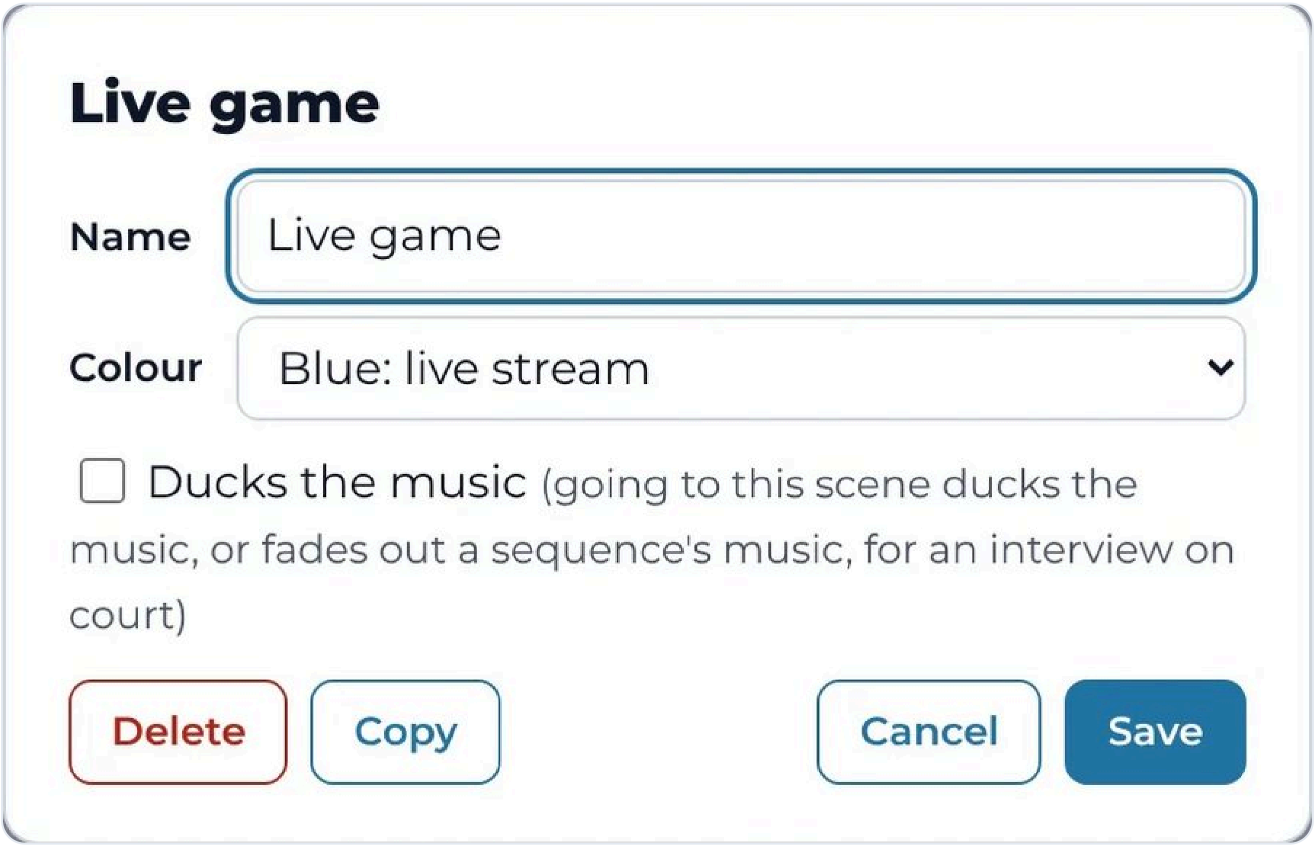
A scene is a saved set of sources on and off. Clicking a scene goes to it, turning sources on and off to match while anything already on stays on, not reloaded. Scenes are listed by colour in the order of a stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads.



▼ SCENES
Add scene
Pre-game
4 on · 28 off
Interview
3 on · 29 off
ON NOW
Live game
6 on · 26 off
Quarter time
4 on · 28 off
Clear
32 off
Full time
4 on · 28 off

The sample league's scenes. Pre-game, Live game, Quarter time, Full time, a quiet Interview scene for an on-court chat, and Clear.

Turn a source on or off by hand after going to a scene and the scene is marked **changed**; the bar under the scenes offers **Back to saved**, **Save over** it, or **Save as new scene**. Each scene tile has three buttons along its foot: the **pencil** opens its settings (name, colour, **Ducks the music** and its switcher actions), **copy** makes a copy beside it, and the **bin** deletes it after asking. None of them changes what's on. A source added later is off in every scene.



Live game

Name: Live game

Colour: Blue: live stream

☐ Ducks the music (going to this scene ducks the music, or fades out a sequence's music, for an interview on court)

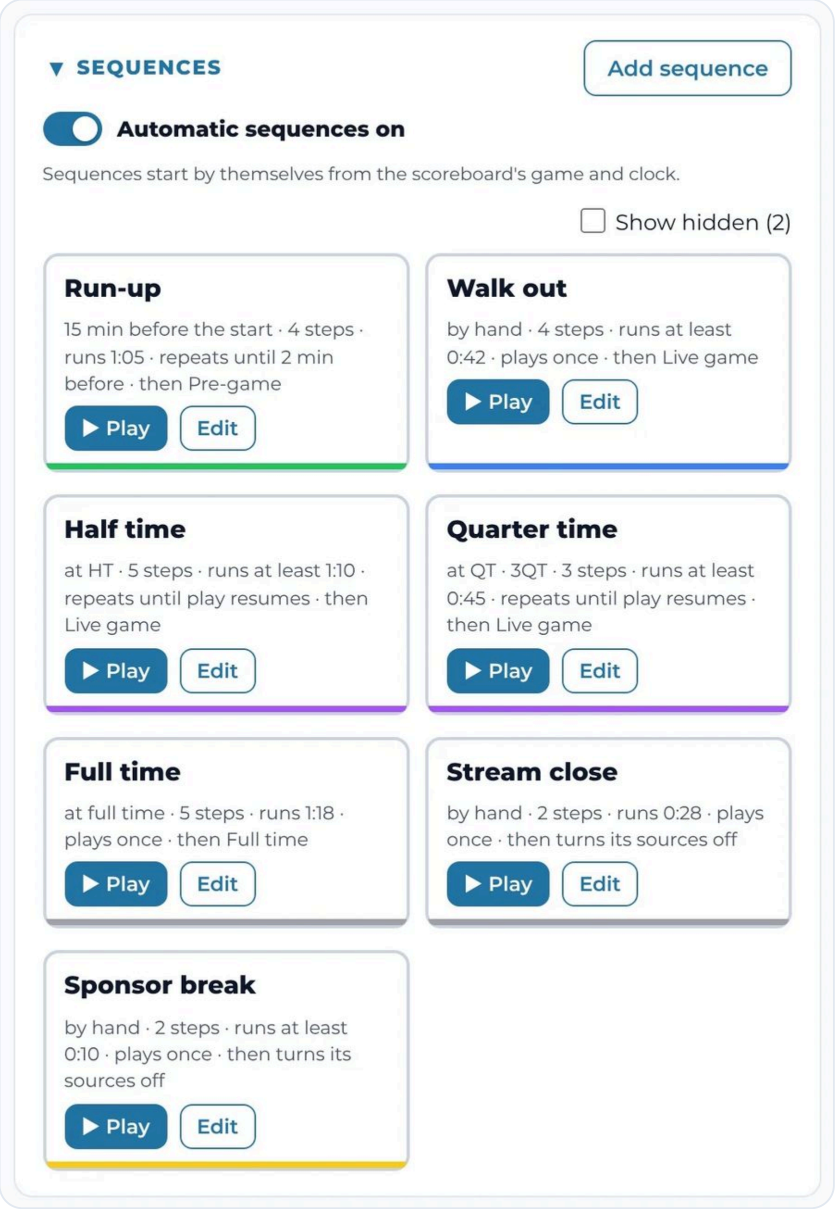
The pencil on a scene: its name, colour group, Ducks the music and switcher actions. Delete and Copy are here too.

SCENES THAT DUCK THE MUSIC

Tick **Ducks the music** on a scene (or a sequence) for an interview on court: whatever music is playing ducks and keeps playing until the next scene or sequence without the tick.

Sequences

A sequence plays the scene's sources one after another. Each step turns on the sources you pick and stays for a set time, until it finishes (its videos end or its animation animates away), until the game moves on (the clock starts, or the next break or full time), or until you press Next. As a step ends, sources the next step doesn't use go off with their own fades.



● Automatic sequences on

Sequences start by themselves from the scoreboard's game and clock.

☐ Show hidden (2)

Run-up
15 min before the start · 4 steps · runs 1:05 · repeats until 2 min before · then Pre-game

Walk out
by hand · 4 steps · runs at least 0:42 · plays once · then Live game

Half time
at HT · 5 steps · runs at least 1:10 · repeats until play resumes · then Live game

Quarter time
at QT · 3QT · 3 steps · runs at least 0:45 · repeats until play resumes · then Live game

Full time
at full time · 5 steps · runs 1:18 · plays once · then Full time

Stream close
by hand · 2 steps · runs 0:28 · plays once · then turns its sources off

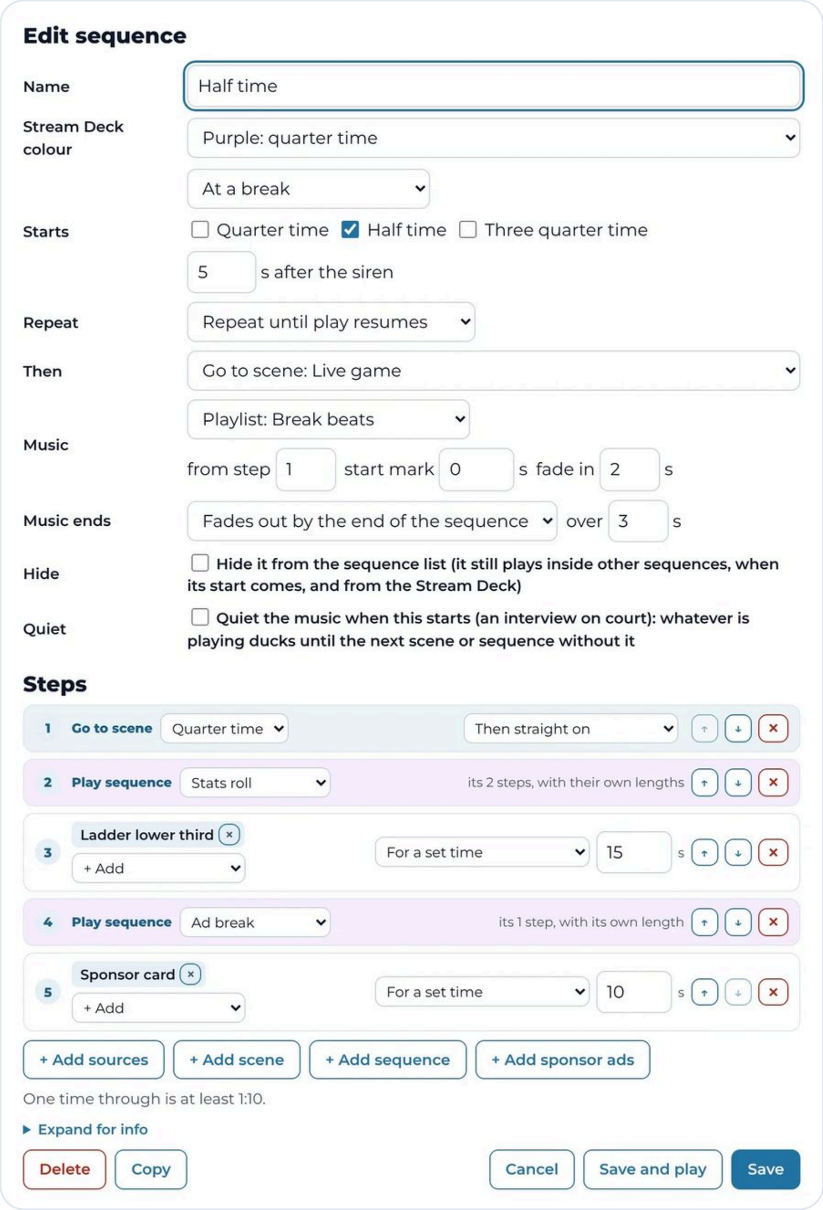
Sponsor break
by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

The sample league's sequences. Each card says when it starts, how many steps it has, how long it runs, whether it repeats, and what it does when it ends.

THREE KINDS OF STEP

- **Sources** (**+ Add sources**): one or more sources on together, for a length.
- **A scene** (**+ Add scene**): a starting point or a reset point. What the scene turns on stays on under the next steps and after the sequence. Start a break sequence with the Live game scene and the scoreboard stays up under the stats.
- **Another sequence** (**+ Add sequence**): its steps play in place. Build a mini sequence once (the sample league's **Break stats** and **Sponsor break**) and use it inside Quarter time, Half time and Full time. A sequence made only of sequences is marked **Master**. Tick **Hide** on a mini sequence to keep it off the list; it still plays inside the others and from the Stream Deck.

+ Add sponsor ads adds a step that plays the next one to five sponsor ads in rotation (Chapter 7).



Edit sequence

Name: Half time

Stream Deck colour: Purple: quarter time

Starts: At a break
☐ Quarter time ☑ Half time ☐ Three quarter time
5 s after the siren

Repeat: Repeat until play resumes

Then: Go to scene: Live game

Music: Playlist: Break beats
from step 1 start mark 0 s fade in 2 s

Music ends: Fades out by the end of the sequence over 3 s

Hide: ☐ Hide it from the sequence list (it still plays inside other sequences, when its start comes, and from the Stream Deck)

Quiet: ☐ Quiet the music when this starts (an interview on court): whatever is playing ducks until the next scene or sequence without it

Steps

1 **Go to scene** Quarter time — Then straight on

2 **Play sequence** Stats roll — its 2 steps, with their own lengths

3 Ladder lower third — For a set time 15 s

4 **Play sequence** Ad break — its 1 step, with its own length

5 Sponsor card — For a set time 10 s

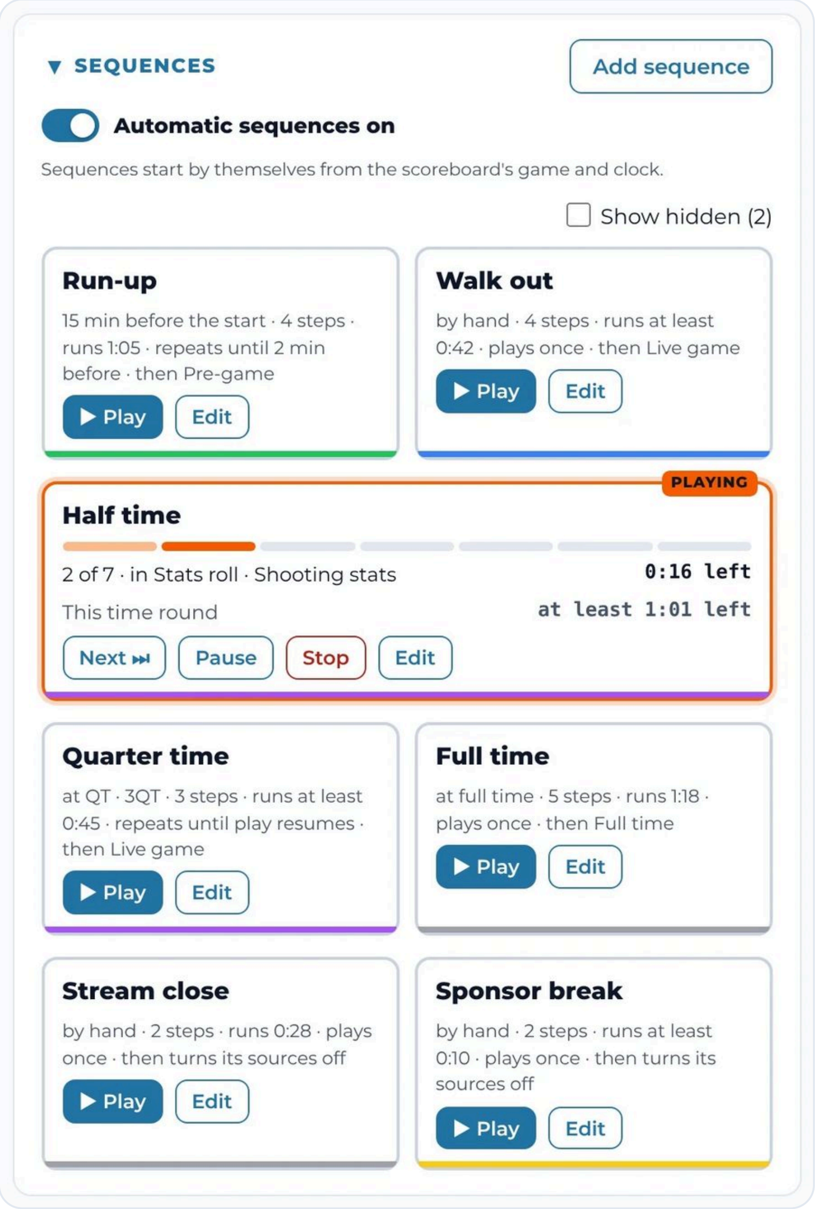
One time through is at least 1:10.

The Half time sequence: starts by itself at half time, repeats until play resumes, plays Break beats, and goes back to the Live game scene when it ends. Its steps are a scene step, two mini sequences and two source steps.

WHEN A SEQUENCE STARTS, REPEATS AND ENDS

SETTING	CHOICES	USED FOR
Starts	By hand · N minutes before the picked game's start (only if Studio was running then; opened later, play it by hand) · at the breaks you tick (QT, HT, 3QT) · at full time, a few seconds after the siren	Run-up 15 minutes out; the breaks and full time from the scoreboard's clock (the PlayHQ scorer's or yours).
Repeat	Play once · until stopped · until play resumes · until N minutes before the start	A break sequence loops until the clock starts again, finishing the step on screen.
Then	Go to a scene · turn off what it turned on · leave everything · play another sequence (one for each output it plays on)	Breaks go back to Live game; Run-up hands over to Walk out.
Music	None · a song or playlist from a step, from a start mark, with a fade in; ends by fading, cutting or carrying on into the next sequence	Warm-up music through the run-up that carries into the walk out.

A sequence that starts by itself waits a few seconds as **Starting**, with **Start now** and **Cancel**: press **Cancel** to hold the break back for a shot after the siren and start it by hand when the ball is dead, or **Stop** (or a scene) to come back from a break early. **Automatic sequences on** above the cards is the one switch for all of that, and it is remembered. Only one sequence plays at a time on each output; going to a scene, playing another sequence or hiding everything stops it without its Then. A sequence only ever turns off sources it turned on itself.



Automatic sequences on

Sequences start by themselves from the scoreboard's game and clock.

☐ Show hidden (2)

Run-up
15 min before the start · 4 steps · runs 1:05 · repeats until 2 min before · then Pre-game

Walk out
by hand · 4 steps · runs at least 0:42 · plays once · then Live game

Half time
2 of 7 · in Stats roll · Shooting stats `0:16 left`
This time round `at least 1:01 left`

Quarter time
at QT · 3QT · 3 steps · runs at least 0:45 · repeats until play resumes · then Live game

Full time
at full time · 5 steps · runs 1:18 · plays once · then Full time

Stream close
by hand · 2 steps · runs 0:28 · plays once · then turns its sources off

Sponsor break
by hand · 2 steps · runs at least 0:10 · plays once · then turns its sources off

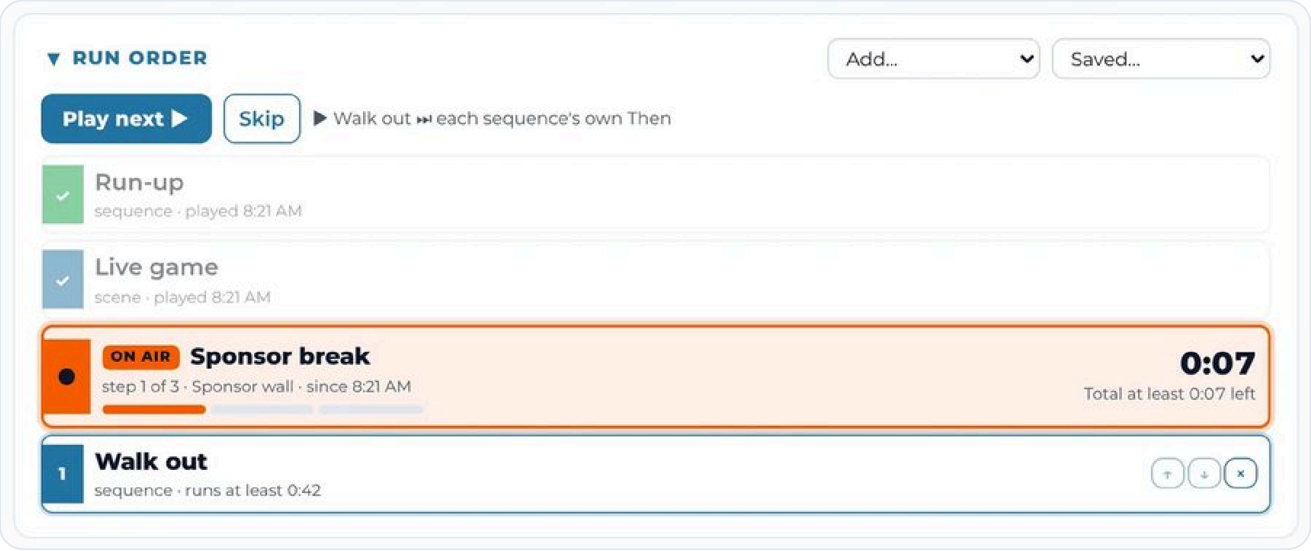
While a sequence plays its card shows a dot per step, the time left on the step and for the whole sequence, with Next, Pause and Stop.

Sequences belong to the competition in use and change with **Use this one**; the sources and scenes are shared. The break lengths on the setup page (quarter time, half time, three quarter time) let the editor say how many times a break sequence fits in each break, and time Commentator stats' **Back in** countdown.

THE RUN ORDER

Above Scenes and Sequences sits the **Run order**: tonight's plan as a numbered list of sequences and scenes, in the order they play. It reads like a playlist: the last two things it has played sit at the top, greyed with a tick and the time they played (any before them fold behind **played earlier tonight**), then what's on air now in orange with **ON AIR**: the sequence playing, with its step, a dot per step, the step's time left in large figures and the whole sequence's under it, or the scene that's on when no sequence plays. Under that come the numbered rows still to play. **Add…** puts a scene or sequence at the end (Run-up, Walk out, Live, Quarter time, Live, Half time…), rows drag or move with their arrows, and × takes one out. **Play next** plays the head now, a sequence or a scene, and **Skip** drops it without playing it (stepping through a sequence is the Sequences panel's own Next). A sequence that ends looks at the run order first and plays what is queued after it; its own **Then** runs only when nothing is queued, so Then is the fallback and a sequence built for one night still works alone on another. Whatever is at the head leaves the list as it starts, however it was started, and moves up to the played rows (a skipped or removed row never played, so it doesn't). On Output 1, a sequence that starts by itself from the game is put at the head with its few seconds' countdown rather than fired blind: **Cancel** takes it out again and **Play next** plays it when the ball is dead. Playing a scene or sequence by hand leaves the list alone unless it was the head, and nothing in the list starts by itself. The status board's **Run order** line and the bar's **Run order** chip say what Play next will do: ▶ the head, ⏭ what follows it. The note under **Play next** says the same. The run order is Output 1's, the production feed, so it shows Output 1's list whichever output tab is picked; the other outputs have none, and a sequence that starts by itself there just starts. A sequence that plays on more than one output says so on its row, "also on Commentary · then Commentary loop": **Play next** starts it on all of them, as its card does, and after it each other output does its own **Then**. When the run order moves on by itself, another output joins the next sequence only if nothing is playing there; one that was playing the sequence that just ended does its own Then.

Saved… keeps tonight's order under a name for next week, loads one in place of what is queued, renames or deletes one, or clears the list (the played rows too; what's on air carries on). A game in the Production Planner can have a saved order that loads as the game comes up on the scoreboard. The played rows are for tonight: a relaunch starts them again. The run order and the saved orders live with the competition's sequences, so a relaunch picks the night up where it was. On the Stream Deck, the plugin's **Run order** key plays what's next, skips it, loads a saved order or clears the list, and names what plays next so you can see what a press will do; without the plugin, `/deck/runorder/next` plays what's next and `/deck/runorder/skip` drops it. The Sequence key's own Next still ends the step on screen.



▶ Walk out ⏭ each sequence's own Then

✓ Run-up
sequence · played 8:21 AM

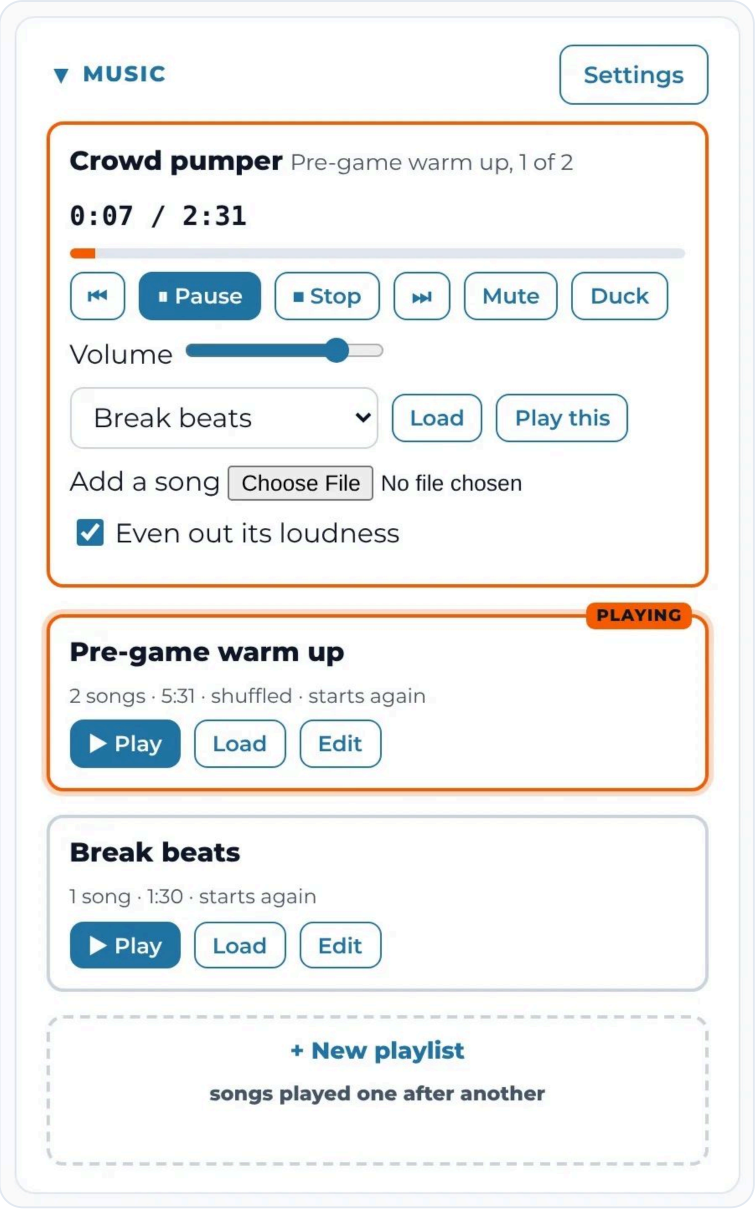
✓ Live game
scene · played 8:21 AM

● ON AIR **Sponsor break**
step 1 of 3 · Sponsor wall · since 8:21 AM
0:07
Total at least 0:07 left

1 **Walk out**
sequence · runs at least 0:42

Partway through the night: Run-up and the Live game scene have played, the Sponsor break is on air with its time left, and Walk out is framed in blue at the head, waiting for Play next.

Music, playlists and mics

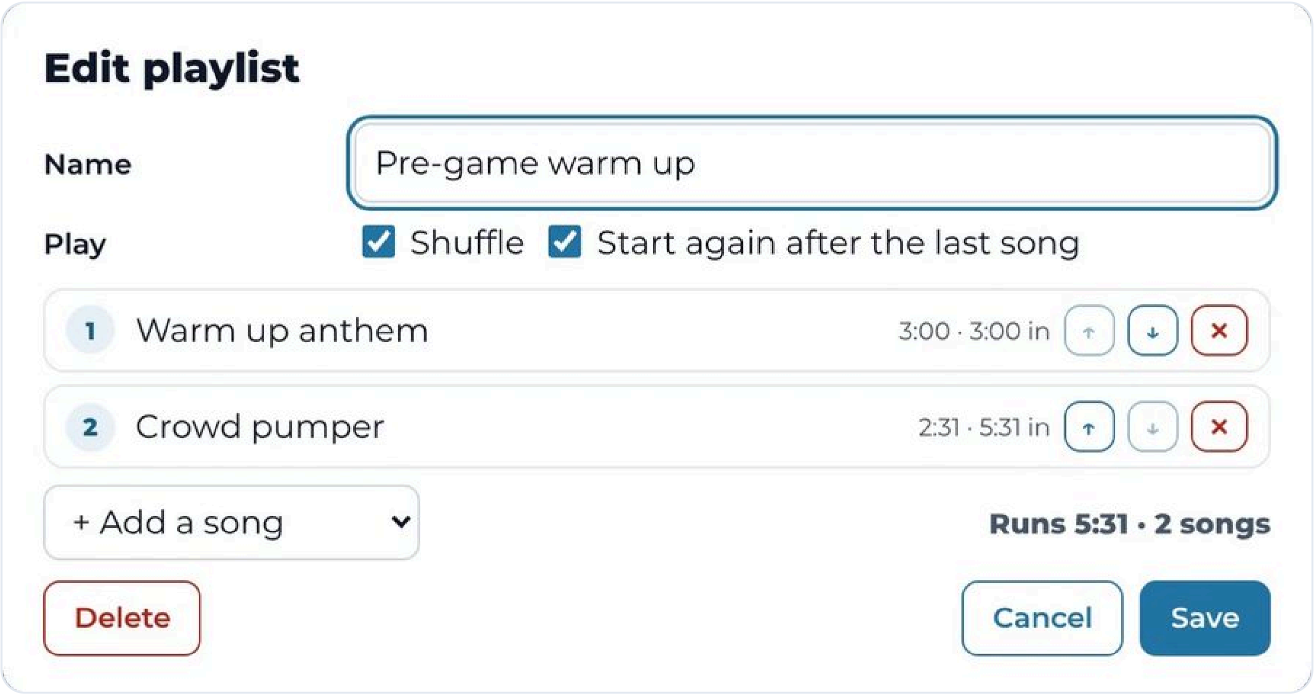

▼ MUSIC
Settings
Crowd pumper Pre-game warm up, 1 of 2
0:07 / 2:31
⏮
■ Pause
■ Stop
⏭
Mute
Duck
Volume
Break beats
Load
Play this
Add a song
Choose File
No file chosen
☑ Even out its loudness
PLAYING
Pre-game warm up
2 songs · 5:31 · shuffled · starts again
▶ Play
Load
Edit
Break beats
1 song · 1:30 · starts again
▶ Play
Load
Edit
+ New playlist
songs played one after another

The Music player and two playlists. Its Settings hold the three fade times (Fade in, Fade out and Quick) and the super shot song.

The Music player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream), one song or a playlist, shuffled or not. Add a song with **Even out its loudness** ticked and it is levelled to match the others. Music plays where Output 1's **Sound goes to** says, never in the Scene View.

- **Play** fades in; **Pause** and **Stop** fade out (slower by default). **Duck** lowers it and keeps it running; **Mute** keeps it running silently.
- While a video with its sound on plays, the music fades out and pauses, then fades back in from where it was, both over the **Quick** time (untick it under **Ducking** to leave the music playing). A themed intro or outro does the same. A video without its Sound ticked leaves the music alone.

- A sequence can bring its own music, and a sequence inside it plays its own while it runs. Going to a scene or another sequence fades a sequence's music out over its Music ends time; going to a sequence with music of its own, the new music fades in as the old fades out.
- The **super shot song** cuts in when the super shot period starts in a super shot division, loops, and fades out when the period ends.



Edit playlist

Name: Pre-game warm up

Play: ☑ Shuffle ☑ Start again after the last song

1 Warm up anthem 3:00 · 3:00 in

2 Crowd pumper 2:31 · 5:31 in

Runs 5:31 · 2 songs

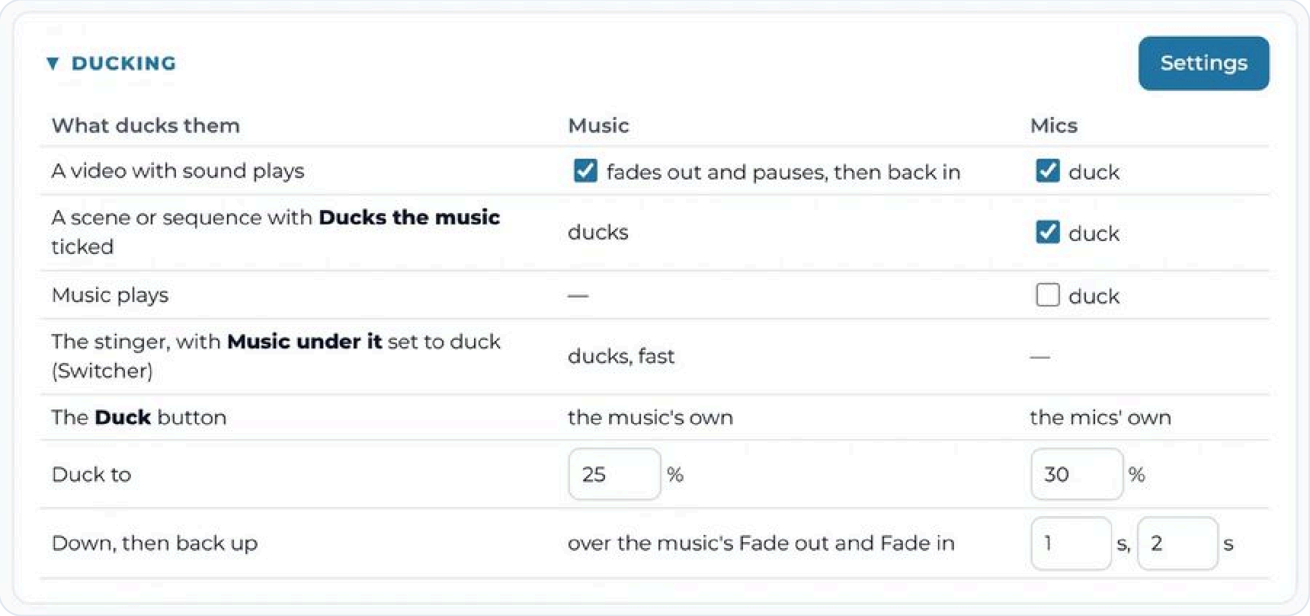
A playlist: songs in order, shuffle, and whether it starts again after the last.

Mics. Wireless effects mics on court and crowd mics go out with the music: plug their receivers into the Mac (directly or through an audio interface) and add each one under **Mics**, with its input and channel (two receivers on one stereo input are two mics, left and right). The mics are Output 1's, like the music: heard while Output 1 is on, never in the Scene View, so a speaker near a mic can't feed back. The first time, the Mac asks whether FirstPass Studio may use the microphone.

- A mic can be plugged into the switcher instead (an ATEM Mini's Mic 1 or Mic 2): the switcher mixes it and Studio moves its fader for the same level, mutes and ducks. Move the fader on the switcher and that becomes the mic's level; Studio never moves it back. Its meter shows the sound coming into the switcher's input, with a mark where its fader is.
- Each mic has a level, a mute, a **Low cut** for wind and rumble and a meter; **Level** and **Mute all** cover every mic, and a limiter stops a shout clipping the stream.
- A mic that hears nothing for 10 seconds while it's listened to (a flat battery, a receiver turned off, a pulled cable) says **Silent for 12 s · check its receiver and battery** in amber on its card, and the status board's Mics line says **Silent**. It clears as soon as sound comes in. A muted mic is never silent, and nor is one nobody is listening to (Output 1 off, or a switcher that sends no levels).

- **A phone as a backup commentator mic.** When a commentator's headset fails, their phone can stand in. Press **+ Phone mic** in Mics and scan the QR code with the phone, on the same Wi-Fi; the first time, the phone asks the Mac (either Allow works). On the phone's **Commentator mic** page, pick the mic and press **Start**: it comes in as a mic named after the phone, marked **phone (backup)**, with the same level, mute, low cut, ducking and Silent warning as the others, mixed into Output 1. A **wired or USB-C** mic plugged into the phone sounds best; a Bluetooth headset's mic sounds like a phone call and arrives about a fifth of a second late. The page keeps the screen awake, says **On air** while Studio hears it (or **Muted in Studio**), and if the connection drops it beeps in the commentator's ears and starts again by itself, while the mic's card and the status board's Mics line say **Dropped**. **Hear Studio** on the phone plays Output 1's sound back in the commentator's headphones: the music, videos and page sounds and the other mics (their co-commentator), never their own voice, with an Off; the phone's own volume buttons set how loud. It's a moment behind the stream (more on Bluetooth headphones), and needs headphones: on the phone's speaker it would come back in through the mic. It needs the Secure phone link turned on (a phone lets a page use its mic only over https), and Output 1 on. Keep the phone on the page with Do Not Disturb on: switching apps or a call stops its mic.
- The mics **duck** while a video with its sound plays and during a scene that ducks the music, and while music plays if you tick it (it starts unticked); each is a tick box under **Ducking**. **Duck** ducks them by hand. A mic marked **Never ducks (an external mixer does it)** stays up: the mixer does the fading.
- **Music for the commentators:** in Output 1's **Sound goes to**, **Add a destination** and pick the commentators' mixer: its sound plays there too, while the SDI or the Mac's sound output carries on as before. Only the mics you tick for that destination are in it, so tick the crowd mic and leave the commentators' own unticked and they won't hear themselves back late. On the mixer (not in Studio), keep what comes in from the Mac out of its audio out back to the Mac (its mix minus), or the music comes back through their mics and goes out twice.
- Stream Deck: `/deck/mics/mute/toggle`, `/deck/mics/duck/toggle`, `/deck/mics/up`, `/deck/mics/down`, and one mic with `/deck/mic/court-mic/mute/toggle`.

Ducking. Everything that takes the music and the mics down is on one card, under Music: a row for each cause and a column each for the music and the mics. Its line says how low each goes and what takes it down; **Settings** opens the card.



What ducks them	Music	Mics
A video with sound plays	☑ fades out and pauses, then back in	☑ duck
A scene or sequence with **Ducks the music** ticked	ducks	☑ duck
Music plays	—	☐ duck
The stinger, with **Music under it** set to duck (Switcher)	ducks, fast	—
The **Duck** button	the music's own	the mics' own
Duck to	25 %	30 %
Down, then back up	over the music's Fade out and Fade in	1 s, 2 s

The Ducking card: what ducks the music and the mics, how low, and how fast.

- **A video with sound:** the music fades out and pauses, then comes back (untick to leave it); the mics duck (a tick).
- **A scene or sequence with Ducks the music ticked:** the music ducks; the mics duck too (a tick).
- **Music playing:** the mics duck under it if ticked (off to start with).
- **The stinger**, with its **Music under it** set to duck on the Switcher card, ducks the music fast. The **Duck** buttons duck the music or the mics by hand.
- **Duck to:** one level for the music (25% to start) and one for the mics (30%). The music goes down over its **Fade out** time and back up over its **Fade in**; the mics have their own down and up times (1 and 2 seconds).

Video output

Settings over the picture opens the **Output settings**, which send the scene to your switcher. Pick one of three routes, then press **Output on** in the row over the picture. While the output is on its settings are locked; turn it off to change them. `/deck/output/on`, `off` and `toggle` do the same from a Stream Deck, and the outputs go dark when Studio goes off air.

ROUTE	WHAT IT NEEDS	WHAT THE SWITCHER GETS
Display	One or two extra displays on the Mac (HDMI into the switcher), set to exactly 1920 × 1080, not HiDPI or scaled	**Chroma**: the scene over a solid colour on one display (green by default, blue or any colour). **Fill and key**: fill on one display, key (white where the scene shows) on the other. A plain M1 drives one extra display; Pro and Max chips drive more.
Blackmagic	An UltraStudio or DeckLink with Blackmagic Desktop Video installed, and the device free	**Fill and key** through the device's external keyer (fill on one SDI out, key on the other), or fill only, at the switcher's 1080p format (1080p50 is picked first). The fill is premultiplied: turn the switcher's Pre Multiplied Key on for clean edges, and leave Straight off (it leaves a light fringe round logos and text). **Sound goes to** the SDI with the fill to start (48 kHz stereo, in time with the picture): turn that switcher input's sound on. The Mac's sound output can be picked instead, or added as well.
NDI®	NDI's free runtime for Mac, and a wired network	Fill and key as two sources, chroma as one, or one source with transparency for receivers that key on NDI's alpha. They appear as this Mac's name with the output's name in brackets (or a name of your own).

HD OR 4K

The **HD** and **4K** buttons over the picture set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate you picked for it (2160p50 for 1080p50). Nothing else changes: scenes, sources, Stream Deck keys, links, NDI names and the switcher key stay as they are, and pressing HD puts everything back exactly as it was.

- **Sharp where Studio draws.** The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so text, shapes, logos drawn by Studio and the animations' sparks and confetti are sharp. Pictures and videos are only as sharp as their files: a 1080p video or a small logo looks softer.
- **Only for 4K gear.** An HD switcher can't take 2160p. A Blackmagic device without the 2160p format (an HD UltraStudio) stays at 1080p and its card says so; a display is left alone, since a 4K screen set to look like 1920 × 1080 already draws at 4K.

- **Pick a 4K format directly.** In 4K a Blackmagic output's video format list also has the device's own 4K formats, such as 2160p50. In HD they're hidden, and the output sends the 1080p format at the same rate.
- **Any graphic's link works at 4K too.** Opened in a browser source or window set to 3840 × 2160, a graphic draws itself at twice the size, with nothing added to its link.
- **Check the Mac keeps up.** A 4K output is about four times the work of an HD one; the performance line under the output says whether it's keeping up.

MORE THAN ONE OUTPUT

One output is all most nights need. To give a camera its own graphics, press **Add an output** beside Output on (up to four). Each output is its own picture with its own route and its own on/off, so a second output can feed an upstream keyer tied to a camera and its graphics cut with that camera, while Output 1 stays the downstream key over everything. Name an output in its card ("Wide camera", say) and that name is used everywhere, including its Stream Deck links.

- **Sources are shared, on/off is per output.** The Sources list follows the output tab picked above the Scene View: each source's On / Off is for that output, in its colour (orange, pink, blue, green for Outputs 1 to 4), and shows or hides it there only. **Hide all** clears that output too. Pick another tab to set another output. A new output starts with everything off.
- **Every output plays its own video sound; the music and the mics are Output 1's.** Each output plays the sound of the videos and animations on it. The music (the player's, a sequence's and the super shot song) and the Mac's mics go out on Output 1 only, so a second output never takes the production's sound, and a sequence played on another output alone plays without its music. A commentators' feed is a destination added to Output 1's **Sound goes to**, with the mics ticked for it.
- **Scenes and sequences are shared; each output plays its own.** Pick an output's tab over the Scene View and clicking a scene or sequence plays it on that output (the sections say which). **Hide all** clears that output only (or the one the Sources list's **Show** is set to), and says which, so clearing a monitor never blanks the stream. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick with how the sequence is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending), show where each one is on. A sequence's **Plays on** ticks (on its card and in its editor) pick the outputs it plays on together: started by itself, or played on any of them, it starts on all of them at once, and each output has its own **Then** for when it ends there, so a quarter-time sequence can go back to the live scene on Output 1 while the commentary monitor goes on to its stats loop. Played on an output it isn't ticked for, it plays there alone (`/deck/output/2/sequence/<name>/only` does that on a ticked one). Stop on a card stops it on that output only. Remove every output it plays on and the sequence stops starting by itself until you tick another, so it never lands on air by surprise. A Sponsor ads step off Output 1 plays its ads in that output's own ad sources. `/deck/scene/<name>` and `/deck/sequence/<name>` play on

Output 1; `/deck/output/2/scene/<name>`, `/deck/output/2/sequence/<name>`, `/deck/output/2/scene/next` and `/deck/output/2/sequence/stop` (and the rest) play on Output 2.

- **Sponsors count on Output 1, and on outputs you tick.** An extra output's card has **Counts for sponsors**, off to start. Tick it only when everything on that output is seen on the stream, like a second downstream key. Leave it off for a camera's upstream key or the commentary monitor: Studio can't tell when they're on air, and the report shouldn't claim time nobody saw. Ads there show whose turn it is without using it up, so the order on air stays fair.
- **The Scene View has a tab per output**, coloured to match, so you can check what each one is sending. Switching tabs is seamless: the picture carries straight on, as the new output's view takes over once it has loaded.
- **Routes can't be shared.** Two outputs can't use the same display, the Blackmagic device or the same NDI name; the earlier output keeps it and the later one says so on its card. Two HDMI displays give either two chroma outputs or one fill and key output. Several NDI outputs can run at once, each with its own name, which suits feeding a switcher's inputs over the network.
- **Stream Deck.** `/deck/output/2/toggle` (or the output's name) turns Output 2 on or off, `/deck/output/all/off` turns them all off, and `/deck/output/2/layer/<source>/toggle` shows or hides a source on Output 2. Source settings lists these links for each output.

EACH OUTPUT COSTS

Every extra output runs its pages once more on the Mac. Try the number of outputs you plan to use on your own Mac before a stream night: while an output is on, its card shows its CPU, memory and frame rate (amber when it can't keep up), with Studio's total under the status, and the log notes them once a minute.

A COMMENTARY MONITOR

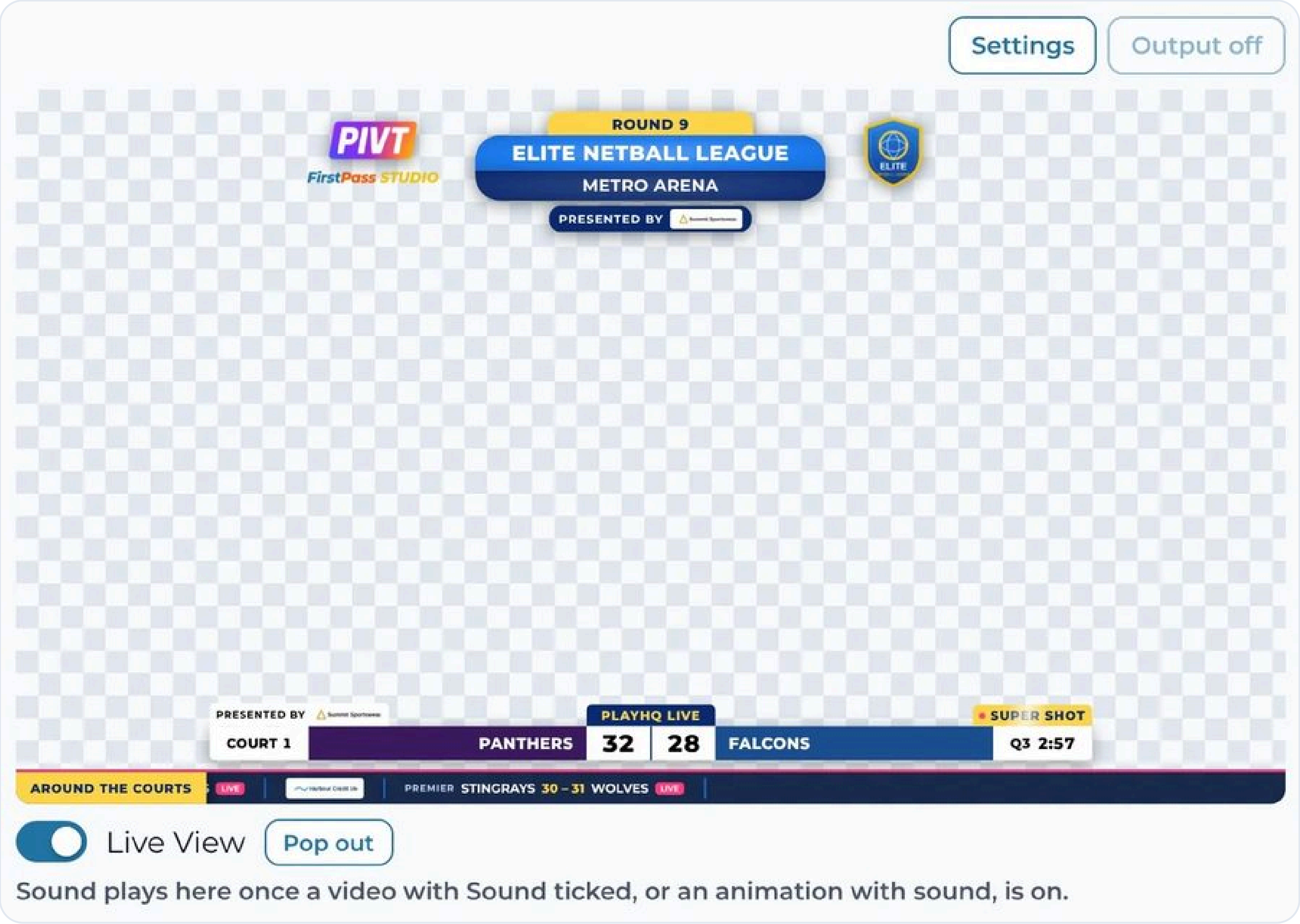
Commentators can have graphics of their own: a monitor at the commentary desk with the live stats they talk to, never seen on the stream. The **Commentator stats** page is made for it: one full screen with both team lists on the left (each team's time since its last goal, and each athlete's games played this season), the game stats big in the middle (each team's logo with its ladder position, record and form under it, head to head under the names, then shooting %, quarters won, best quarter, super shots, runs, biggest lead and lead changes), down the right the game clock with a pill under it for the commentary team (**Starts in 4:32** to the scheduled start, **Back in 2:15 · Quarter time** at a break, counted from the clock reaching 0:00 by the competition's break minutes, then **Due back now** in pink until the clock runs again, and during play the super shot window), who is on the mic (from the Production Planner) and up next, then a list of the latest moments (lead changes, comebacks, super shots, hot streaks), newest at the top, each with its quarter, game time and how long ago it happened (it lists every moment a goal makes, even the ones the stream skips, and opened part way through a game it starts with that game's moments so far), and across the bottom the last 10 goals, the current run in big numbers and the score worm. Its stats refresh themselves every 15 seconds and when the score changes (untick **Auto-refresh** in its options, or pick another frequency), and the clock, moments and goals follow the game as it happens. Other graphics can go on the commentary feed too: the last 10 goals and the scoreboard follow the game live, and the rest show what they read each time they come on.

- **On the same Mac.** Add an output for the commentary monitor (a display, or NDI to a screen on the network) and turn sources on for it with its pills or its Stream Deck keys, such as `/deck/output/commentary/layer/ladder/toggle`. Play scenes and sequences on it from its Scene View tab or its own Stream Deck links while the stream carries on as normal.
- **On a second Mac.** Run Studio on another Mac with the same competition and game, and Follow PlayHQ on. It reads PlayHQ's live scoring for itself, so it needs nothing from the stream Mac. Its Output 1 goes to the commentary monitor, and everything on it is for the commentators.

On the commentary output (or the second Mac), make a **commentaryLoop** sequence that steps through what commentators find useful (game stats, shooting stats, the last 10 goals, the ladder) with **Repeat** set to until stopped, and give the commentators a Stream Deck of their own:

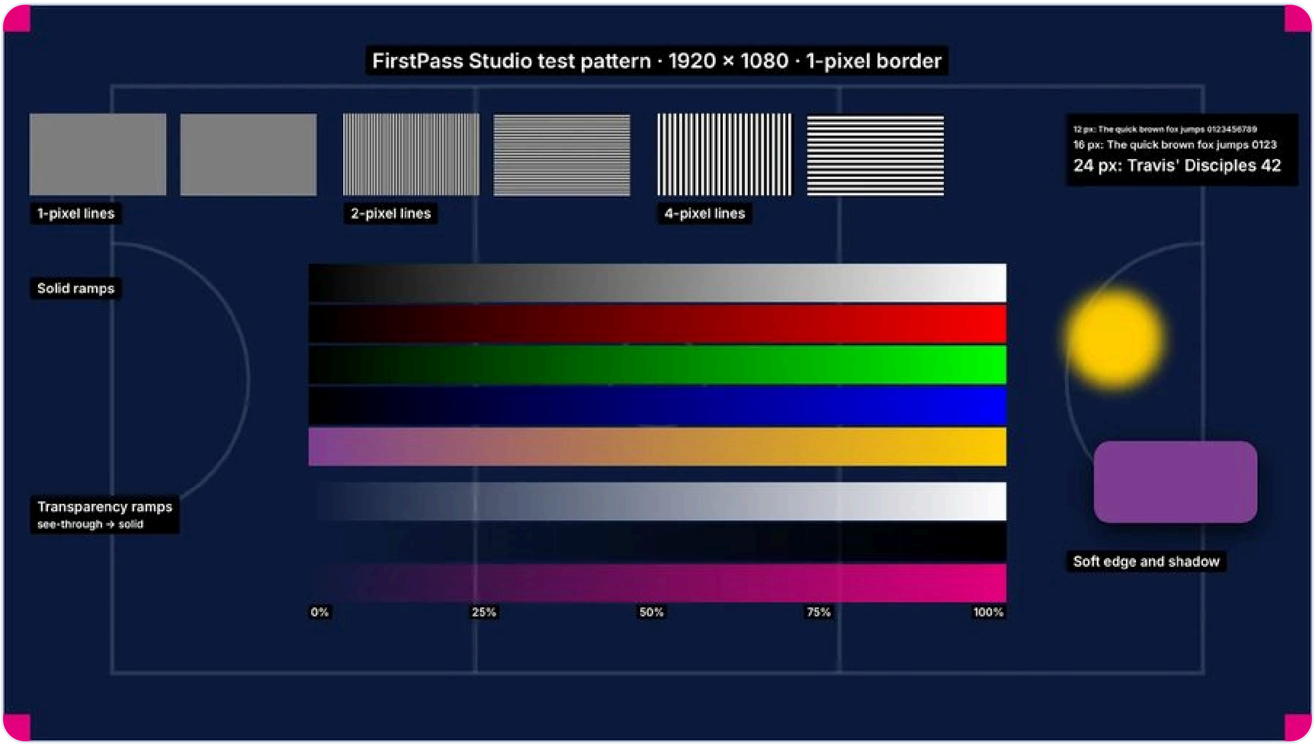
- A key per graphic they want on demand, each going to a scene with that graphic (on the same Mac, `/deck/output/commentary/scene/ladder`). Going to a scene stops the loop on that output only, so their pick stays up and the stream is untouched.
- A **Loop** key to go back to looping when they're done: `/deck/output/commentary/sequence/commentaryloop` on the same Mac, or `/deck/sequence/commentaryloop` on a second Mac. With the FirstPass Studio plugin, use Scene and Sequence keys with **Output** set to the commentary output; they light pink while it's on.

Each output's **Sound goes to** lists where its sound plays: the Mac's sound output to start, or on a Blackmagic output the SDI with the fill; add the Mac's sound output as well, or any of the Mac's sound outputs such as the commentators' mixer. For a screen with no speakers, such as a billboard over NDI, pick **No sound** (last in the list): the output plays no sound anywhere, its card says **Sound: none** and the status board says **No Audio** for it. On Output 1 that silences the music and the mics too, as they go out on Output 1 only. Each output's card says it in a line (Sound: SDI + Mixer (crowd mic); Sound: Mac, not over NDI, since NDI carries no sound), and the status board's **Sound out** rows say it for every output, with what each device plays at and how loud. The **Scene View** on the Output page shows the scene as it is right now, on a checkerboard so see-through parts show; **Pop out** opens it in its own window to move to another screen, with a floating list of On/Off buttons. The Scene View never plays sound: an output's sound plays where its Sound goes to says, on or off air, so nothing plays twice and nothing plays from a page that happens to be open.


Settings
Output off
PIVT
FirstPass STUDIO
ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY
ELITE
PRESENTED BY
COURT 1
PLAYHQ LIVE
PANTHERS
32
28
FALCONS
SUPER SHOT
Q3 2:57
AROUND THE COURTS
LIVE
PREMIER STINGRAYS 30 – 31 WOLVES
LIVE
Live View
Pop out

Sound plays here once a video with Sound ticked, or an animation with sound, is on.

The Scene View: the scene right now. Untick Live View on a busy night, since it runs the scene's pages a second time.


FirstPass Studio test pattern · 1920 × 1080 · 1-pixel border
1-pixel lines
2-pixel lines
4-pixel lines
12 px: The quick brown fox jumps 0123456789
16 px: The quick brown fox jumps 0123
24 px: Travis' Disciples 42
Solid ramps
Transparency ramps
see-through → solid
0%
25%
50%
75%
100%
Soft edge and shadow

The **test pattern** (Add page) checks an output and the switcher's key end to end: a 1-pixel border, fine lines that shimmer if anything is resizing the picture, ramps for banding and soft shapes for the key. Not for air.

CHROMA KEY SAFE

A chroma key cuts out one colour, so anything half see-through fringes. With a chroma output picked, **Chroma key safe** is on: every graphic goes out with solid panels, no glows or soft shadows, and fades become cuts. Pick another output and everything is back as it was. The Output page also lists any competition or club colour close to the key colour.

THE SWITCHER

Studio can also drive a Blackmagic ATEM switcher over the network, so the camera cuts can be pressed from the Output page or a Stream Deck beside the graphics. Open the **Switcher** card on the Output page, press **Set up** and **Look for switchers**: Studio finds the ATEMs on the network the way ATEM Software Control does (macOS asks the first time to let it use the local network) and remembers the one you pick by name. Typing its address works too.

- **Each output's key, checked.** In each output's setup, **Into the switcher as** says which switcher key its graphics come into (usually downstream key 1 for Output 1, upstream keys 2 to 4 for the others, an upstream key for chroma) and which inputs its fill and key are plugged into. Studio reads that key's settings and lists anything that doesn't suit the output, such as the wrong fill source, Pre Multiplied Key off, or Straight ticked on a Blackmagic output. **Set up this key for me** fixes just those, after asking, and leaves the key on or off air as it was.
- **Graphics that go with a camera.** An output on an upstream key can tick the cameras it **goes with**, say a bench graphic with the bench camera. Studio's cuts, autos and stingers then bring its key on with those cameras and take it off with the others, on the same frame. With **Keys follow the camera** on, a cut on the switcher's own panel brings the key a frame or two later. A key turned off by hand stays off until the camera changes. The output's card says **On air with Bench** or **Waiting for Bench**.
- **Inputs in tally colours.** Each input shows by its name from ATEM Software Control, red on program and green on preview. Click one to put it on preview, then **Cut** or **Auto**. Fade to black and each upstream and downstream key are a press away too.
- **Only what your switcher has.** An ATEM Mini or Mini Pro has 4 inputs, one upstream key and one downstream key, so fill and key into the downstream key leaves two cameras; a Mini Extreme has 8 inputs, four upstream keys and two downstream keys. Studio offers what it finds, and says so when a key or link asks for something the switcher doesn't have.
- **Your hands stay on the switcher.** Its own buttons, ATEM Software Control and other controllers keep working, and the card follows them. Studio sends nothing by itself, not even on connecting.
- **If it drops off the network, or is off when Studio connects,** the card turns red within about five seconds and Studio keeps trying. Presses meanwhile are refused rather than saved up, so a cut never lands late.

- **Stinger cuts.** Add the Stinger page as a source and the card's **Stinger** button plays it and cuts the switcher underneath, right in the middle of the bands. It plays on the output on a downstream key, which is Output 1 unless another output that's switched on is. If the cut shows before the bands do (an output's picture can reach the switcher a few frames late, more over NDI), set the **line-up** in frames and press **Test the line-up** before you go live. Each output keeps its own line-up. **Music under it**, beside them, says what the music does as the stinger plays: carries on, ducks for the scene you cut to, or ducks while it plays and comes back up.
- **Camera cuts in sequences and scenes.** A sequence step's 🎥 button, or a scene's 🎥 **Switcher** in its settings (the pencil), adds switcher actions: cut, auto or stinger to a camera, put one on preview, a key on or off, fade to black, or run a macro, as the step starts or a set number of seconds in. So the walk-out sequence can cut to the bench camera as the line-ups come up and bring the scoreboard key back as it ends, with nobody on the switcher. They happen only on Output 1, and if the switcher isn't answering at that moment the action is skipped and logged rather than done late. You can always cut by hand on the switcher itself.
- **Stream Deck.** The plugin's **Switcher** key puts a camera on preview or program, plays a stinger, cuts, autos, turns a key on or off or runs a macro, and lights like the switcher's own buttons: red on program, green on preview. While the switcher isn't answering it goes dark with a yellow dot. Without the plugin, the links are `/deck/switcher/cut`, `auto` and `ftb`, `/deck/switcher/preview/2` (or `preview/Camera%202`), `/deck/switcher/usk/1/toggle`, `/deck/switcher/dsk/1/auto` and `/deck/switcher/macro/1`.

A night, start to finish

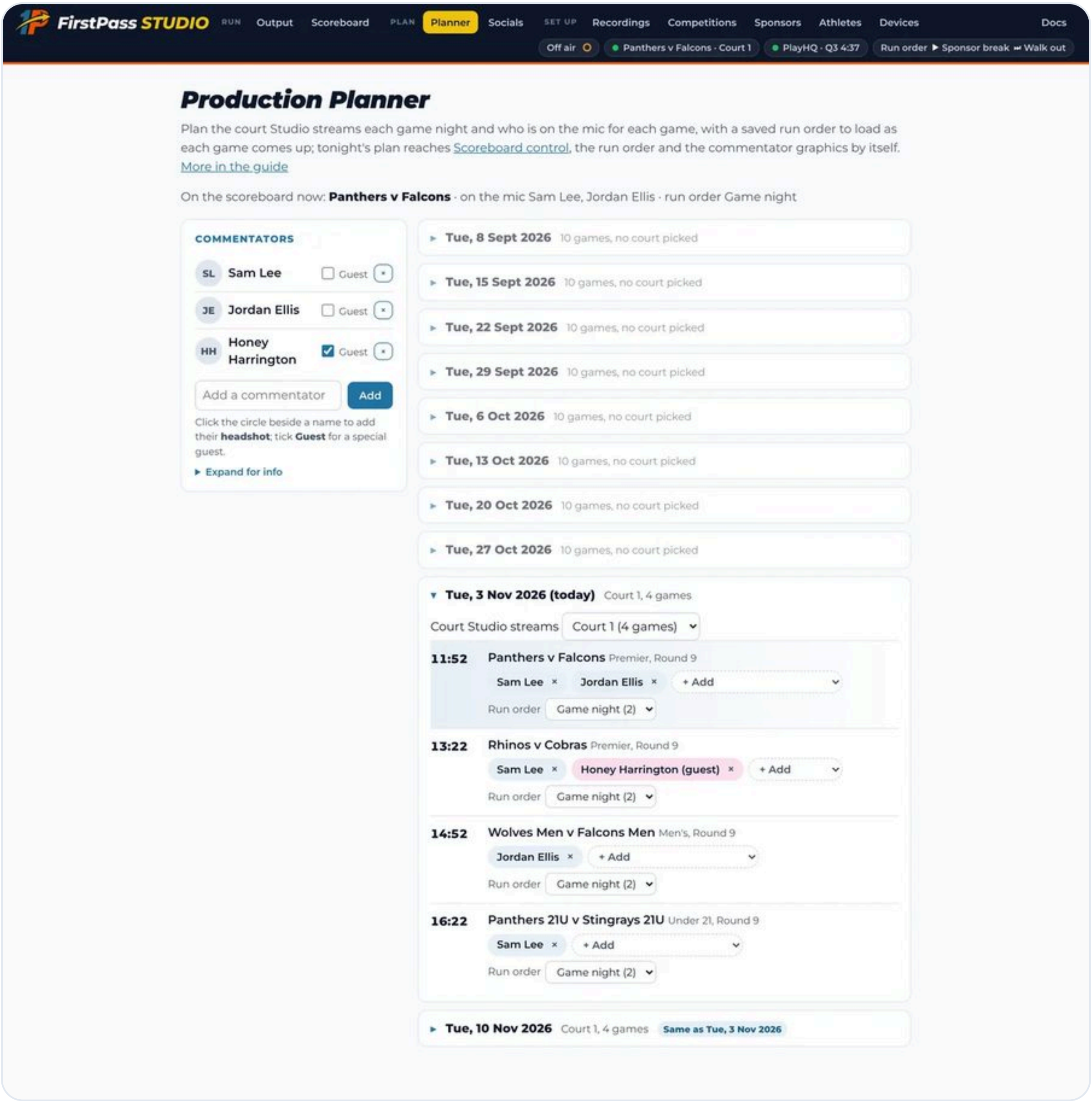
With the sample league's scenes and sequences, a solo operator's evening looks like this. Everything in the right-hand column happens by itself.

ARRIVE	Open the app. Studio goes on air, picks tonight's first game on Court 1 with Follow PlayHQ on. Turn the video output on and check the test pattern on the switcher.
15 MIN OUT	**Run-up** starts by itself: the Pre-game scene, then the sponsor wall, match up and ladder in turn over the warm-up playlist, repeating until 2 minutes before the start.
2 MIN OUT	Run-up hands over to **Walk out**: The Clash smashes the two logos together, the position match ups run through GS to GK, the countdown holds until the clock starts, then the Live game scene comes up and the music fades.
Q1 TO Q4	The scoreboard follows the PlayHQ scorer's score, quarter and clock. Super shots, hot streaks and moments play by themselves. The super shot tab and song mark the last five minutes of each quarter. The operator

	watches the stream and, if they like, keys up a ladder lower third or a lower third for a commentator.
QUARTER TIME	**Quarter time** starts a few seconds after the siren: shooting stats, the last 10 goals and the ladder, then two sponsor ads and the sponsor card, over the break music, round and round until play resumes, then back to Live game.
HALF TIME	**Half time** does the same with Up next and the ladder lower third added, and the house ad if no sponsor has one due.
FULL TIME	**Full time** starts after the final siren: the winner celebration, the full time card, game stats, the score worm, a sponsor break and Up next (which becomes the end-of-night card after the last game).
NEXT GAME	Automatic selection moves the board to the next game on Court 1 as its electronic scoring opens, with the quarter back to Q1 and a full clock. Run-up starts again 15 minutes out.
CLOSE	Press **Stream close**: the Clear scene, then the outro with the night's teams. Turn the output off. The sponsor report has the night's exposure ready to send.

The Production Planner

The **Production Planner** tab plans the season ahead: which court Studio streams each game night, and who is on the mic for each game. Studio streams one court a night; a second court is a second Mac running its own Studio, with its own plan.


FirstPass STUDIO
RUN
Output
Scoreboard
PLAN
Planner
Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 4:37
Run order ▶ Sponsor break ⇻ Walk out
COMMENTATORS
SL Sam Lee ☐ Guest
JE Jordan Ellis ☐ Guest
HH Honey Harrington ☑ Guest
Add a commentator
Add
▸ Expand for info
▸ Tue, 8 Sept 2026 10 games, no court picked
▸ Tue, 15 Sept 2026 10 games, no court picked
▸ Tue, 22 Sept 2026 10 games, no court picked
▸ Tue, 29 Sept 2026 10 games, no court picked
▸ Tue, 6 Oct 2026 10 games, no court picked
▸ Tue, 13 Oct 2026 10 games, no court picked
▸ Tue, 20 Oct 2026 10 games, no court picked
▸ Tue, 27 Oct 2026 10 games, no court picked
▾ Tue, 3 Nov 2026 (today) Court 1, 4 games
11:52 Panthers v Falcons Premier, Round 9
Sam Lee × Jordan Ellis × + Add
13:22 Rhinos v Cobras Premier, Round 9
Sam Lee × Honey Harrington (guest) × + Add
14:52 Wolves Men v Falcons Men Men's, Round 9
Jordan Ellis × + Add
16:22 Panthers 21U v Stingrays 21U Under 21, Round 9
Sam Lee × + Add
▸ Tue, 10 Nov 2026 Court 1, 4 games Same as Tue, 3 Nov 2026

Production Planner

Plan the court Studio streams each game night and who is on the mic for each game, with a saved run order to load as each game comes up; tonight's plan reaches Scoreboard control, the run order and the commentator graphics by itself. More in the guide

On the scoreboard now: **Panthers v Falcons** · on the mic Sam Lee, Jordan Ellis · run order Game night

Click the circle beside a name to add their **headshot**; tick **Guest** for a special guest.

Court Studio streams: Court 1 (4 games)

Run order: Game night (2)

The Production Planner: the commentators down the side, and the season's game nights from PlayHQ, the next one open, with each game on the night's court, who calls it and the run order it loads.

COMMENTATORS

Add the people who call your games to the list on the left. Tick **Guest** for a special guest. The list belongs to the competition; as you type a name, people from your other competitions are suggested, so someone who calls in two leagues is added in a tap. Click the circle beside a name, or drop a photo on the row, to add their **headshot** for the Socials tab's Commentary team picture, the fixture board and the commentators' lower thirds; the small × on it takes it off again.

A NIGHT

Every game night in the season is listed, read from PlayHQ's fixture, by date rather than round. Open one, pick the **Court Studio streams**, and its games on that court appear in time order. Tap **+ Add** on a game to put up to three people on the mic; tap × to take someone off. Everything saves as you go. A night you haven't planned takes the plan of the night before it (the same court, and the same people and run order game by game), marked **Same as** that night, so a season with a regular team is planned once and only the nights that differ need changing. **Same as the night before** drops a night's own plan again.

A RUN ORDER FOR EACH GAME

Each game's **Run order** list holds the run orders saved on the Output page (Saved...). Pick one and it loads into the run order as that game comes up on the scoreboard, whether Studio picked the game or you did, in place of whatever is queued. If that clears anything, the run order panel says so (**Cleared the 2 items queued: the Production Planner loaded "Big game" for Stormz v Titans**) until you press **OK**. What's on air carries on, and it loads once as the game arrives: restarting Studio mid-game doesn't load it again. A game set to **None** leaves the run order alone, and the run order's own buttons work as ever. A run order that gets renamed stays picked; one that gets deleted is flagged on its games.

WHEN THE FIXTURE CHANGES

PlayHQ's fixture changes through a season: a game moves court for availability, or is refixtured to another date. The commentators belong to the time on the court, because that is where the desk and cameras are set up, so they stay put and the game is flagged **Fixture changed** with what was there before. A game that has gone from the court keeps its row, so its commentators can be moved to wherever it went.

ON THE NIGHT

When the night arrives, the plan's court goes to Scoreboard control, which follows it game by game as usual. The **Commentators** graphic, with no names typed in its options, names whoever is planned for the game on the scoreboard, and says **Special guest commentators** when everyone on the mic is a guest. Its **Lower thirds** look puts up a person lower third for each of them from one source, side by side, with a **Special guest** tab over a guest and, if you like, their headshots. **Commentator stats** shows them under the clock, and the **Fixture board** full page puts the whole night's fixture on a screen in the competition room or backstage, with a camera on the streamed court's games and who is on the mic for each: their headshots (or a silhouette) with their names, under an **On the mic** tab beside the teams. Manual control always wins: if you change the court or pick a game by hand on Scoreboard control, the Planner says so at the top, so the commentary team can be moved to match. Swapping someone in the Planner takes effect the next time the graphic comes on.

• Socials

The **Socials** tab turns a game night into pictures for Instagram and the league's other channels, from the same fixture the stream uses, so the posts and the broadcast always agree. Pick the **Game night** (by date, so every division playing that night is in it),

Upcoming or **Results** (a night with results opens on Results), and the sizes: **Square** 1080 × 1080, **Portrait** 1080 × 1350 (the most room in the feed) and **Story** 1080 × 1920. Every picture shows as a preview in each size, in the competition's colours, font and logos. Click a preview to see it as big as the window allows; ← and → step through the others, and **Esc** or a click closes it.

Show narrows the list to the pictures you want, like a scene does on the Output page. **All** lists every picture. **New picture set** makes a set from a name and a tick for each kind of picture (Match up, Form guide, Ladder and so on); it starts with every kind on the list you're looking at ticked. Pick a set and only those kinds are listed, on both Upcoming and Results, and only those are saved; each button shows how many pictures it lists this night. A set picks kinds, not games, so the same set works every week. ✎ on a set changes its name or ticks, or deletes it. Your sets, and the one picked, are remembered for the competition. When more than one division plays that night, **Division** on the same row lists only that division's pictures (its match ups, ladder, top shooters, season draws and so on) along with the ones for the whole night, like the Game night fixture, Results and Byes.

UPCOMING PICTURES

The **Game night fixture** lists every game of the night on every court, by time. A **Match up** for each game puts the two logos side by side with the date, time, court and venue, one for each club to share. A **Form guide** for each game shows both teams' last five results (oldest first, with the scores) and how their meetings this season went. Each division gets a **Round preview**: its games with each team's ladder place and points, the game of the round (the two best placed teams; **Top of the table** when it's 1st v 2nd) outlined, and any team a win could lift into the finals places. **Byes** lists the teams not playing that night, division by division. **Milestones coming up** names athletes whose next game would be their 50th, 100th or another 50; Studio counts this season's PlayHQ line ups plus any games typed in on the Athletes tab (**Games**), so type in earlier seasons' games for it to know. In the last three rounds before a division's finals, its **Finals race** shows each team as in the finals, out, or still a chance, from the ladder and the games left. In the last four rounds, and on finals nights, **Countdown** shows the rounds left before the finals or the days to the grand final. On a division's first night, each of its teams gets a **Season draw** with every game of its season. **On the stream** shows the court Studio streams, with each game's division and who calls it, from the Production Planner. **Commentary team** is a picture for each game someone calls, with that game's commentators side by side with their headshots and the game's time and teams once underneath, to share before the livestream. Someone who calls more than one game is on each of their games' pictures, so the social team can post them one at a time or grouped as they like. Long names shrink together so the row stays balanced, breaking only between words or after a hyphen; anyone without a headshot shows their initials, and a guest gets a **Special guest** pill. The **Polaroid** is a tilted white print for a match photo: drop a photo on it, drag the preview to frame it and set the zoom. With no photo, its window is see-through, so the social team can put their own photo under it in Canva or a phone app.

RESULTS PICTURES

After the night, **Results** shows every finished game's final score, division by division, with the winner in the highlight colour. Each game gets a **Full time** card, with the logos and score big and who won by how much, and a **Polaroid** with the score as its caption and the division and round under it. **Round in review** picks out the night's biggest win, closest game and upsets (a team at least half the ladder, or 4 places, below the team it beat). Each division that played gets its **Ladder**, with the finals line, labelled with the round it is after (that night's, when it is that division's latest night with results; else the latest round played), and on that night each team's move since before it when **Highlight movers** is ticked. Each division also gets its own **Top shooters**, naming its best six that night from each game's line ups (with super shots only where that division plays them), and its **Tournament top shooters**: the best eight over every finished game so far, after this night's round. It's called **Season top shooters** when the fixture runs 15 days or more; pick either name under Game day on Competitions setup to override. On a carnival day with several rounds, set **As at** to an earlier round to count only the games up to it. Tick **Highlight movers** and each shooter shows the places they climbed or fell since the round before, even when both were played that day, with NEW for anyone new to the top eight; the tick is remembered for the competition. **Sharpshooter** ranks each division's most accurate shooters that night (10 attempts or more), **Super shot leaders** (in divisions with super shots) the most super shots that night and over the season, **Team super shots** (in the same divisions) every team's super shots that night and over the season, and **On fire** the shooters whose goals a game over their last three games rose most. **Night in numbers** adds the night up: games, points, super shots, close games, the biggest win, the highest score, the shooting % and the upsets (or, on a night without one, the average winning margin), and, when Studio has every game's whole story, the lead changes, the most goals in a row and the runs of 5 goals. A club with more than one team playing gets a **Club round up** of its teams' results (clubs come from Competitions setup). **On a roll** names the teams unbeaten all season or on three wins or more in a row. Pick a team for its **Team season card**: its wins and losses, ladder place, percentage, biggest win and top shooter so far. On a division's last night, **Season wrap** names its champions (once known), top shooter, sharpest shooter, most super shots and biggest win of the season. A game appears once PlayHQ has its final score; until a night has one, each Results picture says so.

Some pictures are picked by hand on their card. **Player of the match** (one per game) takes a name from either team's line up, with their photo when they approved its use and their shooting that game; type a medal's name, such as an Anzac medal for a special round, and it becomes the title. **Team of the round** (one per division) takes an athlete for each position from that night's line ups. On both Upcoming and Results, **Athlete spotlight** shows one athlete from the Athletes tab (their photo, teams, positions, games played and goals this season) and **Sponsor spotlight** one sponsor big with a thank you, and once the night has been streamed, how long they were on screen. **Thanks for watching** closes the night with the next game night and the sponsors.

For each game Studio streamed, or followed in the background (the night's other games), Results adds a **Game story** (the score quarter by quarter, quarters won, the best quarter, the longest run, the biggest lead and the lead changes) and a **Score worm** (the margin over the game). Studio keeps each game's goals and quarter scores on this Mac as it follows it, so they're there after the scoreboard moves to the next game; a game picked up part way says Studio didn't see every goal. When a second court is streamed by another Mac running Studio, each Mac copies the other's finished games, so the Socials tab on either one has every court's pictures, even on a Mac that was off air for part of the night: while both are on the same network they find each other by themselves and copy every few minutes, and with iCloud Drive on (same Apple account) each finished game also goes into **iCloud Drive/FirstPass Studio/game-stories**, for nights the Macs never saw each other (a game Studio was quit part way through goes there too, half an hour later, at Studio's next look). On a night with copies, Results says how many games came from which Mac and which way. Copying only reads from the other Mac; nothing there changes. When both Macs have the same game, the better one is kept: the one with fewer gaps, then the streamed one over one only followed (a game scored by hand on the stream has goals PlayHQ never had). Where one Mac has a gap (its internet dropped, or it picked the game up part way) and the other Mac saw those goals one by one, the gap is filled with them, and the game's quarters, runs, worm and moments are worked out again; nothing more is asked of PlayHQ, and Results says how many games had gaps filled from which Mac. This happens during the game too: a Mac turned on part way through a game, or whose internet dropped for a while, asks the other Macs on the network for that game so far (every 30 seconds while it has a gap) and fills it there and then, so its worm, runs, streaks and Commentator stats have the whole game, then carries on following the game by itself; the moments it fills in aren't played on air. A game also gets **Moments of the game**: the lead changes, comebacks, runs and super shots Studio saw, in order. Every finished game, streamed or not, gets **Game stats** (team against team from PlayHQ's team counts: goals, super shots, attempts and accuracy, plus quarters won, the longest run, the 3 and 5 goal runs and the biggest lead when Studio streamed or followed it) and a **Shooting duel** (each team's shooters). From the games Studio streamed or followed, the night also gets **Runs of the night** (each game's longest run of goals), **Super shot time** (the points each team scored in each quarter's super shot period, where the clock ran) and **Comeback of the night** (the winner who came back from furthest behind, with the game's worm). Upcoming adds a **Team sheet** for each game (both squads with games played) and, once two teams have met this season, a **Head to head**. **Watch it back** turns the replay's link into a QR code: paste the video's link (it starts with your channel's, from Live viewers on Competitions setup, when there is one).

Finals get their own pictures. After a division's last home and away round, Results adds **Finals bound**, the teams that made the finals with what each one's ladder place earned (a double chance, a week off, an elimination final), and the **Finals draw**, the finals week by week. On each finals night the draw is there again, under Upcoming to post before the games and under Results with the winners filled in, and the Match up, Full time and Results pictures name each final, such as Major semi final or Grand final. Once the grand final has a winner, Results adds **Champions**: the winners' logo, their name and the grand final's score. Call them **Premiers** instead under Finals on Competitions setup. A division gets these when it has a finals format on Competitions setup.

SPONSORS, THE LOGOS AND SAVING

Each picture has a **Sponsor strip** tick, which puts your naming, elite and major sponsors along the bottom and is remembered for next week. Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** sets the look of every picture: **Default** is the stream's own, and Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero and Final each bring their own shapes and lettering, and three netball sets draw on the game itself: Court (the court's lines and circles), Bib (a bib with its position letters) and Goal (the ring, net and post), and five sporting sets bring the kit around it: Bottle, Cones, Sneakers, Stadium and Locker room. Star (☆) the templates you like and tick **Favourites only** to list just those, and the one in use; it's remembered for the competition. Any template works with any colours, and every title stays on one line beside the competition logo, shrinking a little where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes page, and only for athletes ticked **Approved for photo use**. Top shooters shows each shooter's photo automatically. On a Match up, Full time or Polaroid card, pick an athlete by name, narrowing the list by division and team. Hero and Final stand the picked athletes tall, in front of the title; a cut-out PNG with a transparent background looks best. **Colours** gives every picture a different look: **Competition colours** is the stream's own, and the others swap the competition's main, highlight and accent colours around (Accent first, Highlight ground, Accent ground), or go darker (Night), paler (Day) or down to two colours. **New colour set** makes your own from three colours, for a finals night or a sponsor round. A set marked **low contrast** has titles that may be hard to read. Your choice is remembered for the competition. Untick anything you don't want (**Select all** and **Select none** tick or untick every picture at once), then **Save pictures**: they are saved as PNGs in a folder for the night in **Pictures › FirstPass Studio**, which opens in Finder. **Share…** saves them too and opens the Mac's Share menu, to AirDrop them to your phone or send them by Messages or Mail. Tick **As a zip** to also save them as one .zip beside the folder; Share then sends that one file, which is easier to email or upload. Tick **As a carousel** to post them together as one Instagram carousel: the ticked pictures line up above Save in the order they'll post, and you drag one (or use ‹ ›) to move it. A carousel is one size and up to 20 pictures; each is marked 1/6, 2/6 and so on in its top right corner, with an arrow to swipe on (the first says Swipe), and they're saved numbered in that order in a folder of their own. A picture that can't be made, such as Milestones on a night with none in the line ups, is skipped and the rest are still saved; the line beside Save says which were skipped and why, in amber. In a carousel the rest are numbered again without a gap (1/5, 2/5…).

From a phone. The Socials tab works on a phone (or another computer) paired with **Allow everything**. On the Mac, press **Open on your phone** at the top of the Socials tab and scan the QR code with the phone's camera, on the same Wi-Fi as the Mac. The first time, the phone asks the Mac to pair: type a name, press **Ask the Mac** and choose **Allow everything** on the Mac, and the phone goes on to the Socials tab. After that the QR code (or the Studio link with `/socials.html` on the end) opens it straight away; add it to the Home Screen to open it like an app. Pick the pictures as on the Mac, then press **Save to this device**, and the phone shows them: press and hold one and choose **Add to Photos**, or tap **Download** under it. With **As a zip** ticked there's a button for the zip too. Each picture's caption shows above them, with **Copy**. The Mac app has to be running, since it draws them; nothing is left on the Mac, and the pictures stay ready for the phone for an hour. **Share…** isn't offered on a phone, because a phone's browser only shares files from a secure (https) link.

Studio on a phone. On a phone, Studio's bar has just the three phone pages: **Socials**, **Remote** (The Remote on a phone) and **Commentator** (a phone as a backup commentator mic, in Mics). Studio's own link opens Socials there. **Open on your phone** on the Mac has a QR code for each of the three. The Mac's other pages still open on a phone at their own links.

CAPTIONS

Each picture has a **Caption** box under it, for the words that go with the post. Open it, pick a style (**Hype**, **Professional**, **Emotional**, **Neutral** or **Fun**), tick **Short** for one or two sentences, and press **Write caption**. An AI writes it from the same facts the picture shows, and only those: the scores, teams and stats on the picture, never a miss or a low number, in Australian English. Your hashtags go on the end. Read it before you post it. The text is yours to change, or type your own without the AI at all. **Again** writes another, **Undo** brings back what you had, and **Copy** copies it to paste into the post. The next picture starts in the style you picked last. Under the box is who wrote it and about what it cost.

Save pictures also saves each ticked picture's caption, under the picture's name, in **Captions.txt** in the folder with them (and in the zip). On a phone, the captions come back with the pictures, each with **Copy**. The captions are kept until you pick another game night.

Setting it up. The captions are written by Claude, ChatGPT or Gemini, with your club's own API key, so the club pays the AI company for what it uses: usually a fraction of a cent a caption. On the Mac, open the **Captions** card at the top of the Socials tab:

1. **Written by:** pick Claude (Anthropic), ChatGPT (OpenAI) or Gemini (Google).
2. **API key:** press **Get a key**, make a key on the company's site (it asks you to add a payment method), paste it in and press **Save key**. Studio checks the key with the company first, keeps it in the Mac's Keychain and only ever shows its last four characters. **Remove key** takes it off this Mac.
3. **Model:** the models your key can use, each with its price. The smaller, cheaper models (Claude Haiku, GPT mini, Gemini Flash-Lite) write captions well.
4. **Hashtags:** your hashtags and @handles, which go on the end of every caption as typed.

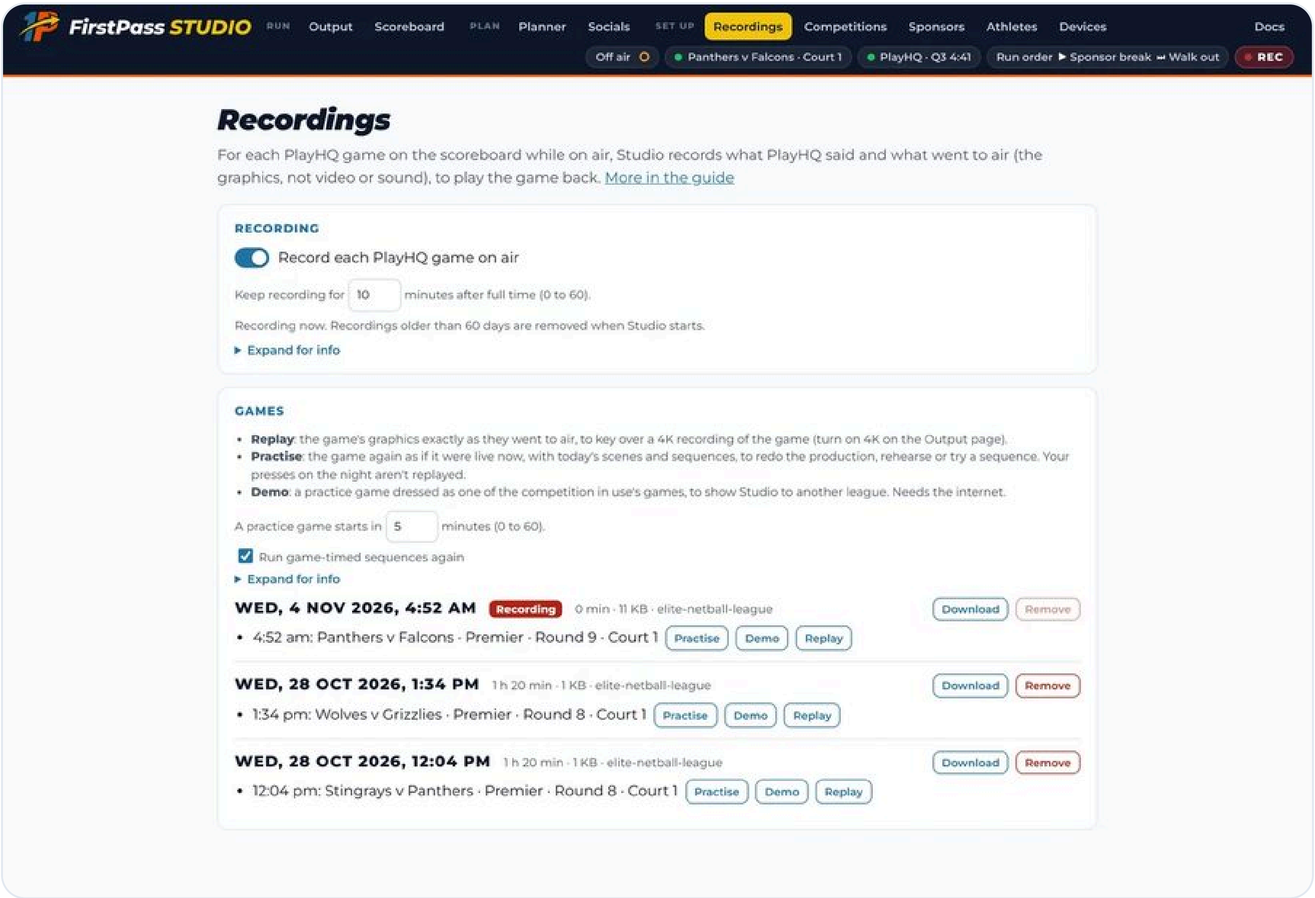
The card shows the model's price and what this month's captions cost, with a link to the company's billing page, which has the real figure. When Studio doesn't know a model's price, the card says so in amber and the captions show the words used instead of a cost. Studio writes one caption at a time, and at most 30 in 10 minutes. The key and the AI settings can only be changed on the Mac; a phone paired with **Allow everything** can write captions with them.

Recordings

While Studio is on air, it records each PlayHQ game on the scoreboard. A recording holds what PlayHQ said (score, clock, quarter, line ups) and what went to air (the scene, the scoreboard and what each graphic showed), with the time of each. It has no video or sound, so a game is about a megabyte. Recording starts when the game is due on the scoreboard (10 minutes before the start, as set by automatic game selection) and stops 10 minutes after full time. It also stops when the scoreboard moves to the next game or Studio goes off air. Recording is on unless you untick **Record each PlayHQ game on air** on the **Recordings** tab, or press **Stop** beside Record on the Output page's status board. Recordings stay on this Mac for 60 days and hold nothing secret.

The Recordings tab lists each recorded game with three buttons. Each plays the game back for a different job:

	REPLAY	PRACTISE	DEMO
Plays	The game's graphics exactly as they went to air	The game again, as if it were live now	The game as one of another league's
Scenes, sequences and settings	The night's, as they were	Today's: change them first to fix the night	Today's, with the other league's teams, athletes, logos and colours
Sequences	Only what went to air	Start by themselves again from the game	As Practise
Your presses on the night	On screen as they went to air	Not replayed: you produce it again	Not replayed
Needs the internet	No	No	Yes
Use it to	Put 4K graphics on a game recorded in 4K, for YouTube	Redo a night's production, rehearse, or try a new sequence	Show Studio to another league



Recordings

For each PlayHQ game on the scoreboard while on air, Studio records what PlayHQ said and what went to air (the graphics, not video or sound), to play the game back. More in the guide

RECORDING

● Record each PlayHQ game on air

Keep recording for 10 minutes after full time (0 to 60).

Recording now. Recordings older than 60 days are removed when Studio starts.

▸ Expand for info

GAMES

- **Replay**: the game's graphics exactly as they went to air, to key over a 4K recording of the game (turn on 4K on the Output page).
- **Practise**: the game again as if it were live now, with today's scenes and sequences, to redo the production, rehearse or try a sequence. Your presses on the night aren't replayed.
- **Demo**: a practice game dressed as one of the competition in use's games, to show Studio to another league. Needs the internet.

A practice game starts in 5 minutes (0 to 60).

☑ Run game-timed sequences again

▸ Expand for info

WED, 4 NOV 2026, 4:52 AM Recording 0 min · 11 KB · elite-netball-league Download Remove

- 4:52 am: Panthers v Falcons · Premier · Round 9 · Court 1 Practise Demo Replay

WED, 28 OCT 2026, 1:34 PM 1 h 20 min · 1 KB · elite-netball-league Download Remove

- 1:34 pm: Wolves v Grizzlies · Premier · Round 8 · Court 1 Practise Demo Replay

WED, 28 OCT 2026, 12:04 PM 1 h 20 min · 1 KB · elite-netball-league Download Remove

- 12:04 pm: Stingrays v Panthers · Premier · Round 8 · Court 1 Practise Demo Replay

The Recordings tab: the recording switch, the practice game settings, and each recorded game with Practise, Demo and Replay.

Whichever one runs, a red strip on every Studio window says so, with **Stop**. Its controls are at the top of the Recordings tab, and the Stream Deck plugin's **Replay** key works them too. Nothing in it counts for your sponsors and it isn't recorded as a game of its own. **Stop** puts the live scoreboard back exactly as it was (a Replay never touches the live scene; a practice game plays on it, as a game night does).

PUT 4K GRAPHICS ON A RECORDED GAME

1. **On the night:** stream at 1080p as usual and record the camera in 4K, on a HyperDeck if you can, with its timecode set to **Time of day**. Studio records the game by itself.
2. **Afterwards:** on the Recordings tab, press **Replay** beside the game. Studio puts that game's graphics back on every output and waits, paused, at the start of the game.
3. On the Output page, turn on **4K**, so the outputs send the graphics at 2160p. Key them over the 4K video in a 4K switcher or recorder. Or add the output's **Output canvas** link (Devices tab, Links) to an editing or streaming program as a web page source at 3840 × 2160.
4. **Line the graphics up with the video.** If the video is on a HyperDeck, set **Follow** to **A HyperDeck's timecode** and type its IP address. The graphics then follow the HyperDeck as you play, stop and scrub it. Another player can do the same over SDI into the UltraStudio (**Timecode into the UltraStudio (SDI)**). With neither, pause the video

on a moment you can see, such as the first centre pass. Pick the same moment in the list, press **Jump there**, then press Play on both together.

5. If the graphics are a little early or late, **Nudge** them a frame or a second. With **Follow**, Studio remembers the sync for that recording.
6. Record the result. Studio adds no video or sound: use the night's programme recording for the sound. Press **Stop** when you're done.

Replay shows the night's data in **today's** version of each graphic, so a graphic fixed since the night shows the fix. A graphic you have removed from the Output page since still plays, because the recording keeps its link. A video or image you've deleted from the media folder comes up blank, because the recording only keeps its name.

REDO A NIGHT'S PRODUCTION

If something went wrong on the night, such as a sequence at the wrong time, a scene that needed changing or a graphic that needed fixing, fix it first. Then run the game again with **Practise**:

1. Fix the scenes, sequences or graphics on the Output page.
2. On the Recordings tab, leave **Run game-timed sequences again** ticked. Set **A practice game starts in** to at least the minutes of your earliest pre-game sequence (a Run-up 15 minutes out needs 15). A pre-game sequence due before the practice starts is skipped, as on a night Studio was opened late.
3. Press **Practise**. The game is due that many minutes from now, and Studio runs as it does on a game night. The scoreboard follows the recorded score and clock, and sequences start by themselves from the game, at the head of Output 1's run order.
4. Produce it as you would live: anything you did by hand on the night, do again. **Jump to** goes to the start, a quarter or full time, and **Pause** stops the clock.

Practise replays only what PlayHQ said, so a game scored by hand rather than live on PlayHQ has no score or clock in a practice. A practice game doesn't follow a player's timecode: to lay it over a video, start them together and use **Pause** and **Nudge** (see below). Practise is also the way to rehearse before the season or try a new sequence.

GRAPHICS ON ANOTHER COURT'S GAME

The night's other games Studio follows in the background (The night's other games) are recorded too, each on its own from Studio's first reading of it until 10 minutes after full time, named **(followed)** and marked **followed, not streamed** on the Recordings tab. Nothing of them went to air, so they have no Replay: play one with **Practise**, and Studio runs it on the scoreboard as if it were live, so you can produce it over somebody's video of that game, from a phone or a camera:

1. Press **Practise** beside the game, then the quarter the video is on (**Q1**, **Q2**...) and **Pause**. Each quarter's button is the moment PlayHQ's scorer started that quarter's clock.

2. Cue the video on that quarter's first centre pass and pause it.
3. Press Play on both together (the Stream Deck's practice **Play** key helps).
4. Watch for a goal: if the score changes before or after the ball goes through, **Nudge** the game a second at a time until they match.

Its recording has what Studio asked PlayHQ that night (the fixture, the ladders, the live scoring); anything else a graphic needs, such as the line ups, is read from PlayHQ when you play it, so it needs the internet, and the line ups show the final numbers.

DEMO GAMES

A demo game is a practice game dressed as another league's game, for showing them Studio in their own colours. Set up their competition and switch to it, then press **Demo** beside one of your recorded games. Pick one of their divisions and a finished game. Studio plays your recording as that game: their teams, logos, colours and athletes, with your recording's score, clock and moments. Athletes are matched by position, so your GS's goals become their GS's. The **Athletes** table shows who each of your athletes is shown as, and you can change any of them. **Swap home and away** lets either team win. **Save demo** keeps the whole setup by name, so on the day it's one click under **Saved demos**. Their competition is read from PlayHQ, so a demo needs the internet.

More in the Reference: **Socials in detail**.

More in the Reference: **Recordings in detail**.

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

IN THIS CHAPTER

1 Two people, not one

2 Switching to manual

3 The score

4 The clock and the quarter

5 Breaks and celebrations by hand

6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

🟦 Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

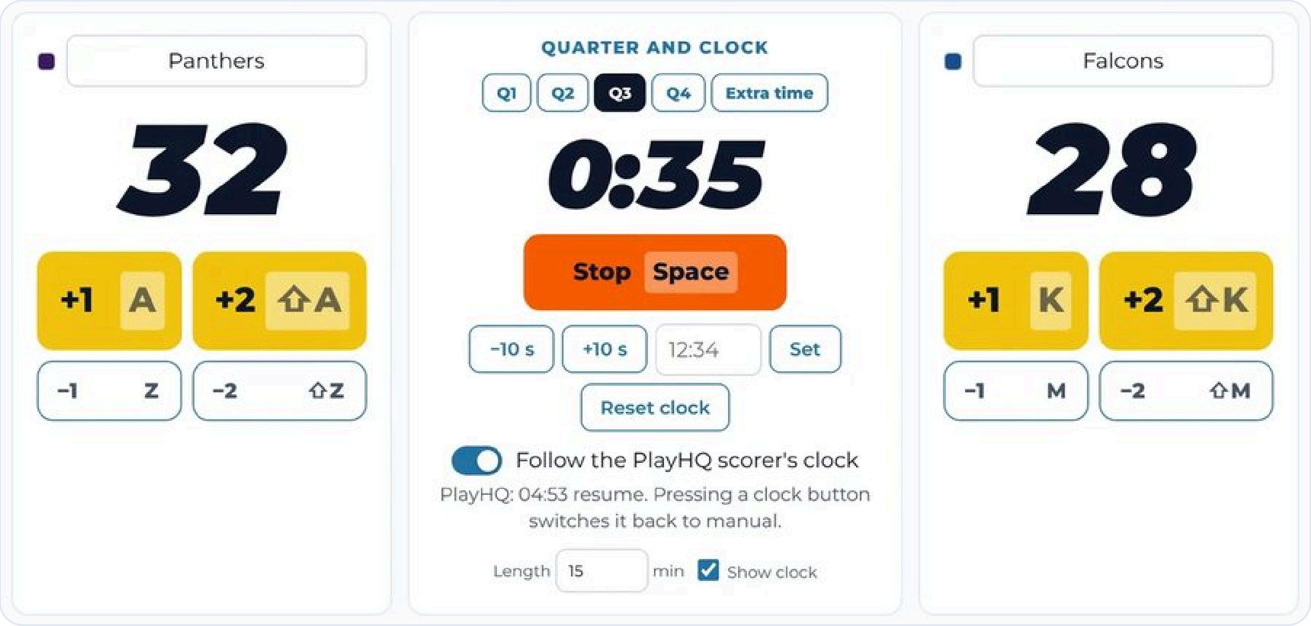
🟧 Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again. If PlayHQ is still behind your score when you press it (the scorebench hasn't caught up yet), the board keeps your score, quarter and clock until PlayHQ catches up or has something new from the scorer, then follows it; the PlayHQ chip reads **your score held** meanwhile. So you can press Follow PlayHQ as soon as the scorebench is back without losing the goals you scored by hand.



● Follow the PlayHQ scorer's clock

PlayHQ: 04:53 resume. Pressing a clock button switches it back to manual.

Length 15 min ☑ Show clock

Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

If the scorebench drops off line instead (its wifi, say), PlayHQ still answers but its score stops moving. Studio spots this two ways: PlayHQ says the game is under way while its live feed says it isn't being scored (or it's five minutes past the start), or nothing new comes from the scorer for four minutes while its clock runs. Then the bar's PlayHQ chip turns purple and reads **PlayHQ not live**, and the status board's PlayHQ line reads **Not live**; hover over either for why. Fix the scorebench if you can, or score by hand until it's back and press Follow PlayHQ: your score stays on the board until PlayHQ catches up. When PlayHQ catches up with several goals at once, the score jumps straight to it, the goals that came together show as a gap on the last 10 goals and the score worm (their order and times aren't known), and no super shot or moment plays for them. If it catches up before the next quarter starts, as it does when the scorebench comes back at quarter time, that quarter's score still counts on the game stats.

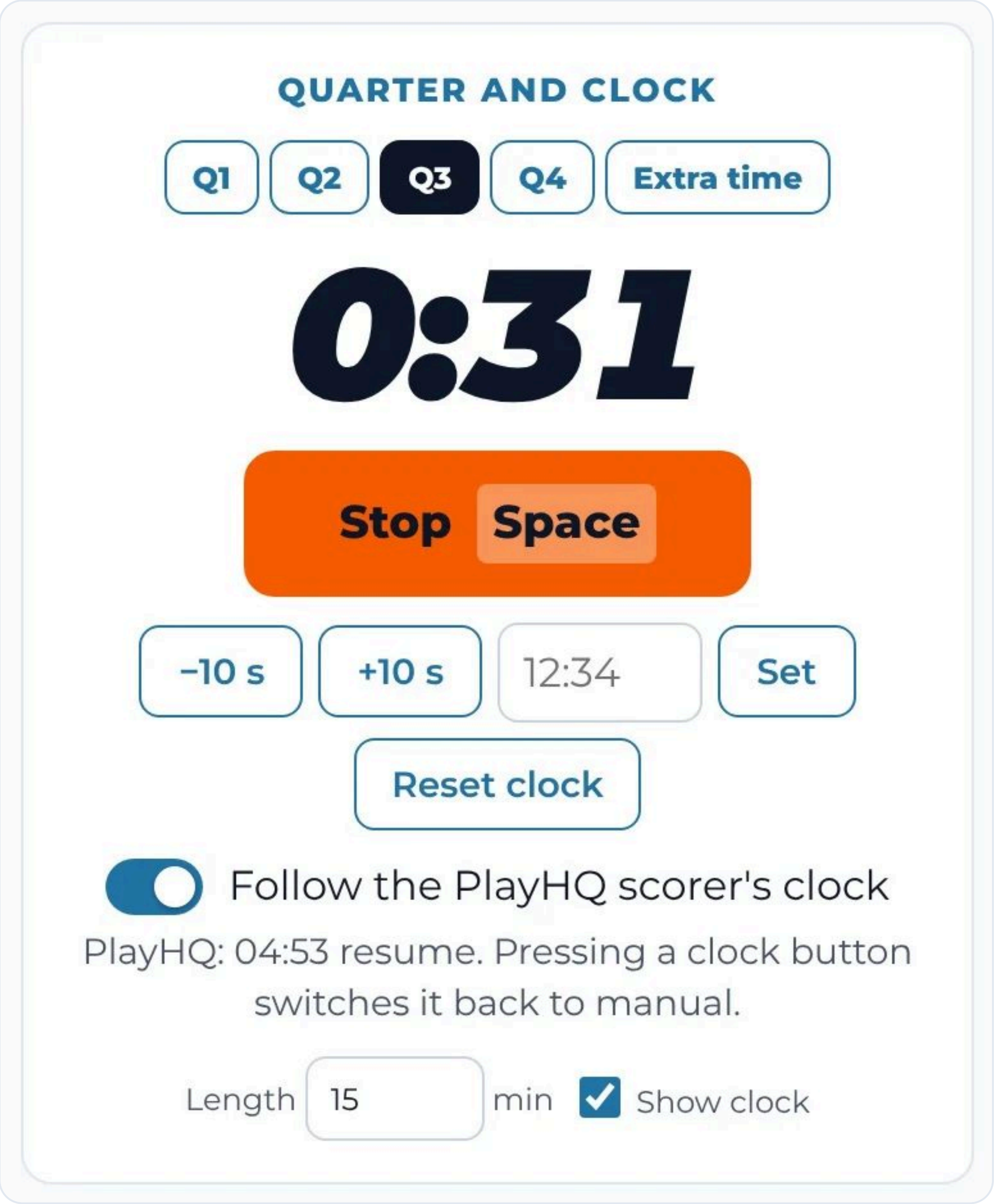
The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.



Follow the PlayHQ scorer's clock

PlayHQ: 04:53 resume. Pressing a clock button switches it back to manual.

Length 15 min ☑ Show clock

Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

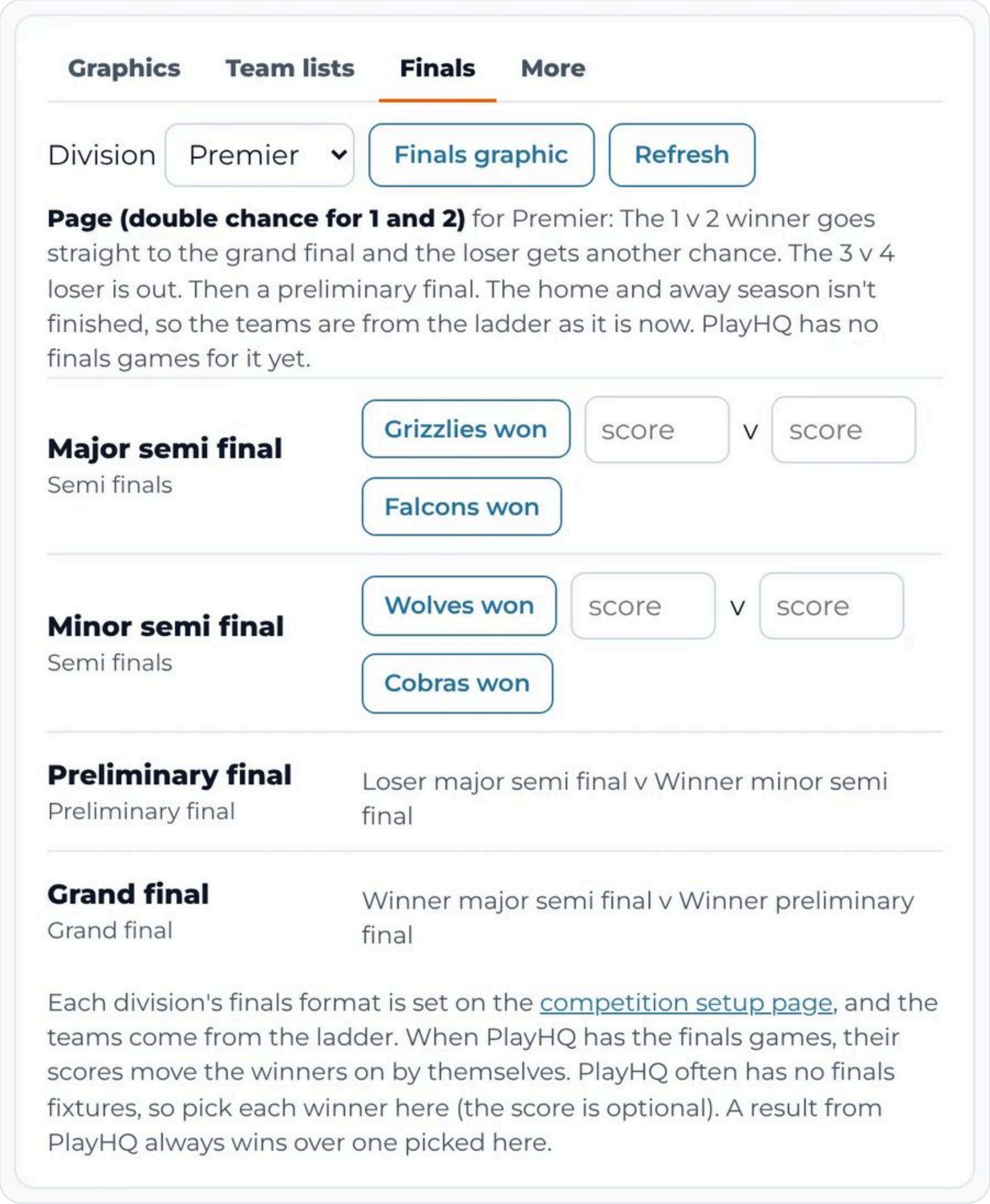
- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time

it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.

- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

Finals

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.


Graphics
Team lists
Finals
More
Division
Premier
Finals graphic
Refresh

Page (double chance for 1 and 2) for Premier: The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. The home and away season isn't finished, so the teams are from the ladder as it is now. PlayHQ has no finals games for it yet.

Major semi final
Semi finals
Grizzlies won
score
v
score
Falcons won

Minor semi final
Semi finals
Wolves won
score
v
score
Cobras won

Preliminary final
Preliminary final
Loser major semi final v Winner minor semi final

Grand final
Grand final
Winner major semi final v Winner preliminary final

Each division's finals format is set on the competition setup page, and the teams come from the ladder. When PlayHQ has the finals games, their scores move the winners on by themselves. PlayHQ often has no finals fixtures, so pick each winner here (the score is optional). A result from PlayHQ always wins over one picked here.

Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

The graphics

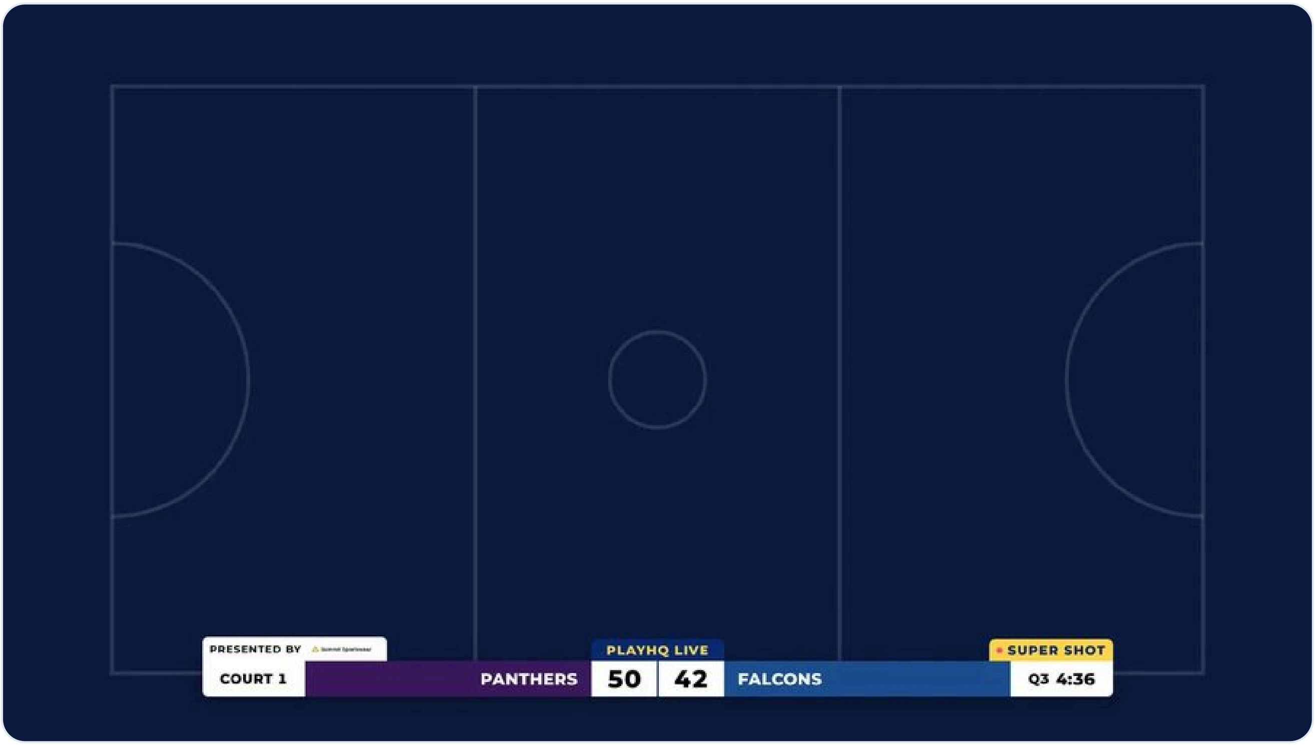
Every graphic Studio draws, grouped by when it is used. Each one is added to the scene with **Add page** on the Output page, and its options appear as fields under its address; the options named here are the ones worth knowing.

IN THIS CHAPTER

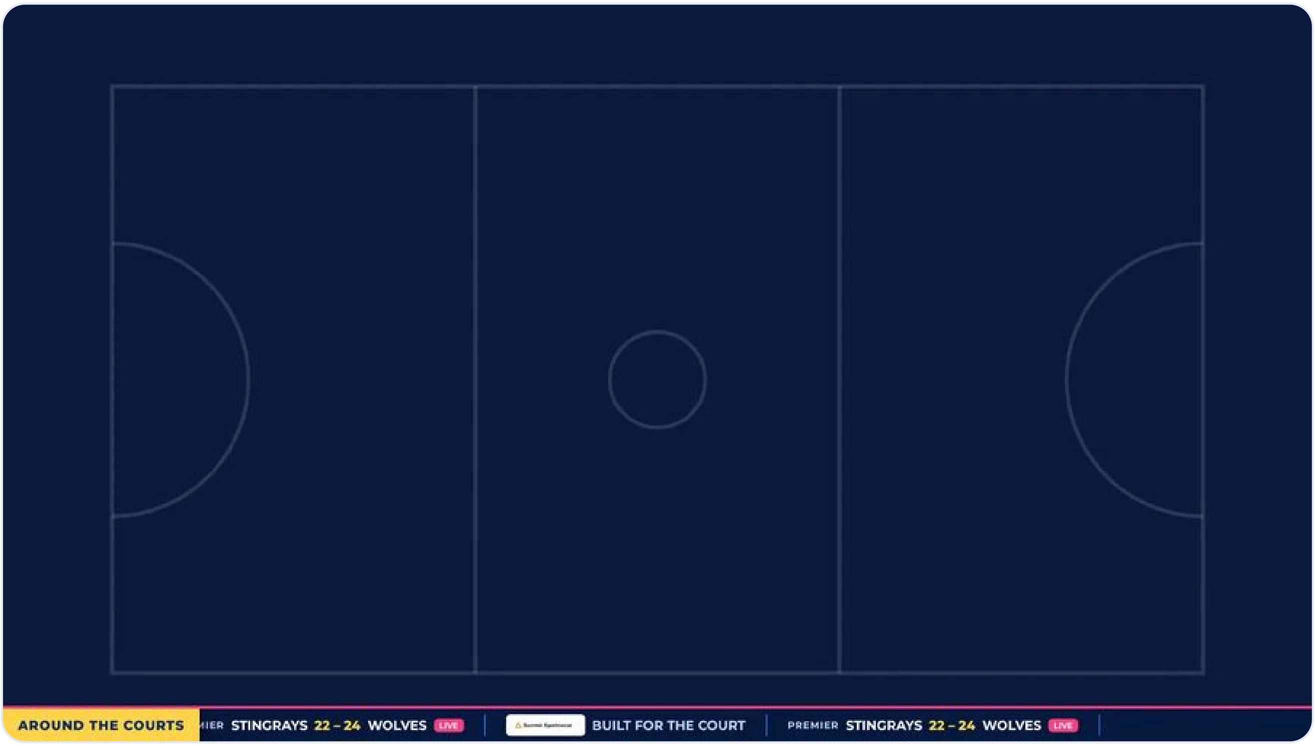
1 Always on
2 Before the game
3 During play
4 After the game
5 The competition
6 The breaks
7 Checking the outputs

All of them follow the game night, game and court picked on Scoreboard control, take the competition's colours, font and logos from its setup, and fit long team names. Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits. Every page, with its link, its options and a Preview button, is also listed under **Links** on the Devices tab.

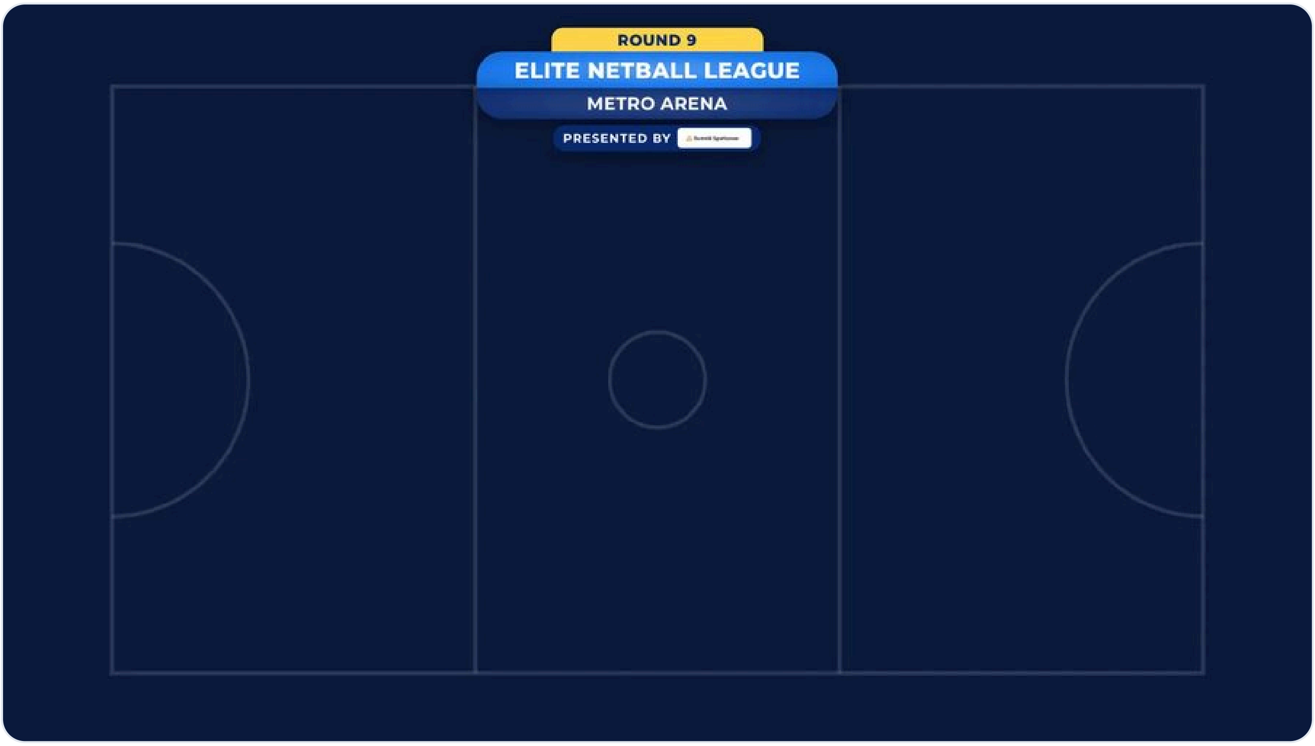
Always on


PRESENTED BY
COURT 1
PANTHERS
PLAYHQ LIVE
50
42
FALCONS
SUPER SHOT
Q3 4:36

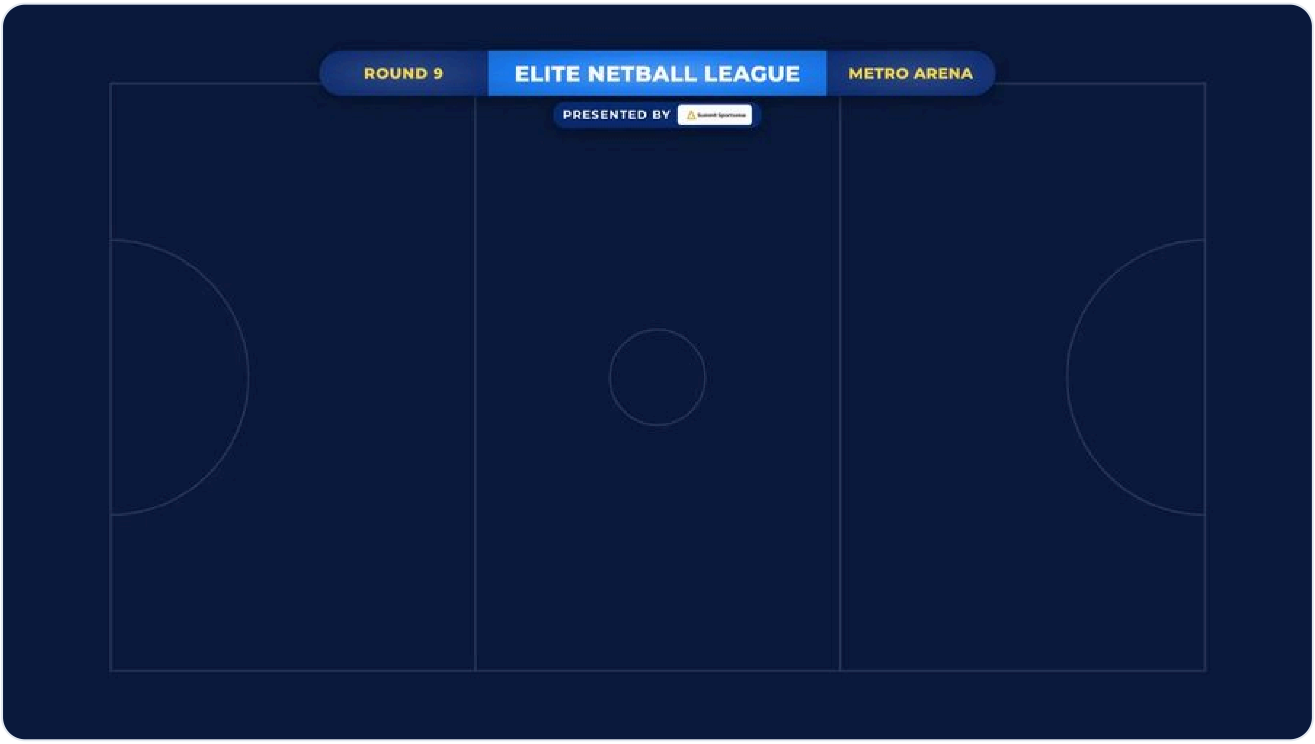
Scoreboard. Court (PlayHQ's name for it, or the one typed on Scoreboard control), both teams with their logos and club colours, the score, the quarter and the clock, centred at the bottom above the ticker. Options: position (top centre, bottom left and so on), super shot tab on or off, and the naming-rights sponsor's tab.


AROUND THE COURTS
IER STINGRAYS 22 – 24 WOLVES LIVE
BUILT FOR THE COURT
PREMIER STINGRAYS 22 – 24 WOLVES LIVE

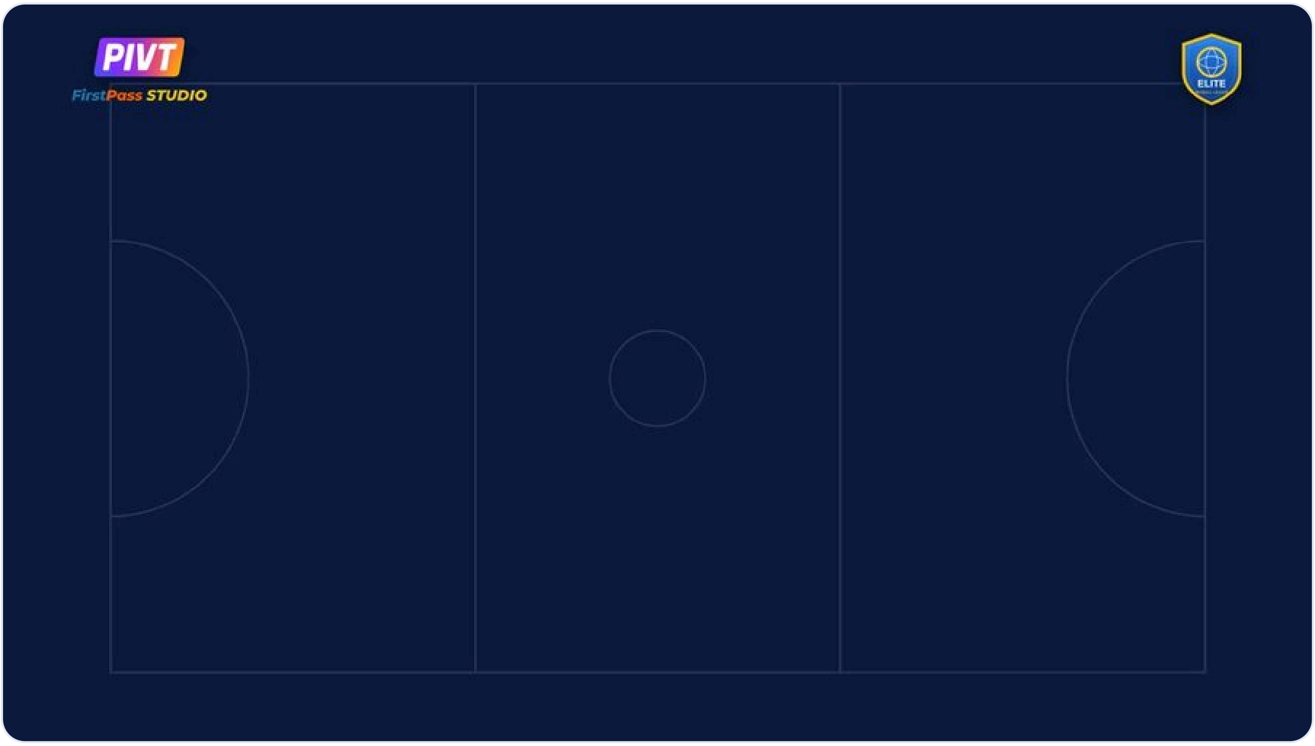
Score ticker. The scores from tonight's other games scrolling along the bottom, leaving out the game on the board, updated every 30 seconds. Speed from the control page or a Stream Deck key; a Partner sponsor's logo takes a slot in the line.


ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY

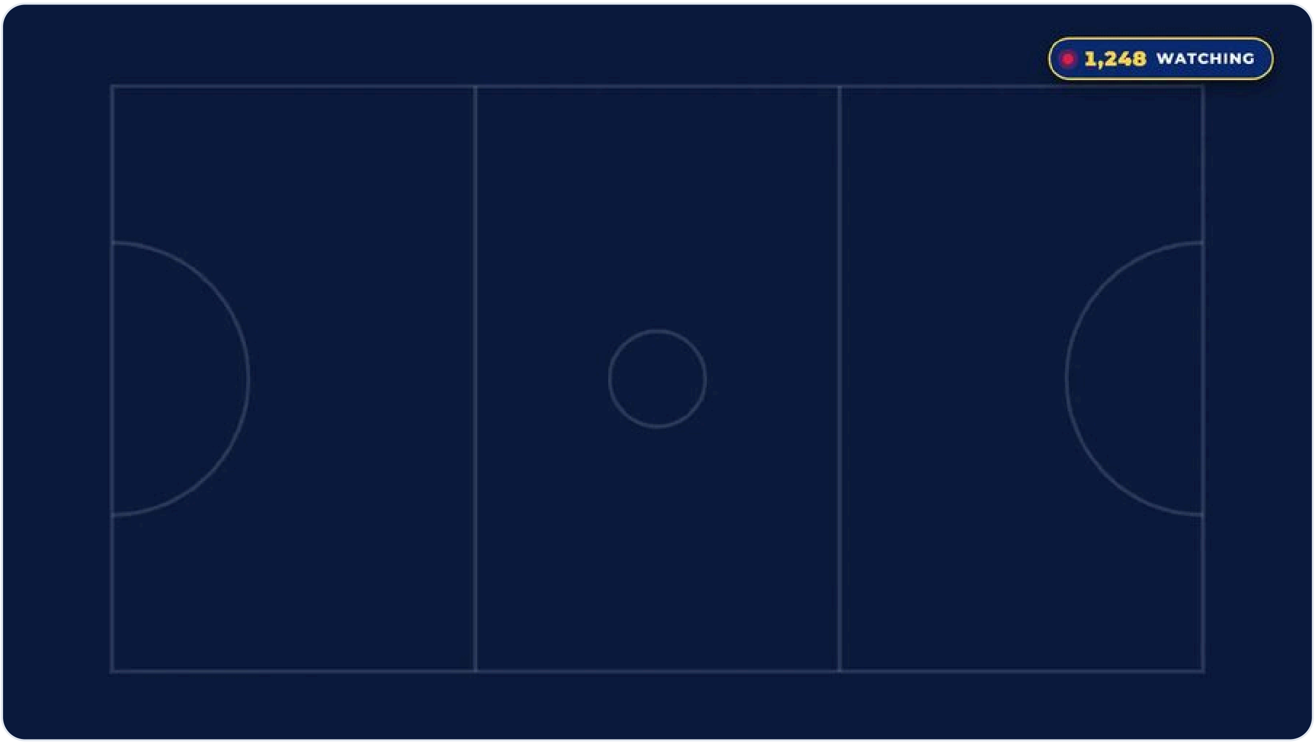
Title. Competition, round and venue in a banner at the top. Options: an emoji either side, the competition logo above it, top or bottom, and the naming-rights sponsor under it.


ROUND 9
ELITE NETBALL LEAGUE
METRO ARENA
PRESENTED BY

Title, wide. The same title as one bar, twice as wide and half as high: round, competition and venue side by side, the name in the middle. Its ends are dark with yellow words, or both yellow, whichever suits the competition's colours. With it, put the commentators box under the title, since the wide bar reaches its corner spot.

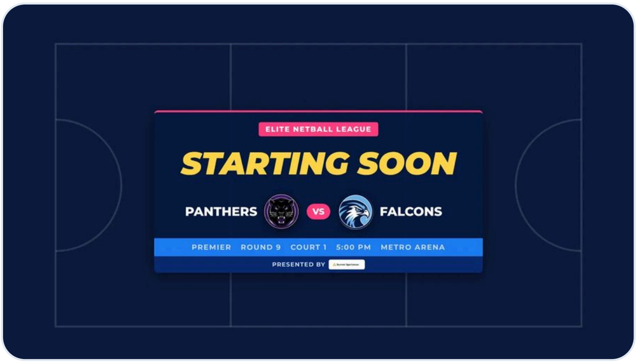


Branding logos. PIVT and FirstPass Studio top left, the competition logo top right, each side 100px in from its edge of the frame so the corners match. Each can be hidden; the gap from the edges (or from the centre instead), height and size adjust the placement.

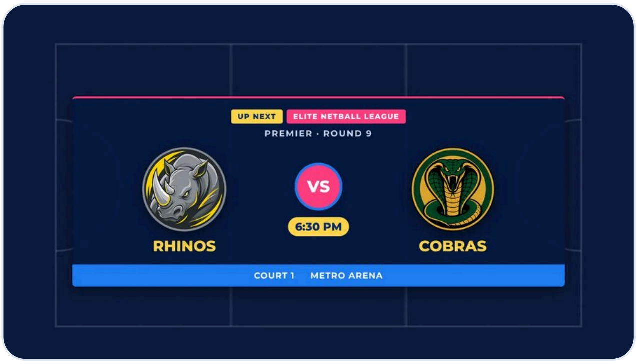


Viewers. "N watching" from the competition's YouTube and Facebook streams, top right, shown only once the count passes the competition's minimum. **Shows: peak** makes it a card for the end of the night.

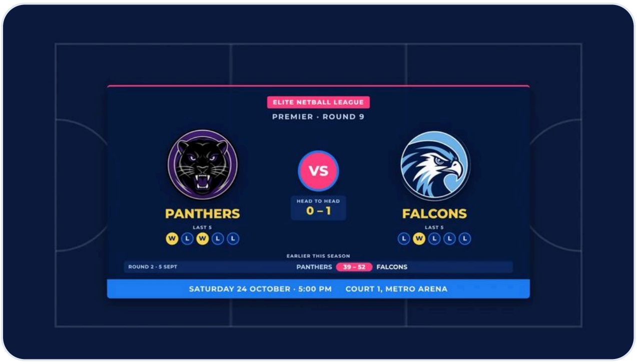
Before the game


ELITE NETBALL LEAGUE
STARTING SOON
PANTHERS
VS
FALCONS
PREMIER
ROUND 9
COURT 1
5:00 PM
METRO ARENA
PRESENTED BY

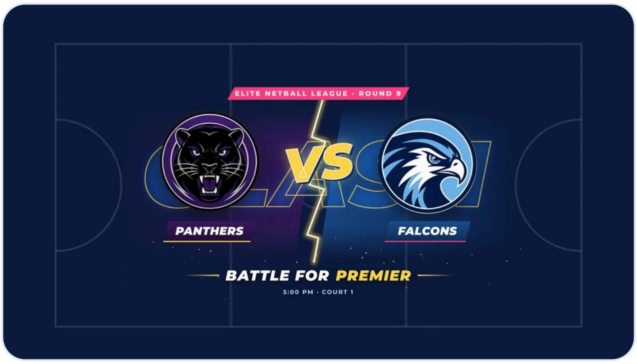
Countdown. Counts down to the picked game's scheduled start, with the two teams. Options: heading, the words at zero, a fade over the last seconds, a sponsor, a **Mini** size (a small timer like the title banner, which can sit where the title does) and a position on a 3 x 3 grid (top, middle or bottom; left, middle or right).


UP NEXT
ELITE NETBALL LEAGUE
PREMIER · ROUND 9
RHINOS
VS
6:30 PM
COBRAS
COURT 1
METRO ARENA

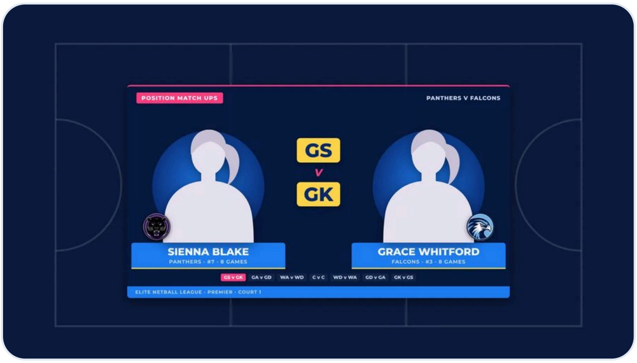
Up next. The next game after the scoreboard's, by time on the same court that day (or any court), whatever round or division it is in, with its time, teams and logos. After the last game it becomes the end-of-night card.


ELITE NETBALL LEAGUE
PREMIER · ROUND 9
VS
HEAD TO HEAD
0 – 1
PANTHERS
LAST 5
W L W L L
FALCONS
LAST 5
L W L L L
EARLIER THIS SEASON
ROUND 2 · 5 SEPT
PANTHERS
39 – 52
FALCONS
SATURDAY 24 OCTOBER · 5:00 PM
COURT 1, METRO ARENA

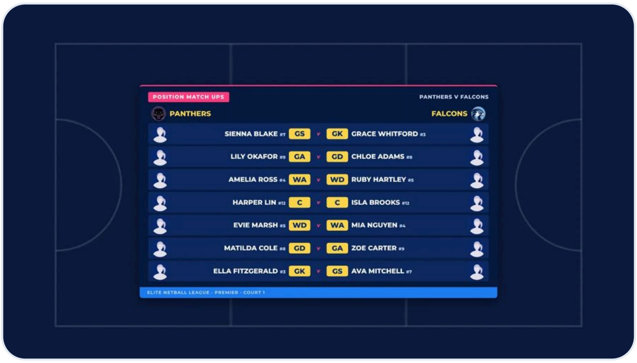
Match up. Both teams side by side with their last five results as form dots and this season's head to head. **Form** and **Head to head** can each be hidden.


ELITE NETBALL LEAGUE · ROUND 9
CLASH
VS
PANTHERS
FALCONS
BATTLE FOR PREMIER
5:00 PM · COURT 1

The Clash. The two teams slam together behind a big word (IMPACT, CLASH, SHOWDOWN, your own) in their club colours. Options: the word, the bottom line, animate away after N seconds.

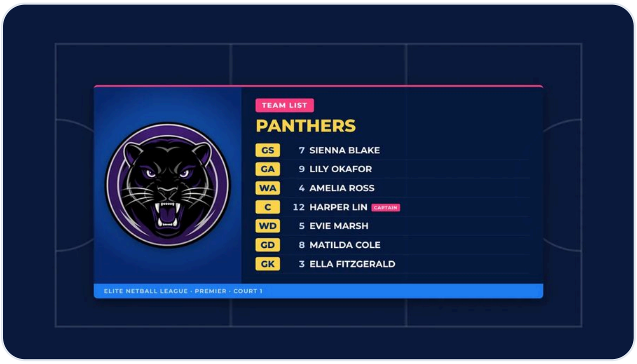

POSITION MATCH UPS
PANTHERS V FALCONS
GS
v
GK
SIENNA BLAKE
PANTHERS · #7 · 8 GAMES
GRACE WHITFORD
FALCONS · #3 · 8 GAMES
GS v GK
GA v GD
WA v WD
C v C
WD v WA
GD v GA
GK v GS
ELITE NETBALL LEAGUE · PREMIER · COURT 1

Position match ups. Who marks whom, position by position, with the athletes' photos from the Athletes page or a silhouette. Options: seconds per pairing, loop or stop, start with GS or GK, one pairing only, animate away at the end.


POSITION MATCH UPS
PANTHERS V FALCONS
ELITE NETBALL LEAGUE · PREMIER · COURT 1

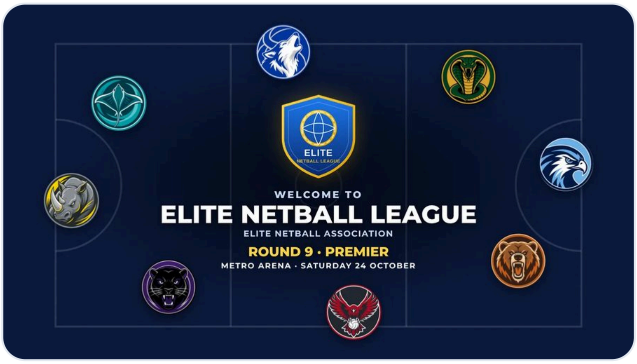
PANTHERS				FALCONS
SIENNA BLAKE #7	GS	v	GK	GRACE WHITFORD #3
LILY OKAFOR #9	GA	v	GD	CHLOE ADAMS #8
AMELIA ROSS #4	WA	v	WD	RUBY HARTLEY #5
HARPER LIN #12	C	v	C	ISLA BROOKS #12
EVIE MARSH #5	WD	v	WA	MIA NGUYEN #4
MATILDA COLE #8	GD	v	GA	ZOE CARTER #9
ELLA FITZGERALD #3	GK	v	GS	AVA MITCHELL #7

Position match ups, all seven. The same page can show every pairing at once as one card.


TEAM LIST
PANTHERS
ELITE NETBALL LEAGUE · PREMIER · COURT 1

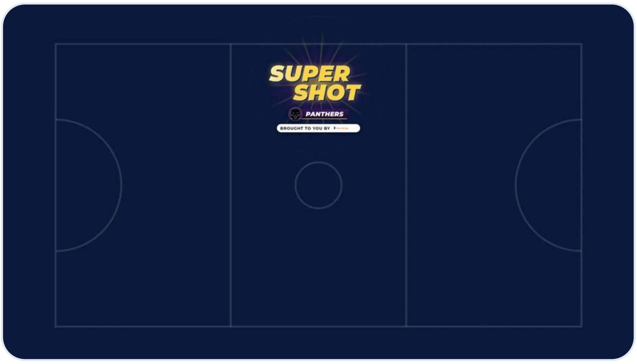
GS	7	SIENNA BLAKE
GA	9	LILY OKAFOR
WA	4	AMELIA ROSS
C	12	HARPER LIN CAPTAIN
WD	5	EVIE MARSH
GD	8	MATILDA COLE
GK	3	ELLA FITZGERALD

Team list. One team's line-up from PlayHQ with positions, captain and bench, warmed up before the game so it shows at once. Add one source per side.

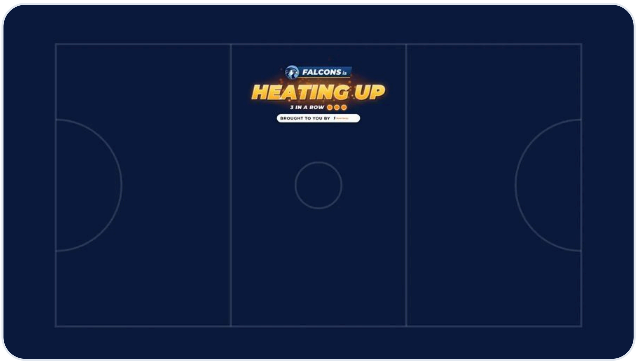

ELITE
NETBALL LEAGUE
WELCOME TO
ELITE NETBALL LEAGUE
ELITE NETBALL ASSOCIATION
ROUND 9 · PREMIER
METRO ARENA · SATURDAY 24 OCTOBER

Intro and outro. Six full-screen styles, about 8 seconds, with the round's teams and a yellow message line. Options: style, teams from a division or round, time on screen, a theme song with its timing and volume, sound effects, a background.

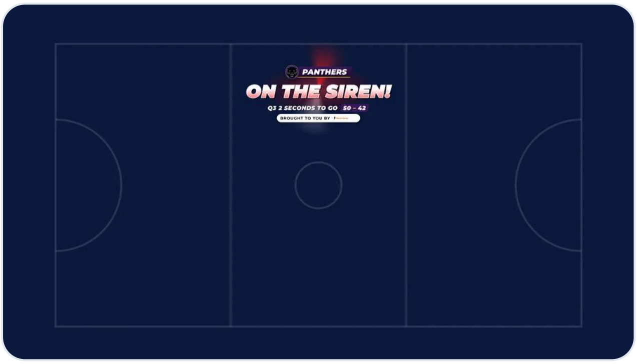
During play


SUPER SHOT
PANTHERS
BROUGHT TO YOU BY

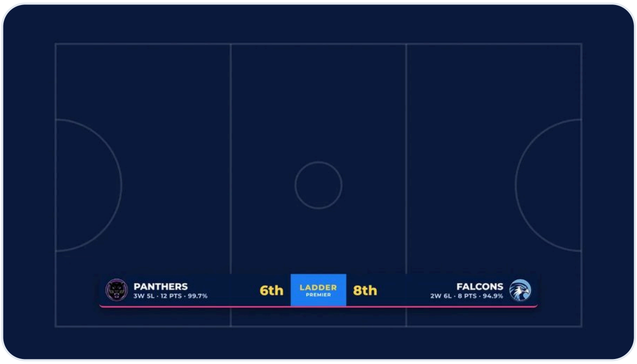
Super shot. Automatic, 3 seconds, the scoring team's logo and colour; the sponsored-moment tag names who brought it. **Plays: home / away / no team** makes a manual one.


FALCONS is
HEATING UP
3 IN A ROW
BROUGHT TO YOU BY

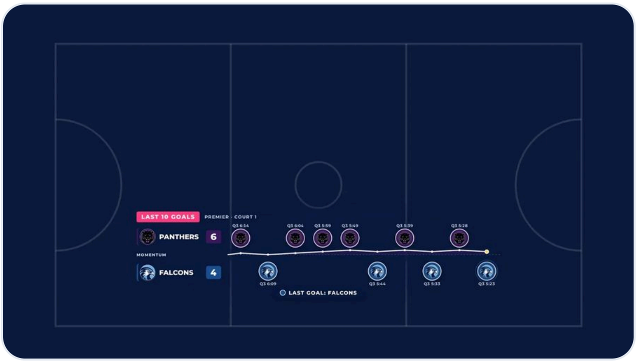
Hot streaks and moments. Heating up, on fire, goal for goal, hits the front, all square, coming back, on the siren, back to back. Automatic from the goal order, or **Plays** picks one to play once.


PANTHERS
ON THE SIREN!
Q3 2 SECONDS TO GO 50 – 42
BROUGHT TO YOU BY

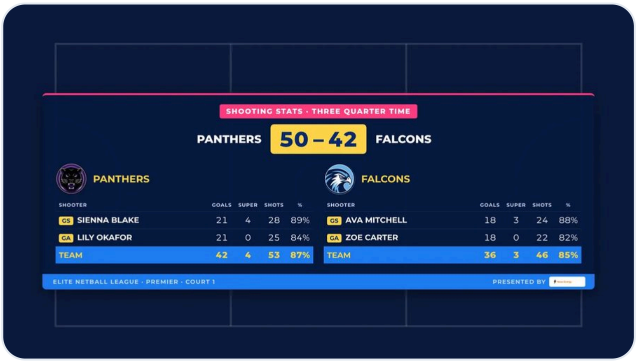
On the siren. A goal in the last seconds of a quarter. Moments are the only animations that need nothing from you; they also take the sponsor's "brought to you by" tag.


PANTHERS
3W 5L · 12 PTS · 99.7%
6th
LADDER
PREMIER
8th
FALCONS
2W 6L · 8 PTS · 94.9%

Ladder lower third. Both teams' ladder position and record as one strip right above the scoreboard, the same width as it, for a glance during play without covering the game.


LAST 10 GOALS
PREMIER · COURT 1
PANTHERS
6
MOMENTUM
FALCONS
4
Q3 6:14
Q3 6:09
Q3 6:04
Q3 5:59
Q3 5:49
Q3 5:44
Q3 5:39
Q3 5:33
Q3 5:28
Q3 5:23
LAST GOAL: FALCONS

Last 10 goals. The game's recent goals oldest to newest with a momentum line and the current run, worked out by Studio from the score, updating while it stays on. Options: horizontal or vertical, how many goals, where it sits (top, middle or bottom, and left, centre or right) and heading.


SHOOTING STATS · THREE QUARTER TIME

PANTHERS 50 – 42 FALCONS

PANTHERS

SHOOTER	GOALS	SUPER	SHOTS	%
GS SIENNA BLAKE	21	4	28	89%
GA LILY OKAFOR	21	0	25	84%
TEAM	42	4	53	87%

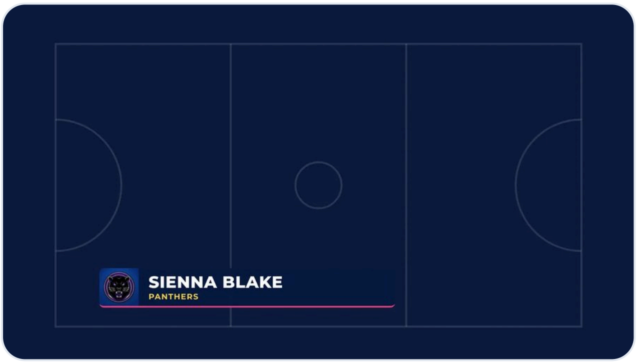
FALCONS

SHOOTER	GOALS	SUPER	SHOTS	%
GS AVA MITCHELL	18	3	24	88%
GA ZOE CARTER	18	0	22	82%
TEAM	36	3	46	85%

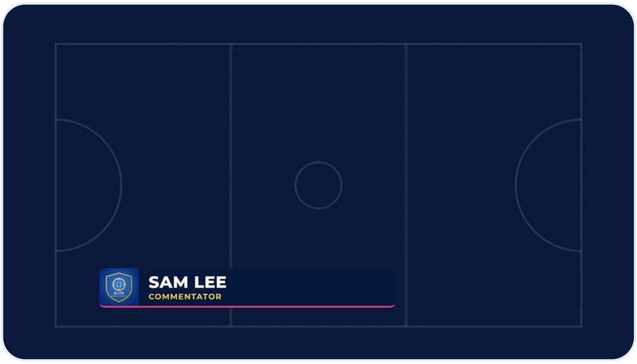
ELITE NETBALL LEAGUE · PREMIER · COURT 1

PRESENTED BY

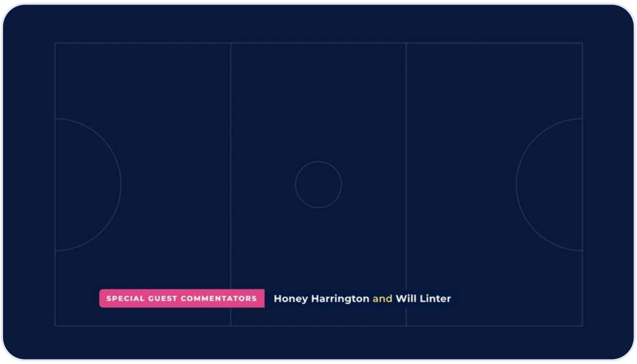
Shooting stats. Goals, super shots and attempts per shooter from PlayHQ's game page, for a break or a quarter-time sequence. Super columns are left off for a division without super shots.


SIENNA BLAKE
PANTHERS

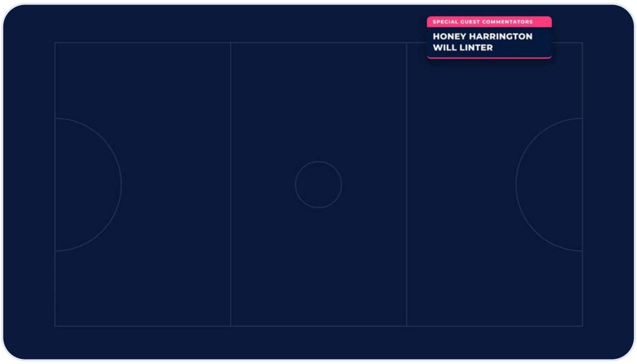
Person lower third, athlete. Name, team and club logo from the athlete register, with their photo if there is one. The Output page suggests names as you type.


SAM LEE
COMMENTATOR

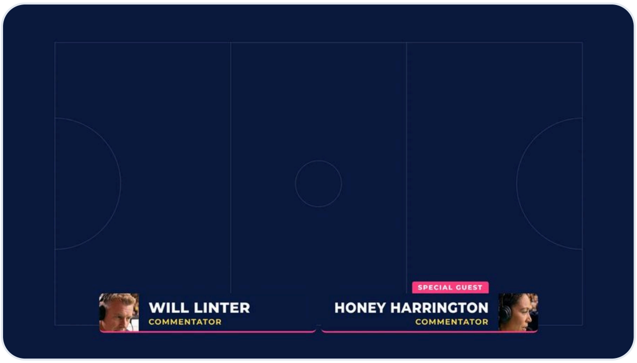
Person lower third, guest. Name, title and the competition logo, for a commentator, an official or a sponsor's guest. Options: a tab above, position left or right, animate away.


SPECIAL GUEST COMMENTATORS
Honey Harrington and Will Linter

Commentators, thin line. The commentary team in one slim strip above the scoreboard: a pink tab, like Special guest commentators, and up to four names joined with "and". With no names typed, whoever the Production Planner has on the mic.

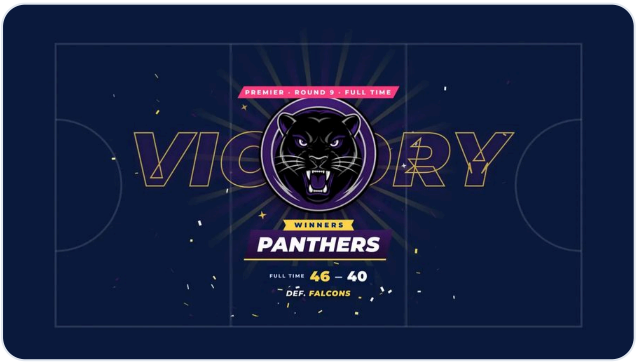

SPECIAL GUEST COMMENTATORS
HONEY HARRINGTON
WILL LINTER

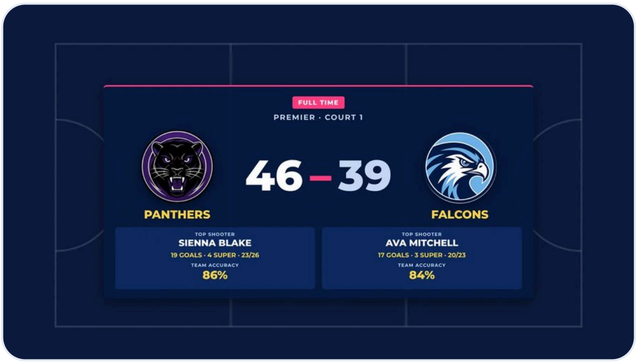
Commentators, box. The same names as an upper third at the top right, just left of the competition logo, a line per name. Options: under the title instead, the tab's words, animate away.


WILL LINTER
COMMENTATOR
SPECIAL GUEST
HONEY HARRINGTON
COMMENTATOR

Commentators, lower thirds. A guest lower third for each person on the mic, from one source: one on the left, two left and right, three across the scoreboard's width. With no names typed, whoever the Production Planner has on, a guest with a Special guest tab. Options: each commentator's headshot in place of the logo, the line under the names, just above the scoreboard instead of at the bottom.

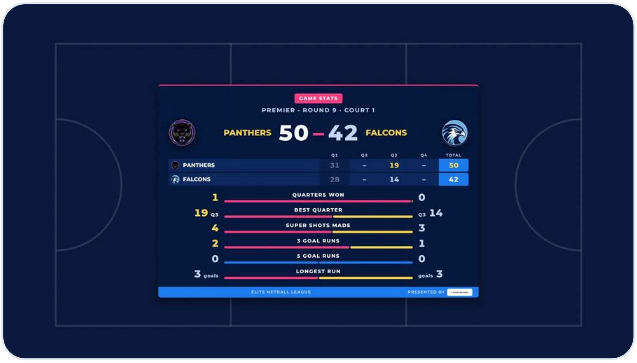
After the game


PREMIER · ROUND 9 · FULL TIME
VIC
ORY
WINNERS
PANTHERS
FULL TIME 46 – 40
DEF. FALCONS

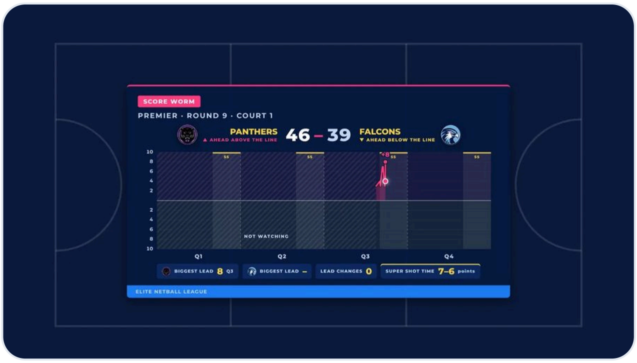

FULL TIME
PREMIER · COURT 1
46 – 39
PANTHERS
FALCONS
TOP SHOOTER
SIENNA BLAKE
19 GOALS · 4 SUPER · 23/26
TEAM ACCURACY
86%
TOP SHOOTER
AVA MITCHELL
17 GOALS · 3 SUPER · 20/23
TEAM ACCURACY
84%

Winner. The winning team, logo, final score and a big word, worked out from the board each time it comes on. Options: force a result, the word, the line under the score, animate away.

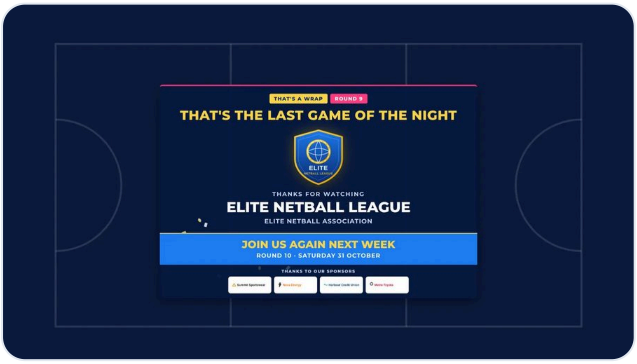
Full time. The final score from the board with each team's top shooter and shooting accuracy from PlayHQ's game stats. **Heading** makes it a half time card.


GAME STATS
PREMIER · ROUND 9 · COURT 1
PANTHERS 50 – 42 FALCONS
QUARTERS WON
BEST QUARTER
SUPER SHOTS MADE
3 GOAL RUNS
5 GOAL RUNS
LONGEST RUN
ELITE NETBALL LEAGUE

Game stats. The score by quarter, tracked by Studio from the board, then team against team: quarters won, best quarter, super shots made, 3 and 5 goal runs and the longest run, with a sponsor's footer.

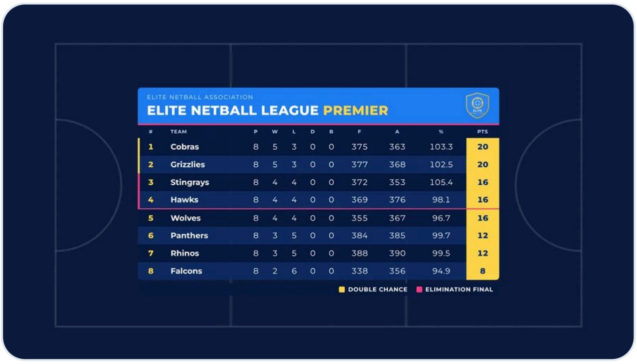

SCORE WORM
PREMIER · ROUND 9 · COURT 1
PANTHERS 46 – 39 FALCONS
NOT WATCHING
Q1
Q2
Q3
Q4
ELITE NETBALL LEAGUE

Score worm. The margin through the game as a line, drawn in over a few seconds. Options: drawing time, club colours or the competition's.


THAT'S A WRAP
ROUND 9
THAT'S THE LAST GAME OF THE NIGHT
THANKS FOR WATCHING
ELITE NETBALL LEAGUE
ELITE NETBALL ASSOCIATION
JOIN US AGAIN NEXT WEEK
ROUND 10 · SATURDAY 31 OCTOBER
THANKS TO OUR SPONSORS

End of night. Up next after the last game: thanks for watching, next week's date or the finals, and the Supporter sponsors in a strip. Options: the invite line, sponsors on or off.

The competition



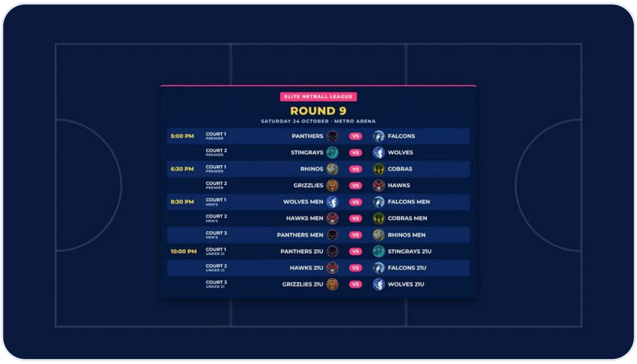
ELITE NETBALL ASSOCIATION

ELITE NETBALL LEAGUE PREMIER

#	TEAM	P	W	L	D	B	F	A	%	PTS
1	Cobras	8	5	3	0	0	375	363	103.3	20
2	Grizzlies	8	5	3	0	0	377	368	102.5	20
3	Stingrays	8	4	4	0	0	372	353	105.4	16
4	Hawks	8	4	4	0	0	369	376	98.1	16
5	Wolves	8	4	4	0	0	355	367	96.7	16
6	Panthers	8	3	5	0	0	384	385	99.7	12
7	Rhinos	8	3	5	0	0	388	390	99.5	12
8	Falcons	8	2	6	0	0	338	356	94.9	8

🟨 DOUBLE CHANCE 🟥 ELIMINATION FINAL

Ladder. The division's ladder with the finals places marked and a key under it. Options: division, rows, logo on or off, light or dark mode logo.



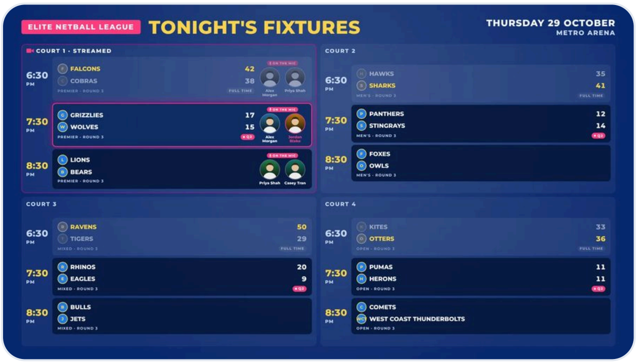
ELITE NETBALL LEAGUE

ROUND 9

SATURDAY 24 OCTOBER · METRO ARENA

5:00 PM	COURT 1 PREMIER	PANTHERS	VS	FALCONS
	COURT 2 PREMIER	STINGRAYS	VS	WOLVES
6:30 PM	COURT 1 PREMIER	RHINOS	VS	COBRAS
	COURT 2 PREMIER	GRIZZLIES	VS	HAWKS
8:30 PM	COURT 1 MEN'S	WOLVES MEN	VS	FALCONS MEN
	COURT 2 MEN'S	HAWKS MEN	VS	COBRAS MEN
	COURT 3 MEN'S	PANTHERS MEN	VS	RHINOS MEN
10:00 PM	COURT 1 UNDER 21	PANTHERS 21U	VS	STINGRAYS 21U
	COURT 2 UNDER 21	HAWKS 21U	VS	FALCONS 21U
	COURT 3 UNDER 21	GRIZZLIES 21U	VS	WOLVES 21U

Round fixture. This round's games across all divisions with day, time and court. **Round** takes next, prev, +2, a number or a name like Final. When divisions are on different rounds, each game shows its own. A finals round PlayHQ has no games for yet shows that week's finals from the finals format, with the time and court TBC.



ELITE NETBALL LEAGUE TONIGHT'S FIXTURES

THURSDAY 29 OCTOBER
METRO ARENA

COURT 1 · STREAMED

6:30 PM
FALCONS 42
COBRAS 38
PREMIER · ROUND 3
FULL TIME
ON THE MIC
Alex Morgan
Priya Shah

7:30 PM
GRIZZLIES 17
WOLVES 15
PREMIER · ROUND 3
● Q2
ON THE MIC
Alex Morgan
Jordan Blake

8:30 PM
LIONS
BEARS
PREMIER · ROUND 3
ON THE MIC
Priya Shah
Casey Tran

COURT 2

6:30 PM
HAWKS 35
SHARKS 41
MEN'S · ROUND 3
FULL TIME

7:30 PM
PANTHERS 12
STINGRAYS 14
MEN'S · ROUND 3
● Q2

8:30 PM
FOXES
OWLS
MEN'S · ROUND 3

COURT 3

6:30 PM
RAVENS 50
TIGERS 29
MIXED · ROUND 3
FULL TIME

7:30 PM
RHINOS 20
EAGLES 9
MIXED · ROUND 3
● Q2

8:30 PM
BULLS
JETS
MIXED · ROUND 3

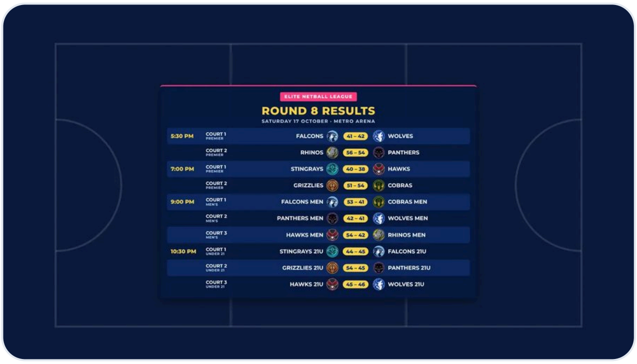
COURT 4

6:30 PM
KITES 33
OTTERS 36
OPEN · ROUND 3
FULL TIME

7:30 PM
PUMAS 11
HERONS 11
OPEN · ROUND 3
● Q2

8:30 PM
COMETS
WEST COAST THUNDERBOLTS
OPEN · ROUND 3

Fixture board (Full Page). The night's whole fixture, a card per court with its games by time, for a screen in the competition room or backstage, or a picture to share before the night (**Tuesday Night's Fixtures**). The streamed court has a camera and its games show the commentators from the Production Planner, **On the mic** with their headshots (a silhouette for anyone without one) where there is room, as name pills on a crowded night; games being played show their live score, and finished ones grey out with the winner in yellow. Not for the stream.



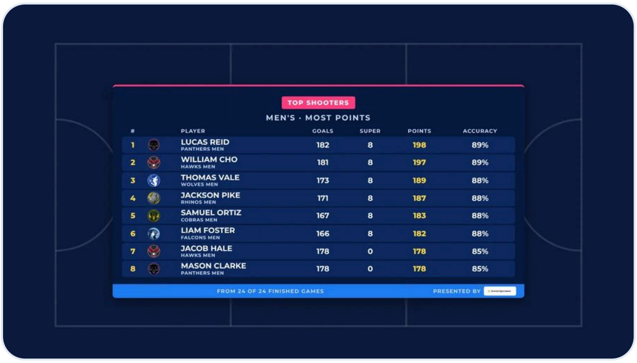
ELITE NETBALL LEAGUE

ROUND 8 RESULTS

SATURDAY 17 OCTOBER · METRO ARENA

5:30 PM	COURT 1 PREMIER	FALCONS	41 – 42	WOLVES
	COURT 2 PREMIER	RHINOS	56 – 54	PANTHERS
7:00 PM	COURT 1 PREMIER	STINGRAYS	40 – 38	HAWKS
	COURT 2 PREMIER	GRIZZLIES	51 – 54	COBRAS
9:00 PM	COURT 1 MEN'S	FALCONS MEN	53 – 41	COBRAS MEN
	COURT 2 MEN'S	PANTHERS MEN	42 – 41	WOLVES MEN
	COURT 3 MEN'S	HAWKS MEN	54 – 42	RHINOS MEN
10:30 PM	COURT 1 UNDER 21	STINGRAYS 21U	44 – 45	FALCONS 21U
	COURT 2 UNDER 21	GRIZZLIES 21U	54 – 45	PANTHERS 21U
	COURT 3 UNDER 21	HAWKS 21U	45 – 46	WOLVES 21U

Results. The same round once its games are final, with scores. The same round option applies.



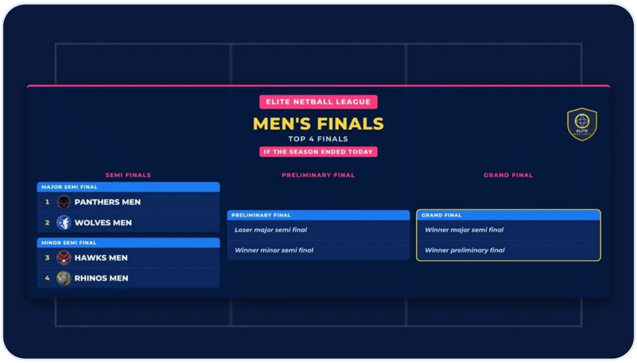
TOP SHOOTERS

MEN'S · MOST POINTS

#	PLAYER	GOALS	SUPER	POINTS	ACCURACY
1	LUCAS REID PANTHERS MEN	182	8	198	89%
2	WILLIAM CHO HAWKS MEN	181	8	197	89%
3	THOMAS VALE WOLVES MEN	173	8	189	88%
4	JACKSON PIKE RHINOS MEN	171	8	187	88%
5	SAMUEL ORTIZ COBRAS MEN	167	8	183	88%
6	LIAM FOSTER FALCONS MEN	166	8	182	88%
7	JACOB HALE HAWKS MEN	178	0	178	85%
8	MASON CLARKE PANTHERS MEN	178	0	178	85%

FROM 24 OF 24 FINISHED GAMES PRESENTED BY

Top shooters. The season's leading shooters for a division. Options: order by goals or average, how many shooters, a sponsor footer.



ELITE NETBALL LEAGUE

MEN'S FINALS

TOP 4 FINALS

IF THE SEASON ENDED TODAY

SEMI FINALS

MAJOR SEMI FINAL
1 PANTHERS MEN
2 WOLVES MEN

MINOR SEMI FINAL
3 HAWKS MEN
4 RHINOS MEN

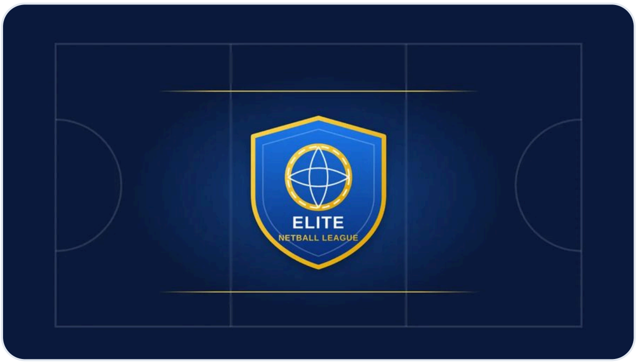
PRELIMINARY FINAL

PRELIMINARY FINAL
Loser major semi final
Winner minor semi final

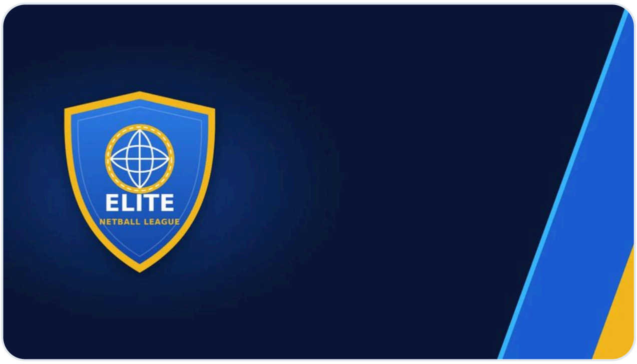
GRAND FINAL

GRAND FINAL
Winner major semi final
Winner preliminary final

Finals. Who plays whom week by week in the division's finals format, with ladder positions and scores; "If the season ended today" until the home-and-away rounds finish. Options: one week only, the heading and the line under it.

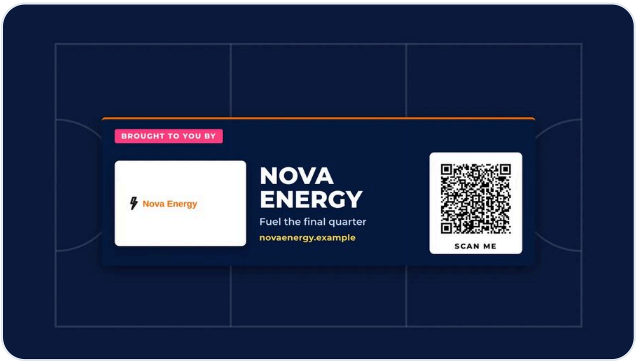


Competition logo animation. Seventeen templates (pro, coin, tiles, glitch, badge, lightning, court, bibs, net and more), about 5 seconds, with optional words and a smaller line.

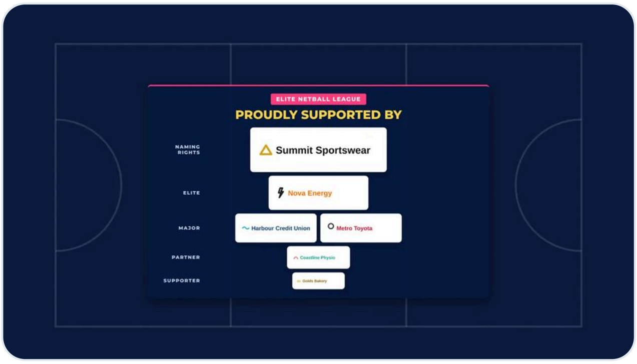


Stinger. A 1.5 second transition: bands in the competition's colours sweep across and cover the whole picture with the competition logo for half a second, so a cut can happen underneath, then carry on off the other side. Options: hold, length, logo on or off, light or dark mode logo, whoosh volume, and the music ducked for the next scene or while it plays.

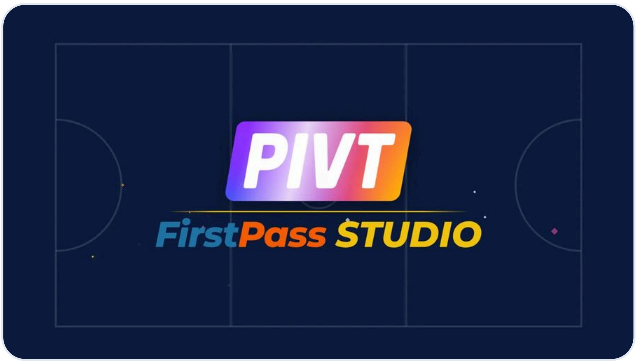
The breaks


BROUGHT TO YOU BY
Nova Energy
NOVA ENERGY
Fuel the final quarter
novaenergy.example
SCAN ME

Sponsor card. One sponsor's logo, tagline, offer and a QR code to their website, taking turns by tier. Options: one sponsor always, which tiers rotate, animate away.

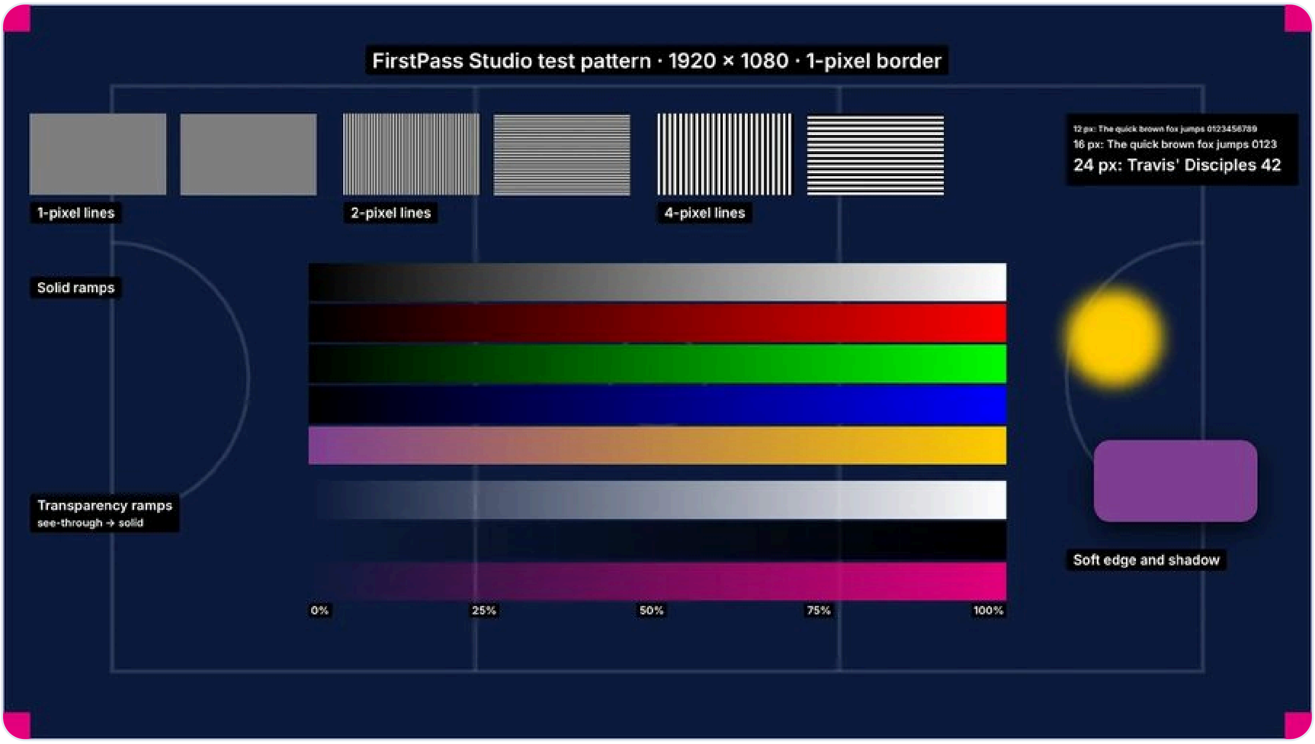

ELITE NETBALL LEAGUE
PROUDLY SUPPORTED BY
NAMING RIGHTS
Summit Sportswear
ELITE
Nova Energy
MAJOR
Harbour Credit Union
Metro Toyota
PARTNER
Coastline Physio
SUPPORTER
Golds Bakery

Sponsor wall. Every sponsor by tier on one card, for the run-up and the close. Options: which tiers, names under the logos.


PIVT
FirstPass STUDIO

PIVT and FirstPass Studio ad. A 10-second house ad for a break with nothing sold. A Sponsor ads step plays it when no sponsor's ad is due, if **House ad** is ticked.

Checking the outputs


FirstPass Studio test pattern · 1920 × 1080 · 1-pixel border
1-pixel lines
2-pixel lines
4-pixel lines
12 px: The quick brown fox jumps 0123456789
16 px: The quick brown fox jumps 0123
24 px: Travis' Disciples 42
Solid ramps
Transparency ramps
see-through → solid
Soft edge and shadow
0%
25%
50%
75%
100%

Test pattern. Borders, fine lines, ramps and soft shapes to check an output, the switcher's scaling and its key end to end before the stream. Never for air.

CHAPTER 6 OF 11

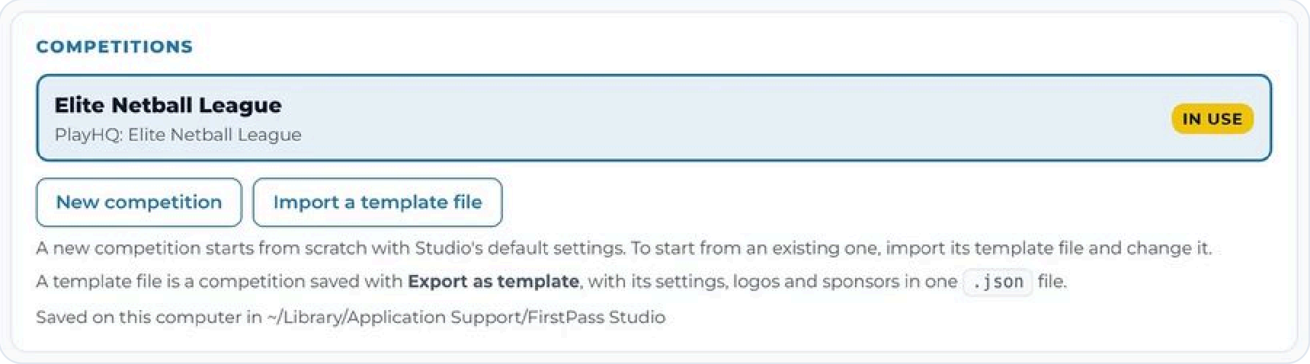
Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

IN THIS CHAPTER

1 PlayHQ
2 Names on the graphics
3 Game rules
4 Finals
5 Game night
6 Colours and font
7 Logos
8 Live viewers
9 Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.



A new competition starts from scratch with Studio's default settings. To start from an existing one, import its template file and change it.

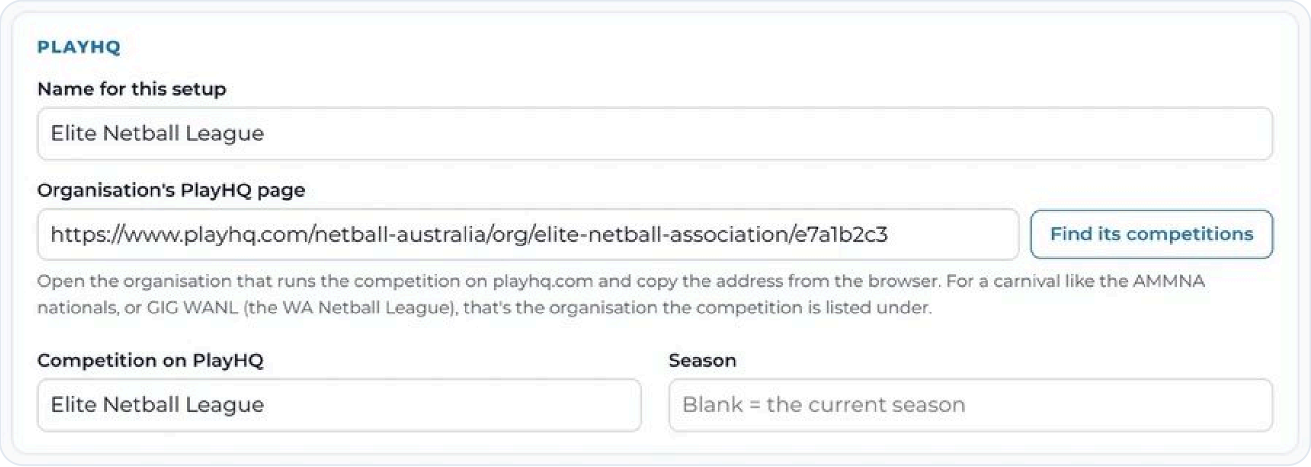
A template file is a competition saved with **Export as template**, with its settings, logos and sponsors in one `.json` file.

Saved on this computer in ~/Library/Application Support/FirstPass Studio

The competitions list. Start from a template file (the repository has Thunder League, AMMNA Nationals, GIG WANL and West Coast Thunder) or press New competition for Studio's defaults.

PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.



PLAYHQ

Name for this setup: Elite Netball League

Organisation's PlayHQ page: https://www.playhq.com/netball-australia/org/elite-netball-association/e7a1b2c3

Open the organisation that runs the competition on playhq.com and copy the address from the browser. For a carnival like the AMMNA nationals, or GIG WANL (the WA Netball League), that's the organisation the competition is listed under.

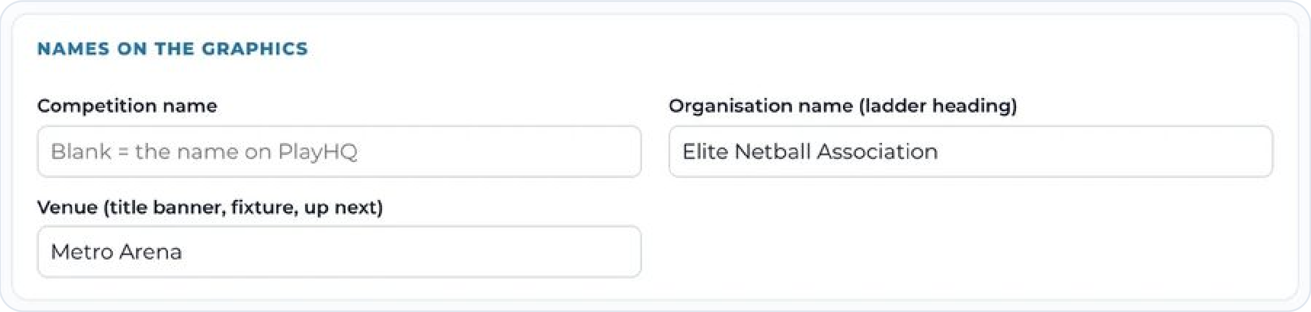
Competition on PlayHQ: Elite Netball League

Season: Blank = the current season

The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.



NAMES ON THE GRAPHICS

Competition name: Blank = the name on PlayHQ

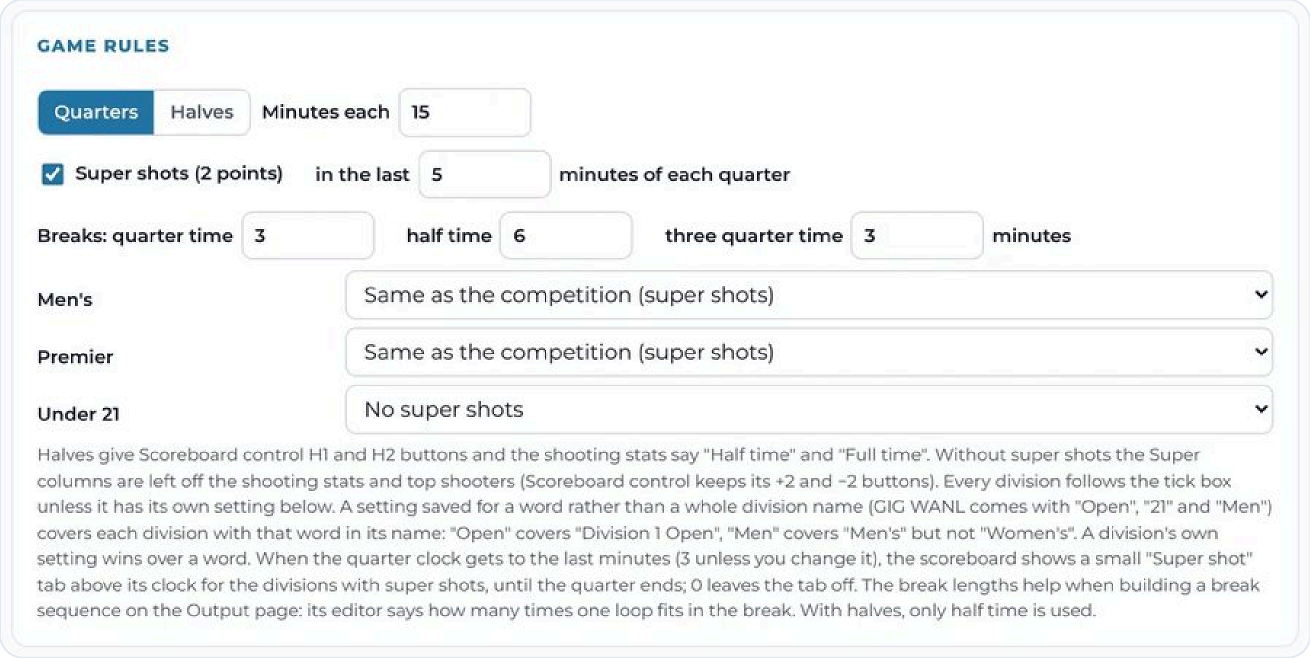
Organisation name (ladder heading): Elite Netball Association

Venue (title banner, fixture, up next): Metro Arena

Names on the graphics.

Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.



GAME RULES

Minutes each: 15

☑ Super shots (2 points) in the last 5 minutes of each quarter

Breaks: quarter time 3 half time 6 three quarter time 3 minutes

Men's: Same as the competition (super shots)

Premier: Same as the competition (super shots)

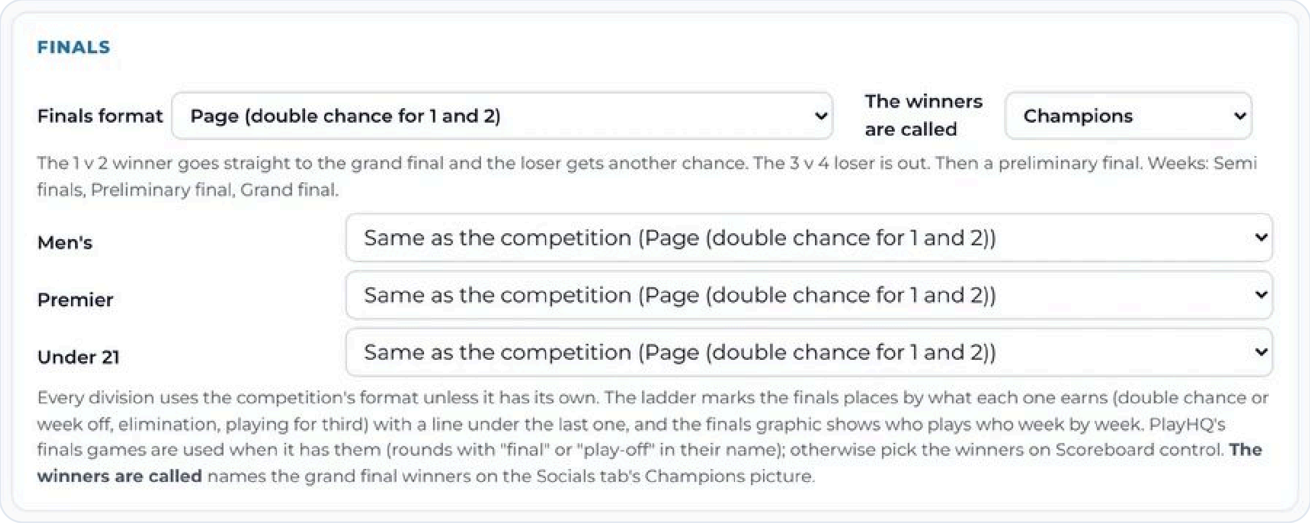
Under 21: No super shots

Halves give Scoreboard control H1 and H2 buttons and the shooting stats say "Half time" and "Full time". Without super shots the Super columns are left off the shooting stats and top shooters (Scoreboard control keeps its +2 and −2 buttons). Every division follows the tick box unless it has its own setting below. A setting saved for a word rather than a whole division name (GIG WANL comes with "Open", "21" and "Men") covers each division with that word in its name: "Open" covers "Division 1 Open", "Men" covers "Men's" but not "Women's". A division's own setting wins over a word. When the quarter clock gets to the last minutes (3 unless you change it), the scoreboard shows a small "Super shot" tab above its clock for the divisions with super shots, until the quarter ends; 0 leaves the tab off. The break lengths help when building a break sequence on the Output page: its editor says how many times one loop fits in the break. With halves, only half time is used.

Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control. **The winners are called** picks Champions or Premiers for the Socials tab's picture of the grand final winners.



FINALS

Finals format: Page (double chance for 1 and 2)

The winners are called: Champions

The 1 v 2 winner goes straight to the grand final and the loser gets another chance. The 3 v 4 loser is out. Then a preliminary final. Weeks: Semi finals, Preliminary final, Grand final.

Men's: Same as the competition (Page (double chance for 1 and 2))

Premier: Same as the competition (Page (double chance for 1 and 2))

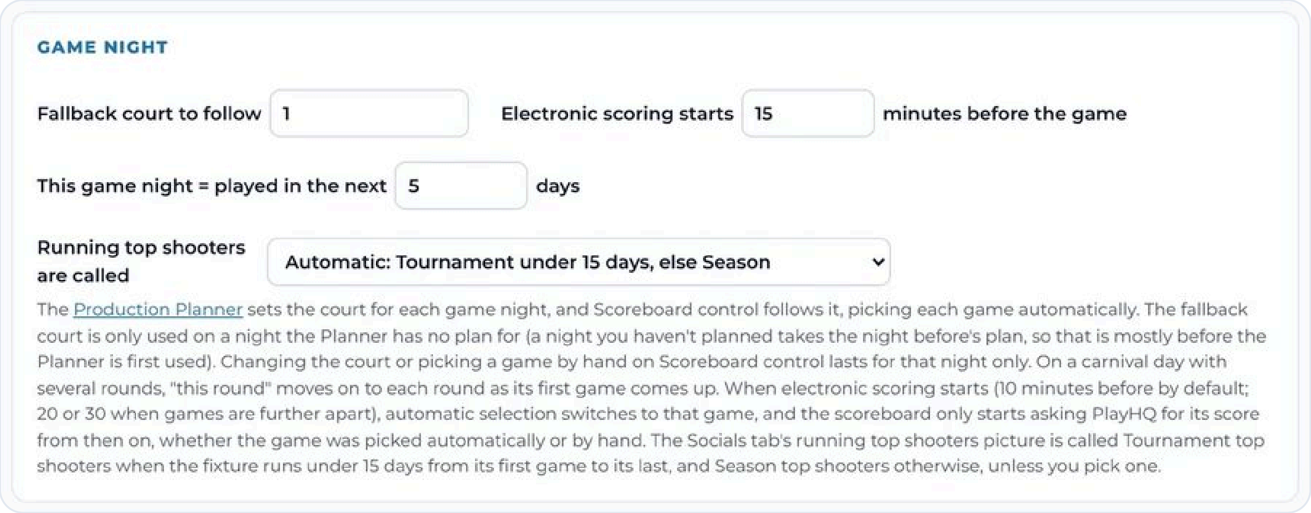
Under 21: Same as the competition (Page (double chance for 1 and 2))

Every division uses the competition's format unless it has its own. The ladder marks the finals places by what each one earns (double chance or week off, elimination, playing for third) with a line under the last one, and the finals graphic shows who plays who week by week. PlayHQ's finals games are used when it has them (rounds with "final" or "play-off" in their name); otherwise pick the winners on Scoreboard control. **The winners are called** names the grand final winners on the Socials tab's Champions picture.

Finals formats, per division where they differ.

Game night

The **Fallback court to follow**, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. The Production Planner sets each game night's court; the fallback is only used on a night with no plan, which is mostly before the Planner is first used, since a night you haven't planned takes the night before's plan. Every game night starts with the game picked automatically. Scoreboard control can change the court, pick a game by hand or pick another game night, for that night only. On a carnival day with several rounds, this round moves on as each round's first game comes up. **Running top shooters are called** sets the Socials picture's name: Automatic calls it Tournament top shooters when the fixture runs under 15 days and Season top shooters otherwise.



GAME NIGHT

Fallback court to follow: 1

Electronic scoring starts: 15 minutes before the game

This game night = played in the next: 5 days

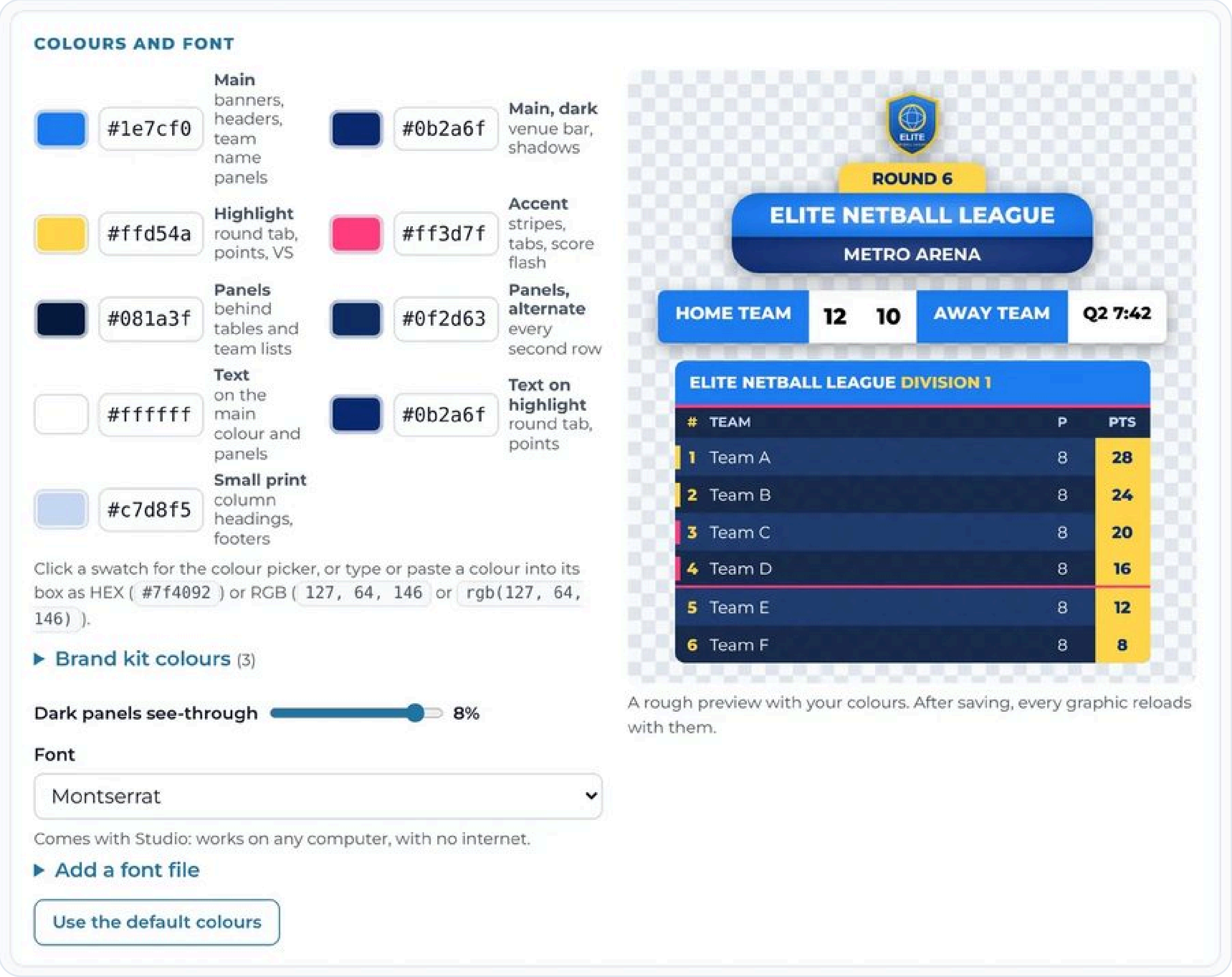
Running top shooters are called: Automatic: Tournament under 15 days, else Season

The Production Planner sets the court for each game night, and Scoreboard control follows it, picking each game automatically. The fallback court is only used on a night the Planner has no plan for (a night you haven't planned takes the night before's plan, so that is mostly before the Planner is first used). Changing the court or picking a game by hand on Scoreboard control lasts for that night only. On a carnival day with several rounds, "this round" moves on to each round as its first game comes up. When electronic scoring starts (10 minutes before by default; 20 or 30 when games are further apart), automatic selection switches to that game, and the scoreboard only starts asking PlayHQ for its score from then on, whether the game was picked automatically or by hand. The Socials tab's running top shooters picture is called Tournament top shooters when the fixture runs under 15 days from its first game to its last, and Season top shooters otherwise, unless you pick one.

Game night.

Colours and font

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.



COLOURS AND FONT

🟦	`#1e7cf0`	**Main** banners, headers, team name panels	🟦	`#0b2a6f`	**Main, dark** venue bar, shadows
🟨	`#ffd54a`	**Highlight** round tab, points, VS	🟥	`#ff3d7f`	**Accent** stripes, tabs, score flash
⬛	`#081a3f`	**Panels** behind tables and team lists	🟦	`#0f2d63`	**Panels, alternate** every second row
⬜	`#ffffff`	**Text** on the main colour and panels	🟦	`#0b2a6f`	**Text on highlight** round tab, points
🟦	`#c7d8f5`	**Small print** column headings, footers			

Click a swatch for the colour picker, or type or paste a colour into its box as HEX (`#7f4092`) or RGB (`127, 64, 146` or `rgb(127, 64, 146)`).

▸ Brand kit colours (3)

Dark panels see-through 8%

Font

Montserrat

Comes with Studio: works on any computer, with no internet.

▸ Add a font file

Use the default colours

ELITE NETBALL LEAGUE DIVISION 1

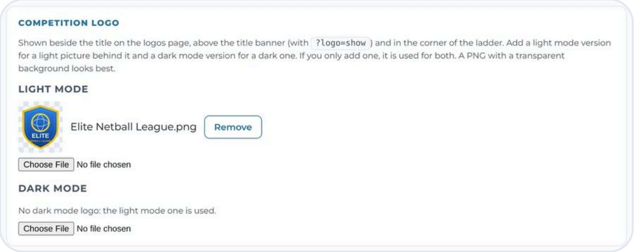
#	TEAM	P	PTS
1	Team A	8	28
2	Team B	8	24
3	Team C	8	20
4	Team D	8	16
5	Team E	8	12
6	Team F	8	8

A rough preview with your colours. After saving, every graphic reloads with them.

Colours and font, with the preview.

Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 21U" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.



COMPETITION LOGO

Shown beside the title on the logos page, above the title banner (with `?logo=show`) and in the corner of the ladder. Add a light mode version for a light picture behind it and a dark mode version for a dark one. If you only add one, it is used for both. A PNG with a transparent background looks best.

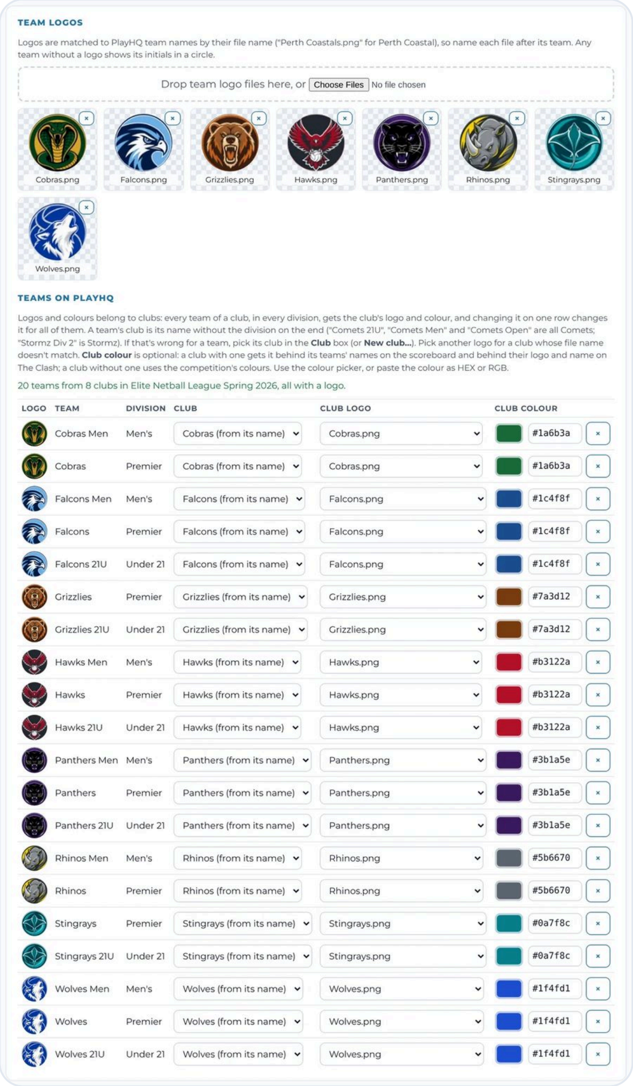
LIGHT MODE

Elite Netball League.png

DARK MODE

No dark mode logo: the light mode one is used.

Competition logo, light and dark mode.



TEAM LOGOS

Logos are matched to PlayHQ team names by their file name ("Perth Coastals.png" for Perth Coastal), so name each file after its team. Any team without a logo shows its initials in a circle.

Drop team logo files here, or

TEAMS ON PLAYHQ

Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, and changing it on one row changes it for all of them. A team's club is its name without the division on the end ("Comets 21U", "Comets Men" and "Comets Open" are all Comets; "Stormz Div 2" is Stormz). If that's wrong for a team, pick its club in the **Club** box (or **New club...**). Pick another logo for a club whose file name doesn't match. **Club colour** is optional: a club with one gets it behind its teams' names on the scoreboard and behind their logo and name on The Clash; a club without one uses the competition's colours. Use the colour picker, or paste the colour as HEX or RGB.

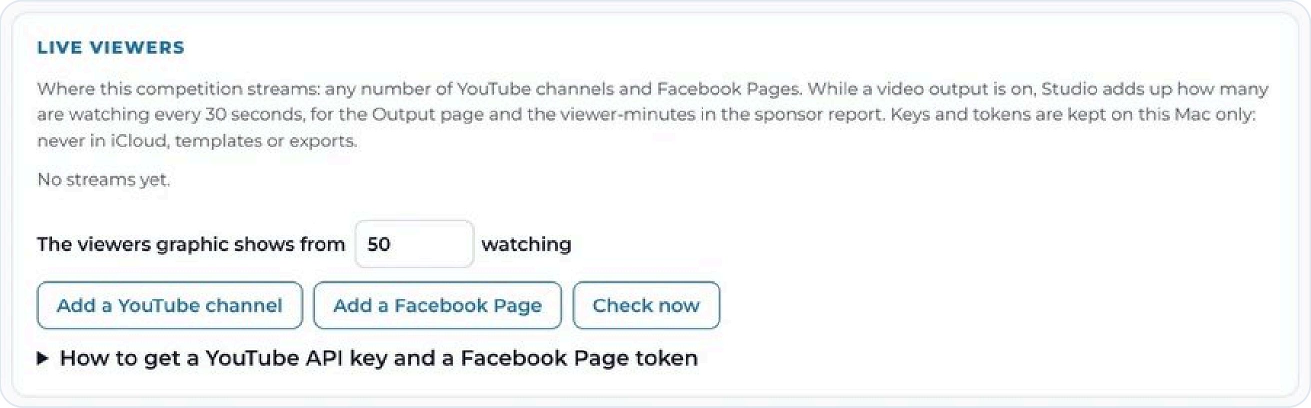
20 teams from 8 clubs in Elite Netball League Spring 2026, all with a logo.

LOGO	TEAM	DIVISION	CLUB	CLUB LOGO	CLUB COLOUR
	Cobras Men	Men's	Cobras (from its name)	Cobras.png	#1a6b3a
	Cobras	Premier	Cobras (from its name)	Cobras.png	#1a6b3a
	Falcons Men	Men's	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons	Premier	Falcons (from its name)	Falcons.png	#1c4f8f
	Falcons 21U	Under 21	Falcons (from its name)	Falcons.png	#1c4f8f
	Grizzlies	Premier	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Grizzlies 21U	Under 21	Grizzlies (from its name)	Grizzlies.png	#7a3d12
	Hawks Men	Men's	Hawks (from its name)	Hawks.png	#b3122a
	Hawks	Premier	Hawks (from its name)	Hawks.png	#b3122a
	Hawks 21U	Under 21	Hawks (from its name)	Hawks.png	#b3122a
	Panthers Men	Men's	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers	Premier	Panthers (from its name)	Panthers.png	#3b1a5e
	Panthers 21U	Under 21	Panthers (from its name)	Panthers.png	#3b1a5e
	Rhinos Men	Men's	Rhinos (from its name)	Rhinos.png	#5b6670
	Rhinos	Premier	Rhinos (from its name)	Rhinos.png	#5b6670
	Stingrays	Premier	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Stingrays 21U	Under 21	Stingrays (from its name)	Stingrays.png	#0a7f8c
	Wolves Men	Men's	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves	Premier	Wolves (from its name)	Wolves.png	#1f4fd1
	Wolves 21U	Under 21	Wolves (from its name)	Wolves.png	#1f4fd1

Team logos and the teams on PlayHQ, grouped by club with a club colour each.

Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.



LIVE VIEWERS

Where this competition streams: any number of YouTube channels and Facebook Pages. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page and the viewer-minutes in the sponsor report. Keys and tokens are kept on this Mac only: never in iCloud, templates or exports.

No streams yet.

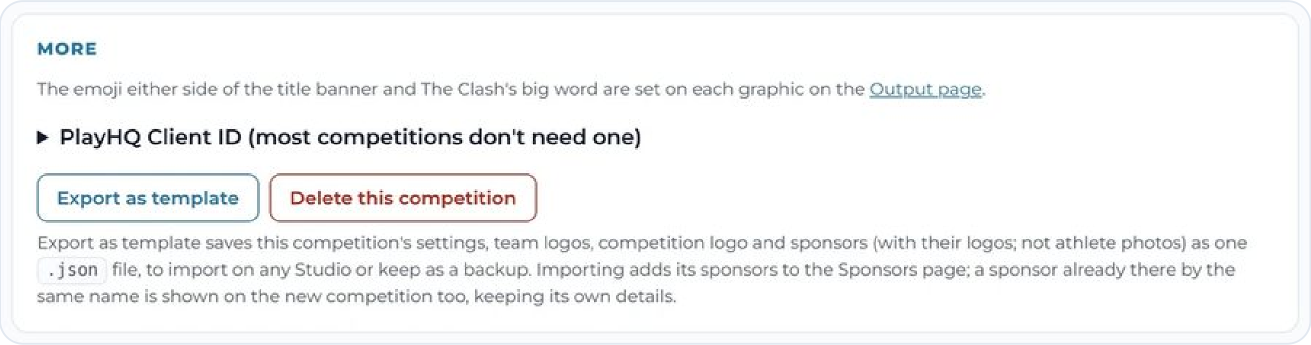
The viewers graphic shows from 50 watching

Add a YouTube channel
Add a Facebook Page
Check now
▶ How to get a YouTube API key and a Facebook Page token

Live viewers.

Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **Import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.



MORE

The emoji either side of the title banner and The Clash's big word are set on each graphic on the Output page.

▶ PlayHQ Client ID (most competitions don't need one)
Export as template
Delete this competition

Export as template saves this competition's settings, team logos, competition logo and sponsors (with their logos; not athlete photos) as one `.json` file, to import on any Studio or keep as a backup. Importing adds its sponsors to the Sponsors page; a sponsor already there by the same name is shown on the new competition too, keeping its own details.

The More section: Client ID, export and delete.

More in the Reference: Competitions setup in detail.

CHAPTER 7 OF 11 ●●●●●●●○○○○

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

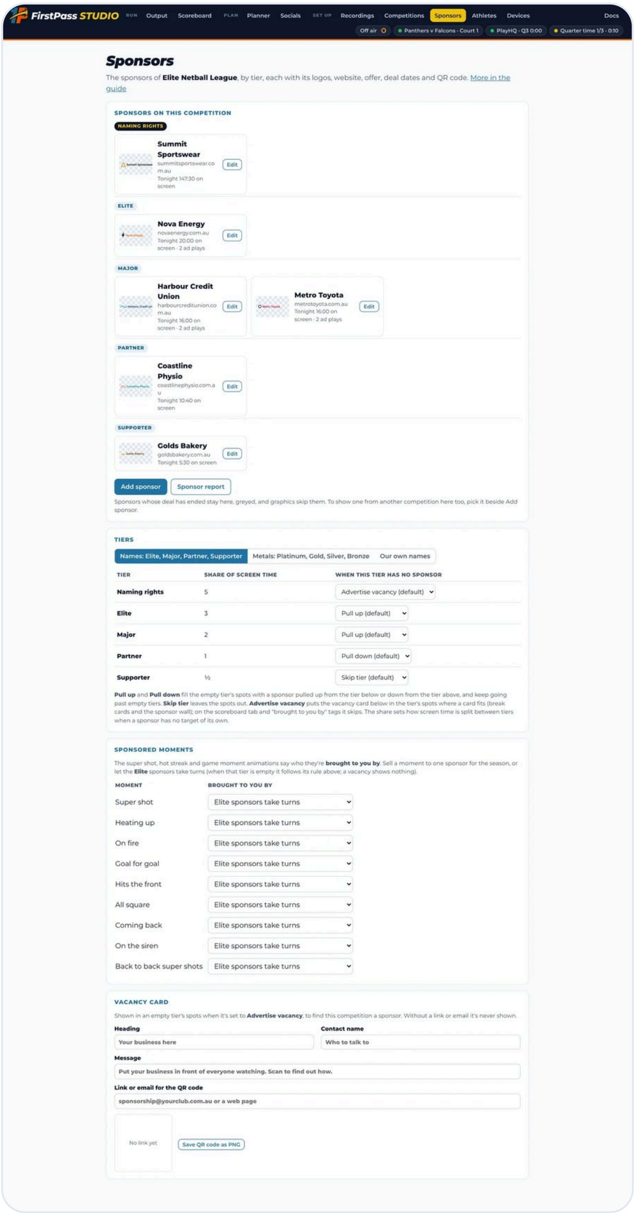
IN THIS CHAPTER

1 Tiers

2 A sponsor

3 Spots and moments

4 Proof of play


FirstPass STUDIO
RUN
Output
Scoreboard
PLAN
Planner
Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 0:00
Quarter time 1/3 · 0:10
Edit
Add sponsor
Sponsor report
Names: Elite, Major, Partner, Supporter
Metals: Platinum, Gold, Silver, Bronze
Our own names
No link yet
Save QR code as PNG

Sponsors

The sponsors of **Elite Netball League**, by tier, each with its logos, website, offer, deal dates and QR code. More in the guide

SPONSORS ON THIS COMPETITION

NAMING RIGHTS

Summit Sportswear
summitsportswear.com.au
Tonight 147:30 on screen

ELITE

Nova Energy
novaenergy.com.au
Tonight 20:00 on screen · 2 ad plays

MAJOR

Harbour Credit Union
harbourcreditunion.com.au
Tonight 16:00 on screen · 2 ad plays

Metro Toyota
metrotoyota.com.au
Tonight 16:00 on screen · 2 ad plays

PARTNER

Coastline Physio
coastlinephysio.com.au
Tonight 10:40 on screen

SUPPORTER

Golds Bakery
goldsbakery.com.au
Tonight 5:30 on screen

Sponsors whose deal has ended stay here, greyed, and graphics skip them. To show one from another competition here too, pick it beside Add sponsor.

TIERS

TIER	SHARE OF SCREEN TIME	WHEN THIS TIER HAS NO SPONSOR
Naming rights	5	Advertise vacancy (default)
Elite	3	Pull up (default)
Major	2	Pull up (default)
Partner	1	Pull down (default)
Supporter	½	Skip tier (default)

Pull up and **Pull down** fill the empty tier's spots with a sponsor pulled up from the tier below or down from the tier above, and keep going past empty tiers. **Skip tier** leaves the spots out. **Advertise vacancy** puts the vacancy card below in the tier's spots where a card fits (break cards and the sponsor wall); on the scoreboard tab and "brought to you by" tags it skips. The share sets how screen time is split between tiers when a sponsor has no target of its own.

SPONSORED MOMENTS

The super shot, hot streak and game moment animations say who they're **brought to you by**. Sell a moment to one sponsor for the season, or let the **Elite** sponsors take turns (when that tier is empty it follows its rule above; a vacancy shows nothing).

MOMENT	BROUGHT TO YOU BY
Super shot	Elite sponsors take turns
Heating up	Elite sponsors take turns
On fire	Elite sponsors take turns
Goal for goal	Elite sponsors take turns
Hits the front	Elite sponsors take turns
All square	Elite sponsors take turns
Coming back	Elite sponsors take turns
On the siren	Elite sponsors take turns
Back to back super shots	Elite sponsors take turns

VACANCY CARD

Shown in an empty tier's spots when it's set to **Advertise vacancy**, to find this competition a sponsor. Without a link or email it's never shown.

Heading: Your business here
Contact name: Who to talk to
Message: Put your business in front of everyone watching. Scan to find out how.
Link or email for the QR code: sponsorship@yourclub.com.au or a web page

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

Tiers

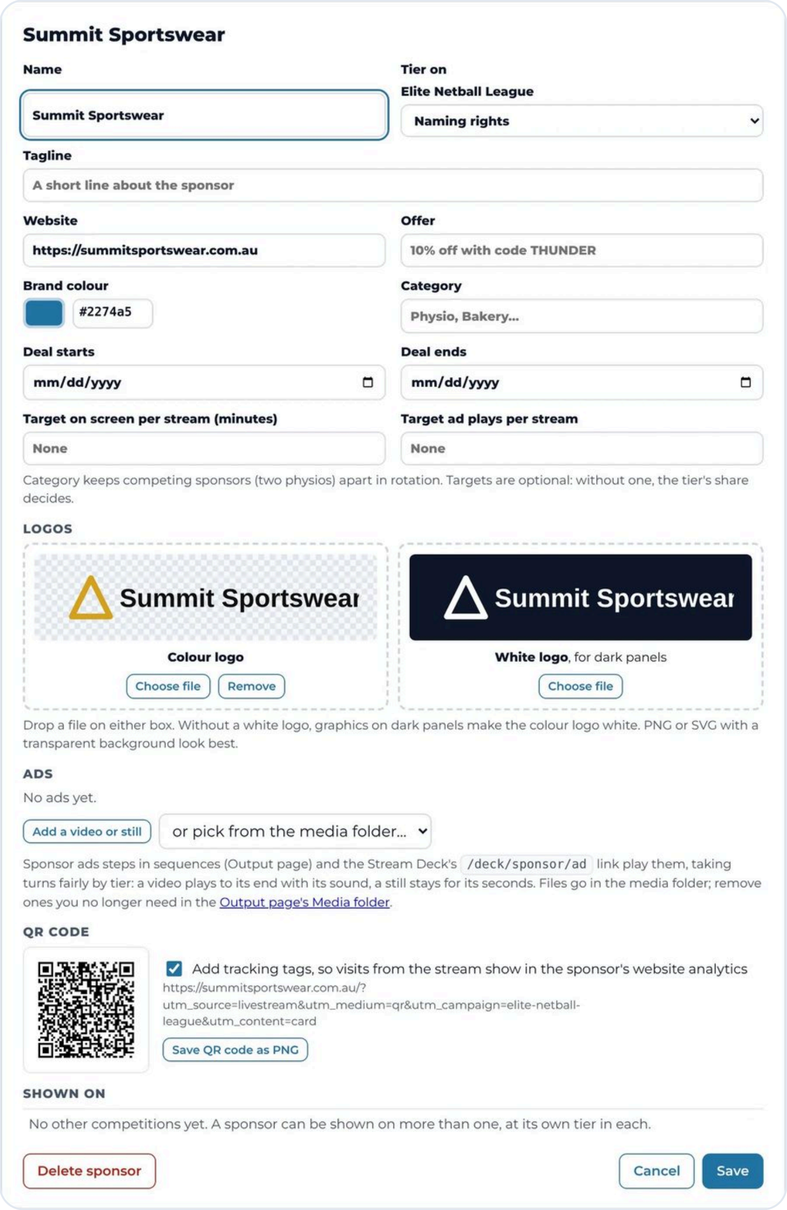
Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.



Summit Sportswear

Name: Summit Sportswear

Tier on Elite Netball League: Naming rights

Tagline: A short line about the sponsor

Website: https://summitsportswear.com.au

Offer: 10% off with code THUNDER

Brand colour: #2274a5

Category: Physio, Bakery...

Deal starts: mm/dd/yyyy

Deal ends: mm/dd/yyyy

Target on screen per stream (minutes): None

Target ad plays per stream: None

Category keeps competing sponsors (two physios) apart in rotation. Targets are optional: without one, the tier's share decides.

LOGOS

Colour logo

White logo, for dark panels

Drop a file on either box. Without a white logo, graphics on dark panels make the colour logo white. PNG or SVG with a transparent background look best.

ADS

No ads yet.

Sponsor ads steps in sequences (Output page) and the Stream Deck's `/deck/sponsor/ad` link play them, taking turns fairly by tier: a video plays to its end with its sound, a still stays for its seconds. Files go in the media folder; remove ones you no longer need in the Output page's Media folder.

QR CODE

☑ Add tracking tags, so visits from the stream show in the sponsor's website analytics

https://summitsportswear.com.au/?utm_source=livestream&utm_medium=qr&utm_campaign=elite-netball-league&utm_content=card

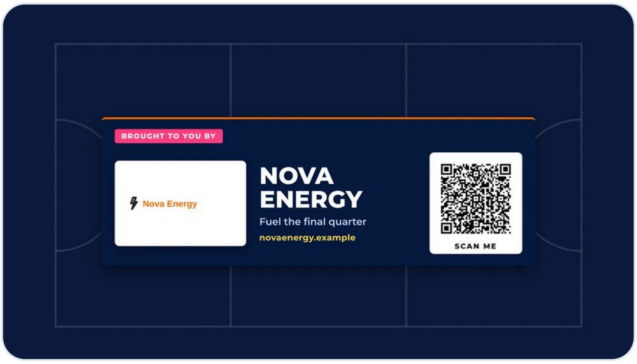
SHOWN ON

No other competitions yet. A sponsor can be shown on more than one, at its own tier in each.

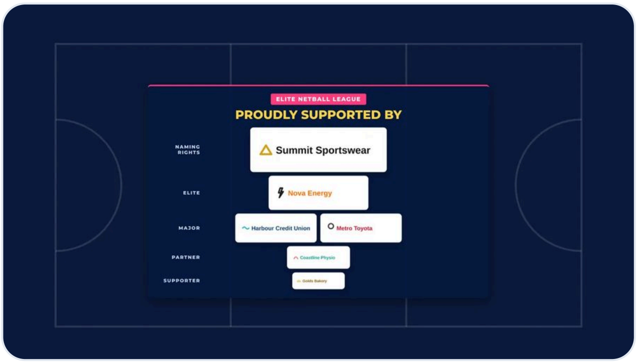
The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

Spots and moments


BROUGHT TO YOU BY
Nova Energy
NOVA ENERGY
Fuel the final quarter
novaenergy.example
SCAN ME

The sponsor card in a break, with its QR code.

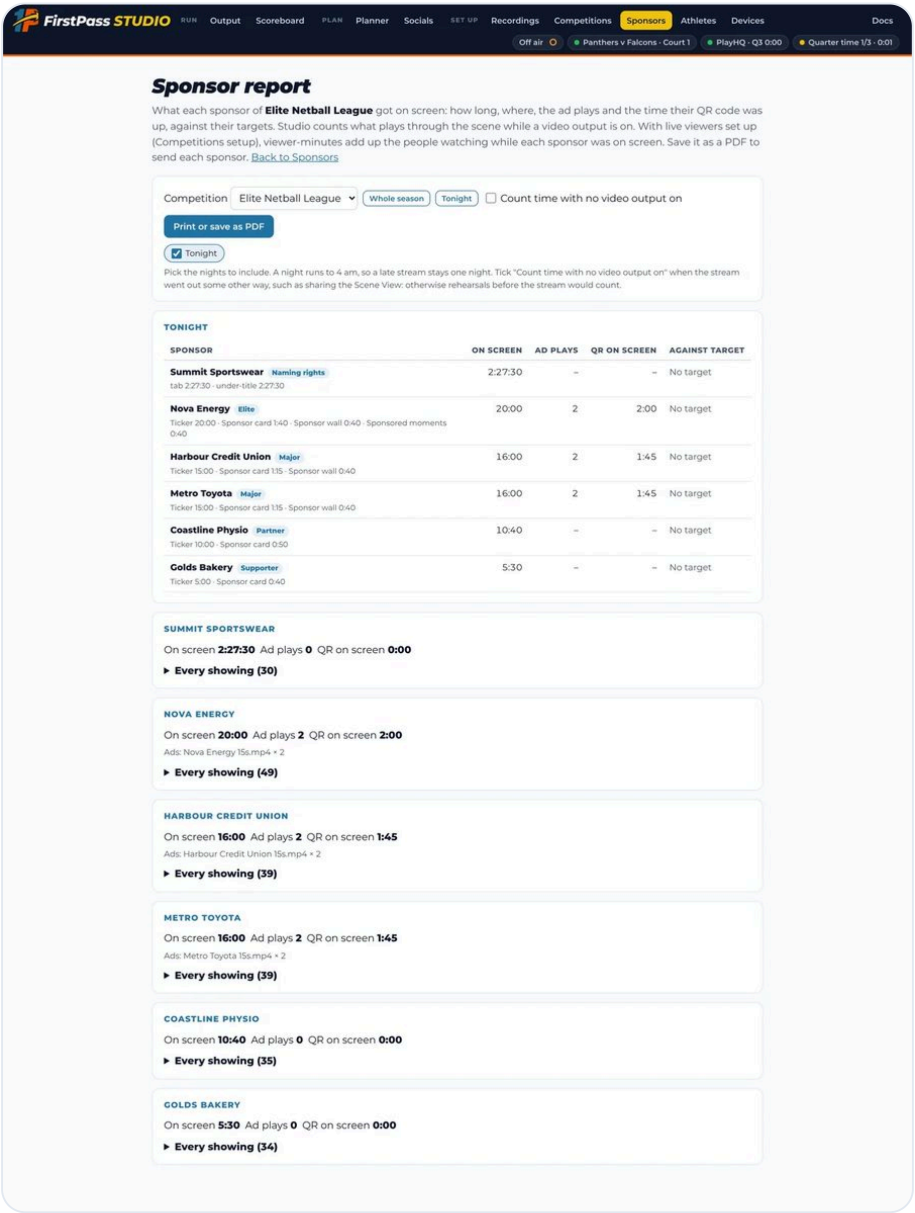

ELITE NETBALL LEAGUE
PROUDLY SUPPORTED BY
NAMING RIGHTS
Summit Sportswear
ELITE
Nova Energy
MAJOR
Harbour Credit Union
Metro Toyota
PARTNER
Coastline Physio
SUPPORTER
Golds Bakery

The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

Proof of play

While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.


FirstPass STUDIO
RUN
Output
Scoreboard
PLAN
Planner
Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 0:00
Quarter time 1/3 · 0:01
Whole season
Tonight
Print or save as PDF

Sponsor report

What each sponsor of **Elite Netball League** got on screen: how long, where, the ad plays and the time their QR code was up, against their targets. Studio counts what plays through the scene while a video output is on. With live viewers set up (Competitions setup), viewer-minutes add up the people watching while each sponsor was on screen. Save it as a PDF to send each sponsor. Back to Sponsors

Competition: Elite Netball League

☐ Count time with no video output on

☑ Tonight

Pick the nights to include. A night runs to 4 am, so a late stream stays one night. Tick "Count time with no video output on" when the stream went out some other way, such as sharing the Scene View: otherwise rehearsals before the stream would count.

TONIGHT

SPONSOR	ON SCREEN	AD PLAYS	QR ON SCREEN	AGAINST TARGET
Summit Sportswear Naming rights tab 2:27:30 · under-title 2:27:30	2:27:30	–	–	No target
Nova Energy Elite Ticker 20:00 · Sponsor card 1:40 · Sponsor wall 0:40 · Sponsored moments 0:40	20:00	2	2:00	No target
Harbour Credit Union Major Ticker 15:00 · Sponsor card 1:15 · Sponsor wall 0:40	16:00	2	1:45	No target
Metro Toyota Major Ticker 15:00 · Sponsor card 1:15 · Sponsor wall 0:40	16:00	2	1:45	No target
Coastline Physio Partner Ticker 10:00 · Sponsor card 0:50	10:40	–	–	No target
Golds Bakery Supporter Ticker 5:00 · Sponsor card 0:40	5:30	–	–	No target

SUMMIT SPORTSWEAR

On screen **2:27:30** Ad plays **0** QR on screen **0:00**

▶ **Every showing (30)**

NOVA ENERGY

On screen **20:00** Ad plays **2** QR on screen **2:00**

Ads: Nova Energy 15s.mp4 × 2

▶ **Every showing (49)**

HARBOUR CREDIT UNION

On screen **16:00** Ad plays **2** QR on screen **1:45**

Ads: Harbour Credit Union 15s.mp4 × 2

▶ **Every showing (39)**

METRO TOYOTA

On screen **16:00** Ad plays **2** QR on screen **1:45**

Ads: Metro Toyota 15s.mp4 × 2

▶ **Every showing (39)**

COASTLINE PHYSIO

On screen **10:40** Ad plays **0** QR on screen **0:00**

▶ **Every showing (35)**

GOLDS BAKERY

On screen **5:30** Ad plays **0** QR on screen **0:00**

▶ **Every showing (34)**

The sponsor report for the sample night.

More in the Reference: **Sponsors in detail**.

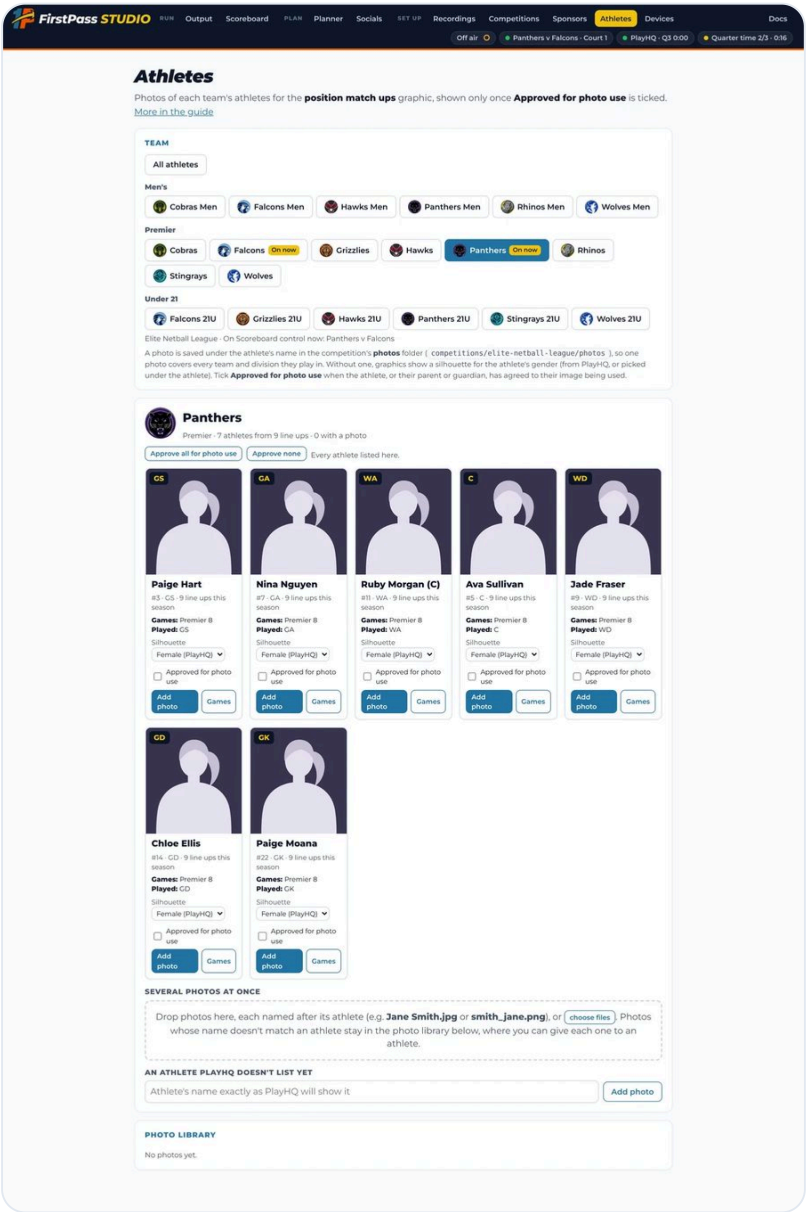
CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

IN THIS CHAPTER

1 Photos
2 Positions and games played



FirstPass STUDIO
RUN
Output
Scoreboard
PLAN
Planner
Socials
SET UP
Recordings
Competitions
Sponsors
Athletes
Devices
Docs
Off air
Panthers v Falcons · Court 1
PlayHQ · Q3 0:00
Quarter time 2/3 · 0:16
TEAM
All athletes
Men's
Cobras Men
Falcons Men
Hawks Men
Panthers Men
Rhinos Men
Wolves Men
Premier
Cobras
Falcons
On now
Grizzlies
Hawks
Panthers
On now
Rhinos
Stingrays
Wolves
Under 21
Falcons 21U
Grizzlies 21U
Hawks 21U
Panthers 21U
Stingrays 21U
Wolves 21U
Approve all for photo use
Approve none
GS
GA
WA
C
WD
GD
GK
Add photo
Games
choose files
Add photo

Athletes

Photos of each team's athletes for the **position match ups** graphic, shown only once **Approved for photo use** is ticked.
More in the guide

Elite Netball League · On Scoreboard control now: Panthers v Falcons

A photo is saved under the athlete's name in the competition's **photos** folder (`competitions/elite-netball-league/photos`), so one photo covers every team and division they play in. Without one, graphics show a silhouette for the athlete's gender (from PlayHQ, or picked under the athlete). Tick **Approved for photo use** when the athlete, or their parent or guardian, has agreed to their image being used.

Panthers

Premier · 7 athletes from 9 line ups · 0 with a photo

Every athlete listed here.

Paige Hart
#3 · GS · 9 line ups this season
Games: Premier 8
Played: GS
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Nina Nguyen
#7 · GA · 9 line ups this season
Games: Premier 8
Played: GA
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Ruby Morgan (C)
#11 · WA · 9 line ups this season
Games: Premier 8
Played: WA
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Ava Sullivan
#5 · C · 9 line ups this season
Games: Premier 8
Played: C
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Jade Fraser
#9 · WD · 9 line ups this season
Games: Premier 8
Played: WD
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Chloe Ellis
#14 · GD · 9 line ups this season
Games: Premier 8
Played: GD
Silhouette: Female (PlayHQ)
☐ Approved for photo use

Paige Moana
#22 · GK · 9 line ups this season
Games: Premier 8
Played: GK
Silhouette: Female (PlayHQ)
☐ Approved for photo use

SEVERAL PHOTOS AT ONCE

Drop photos here, each named after its athlete (e.g. **Jane Smith.jpg** or **smith_jane.png**), or choose files. Photos whose name doesn't match an athlete stay in the photo library below, where you can give each one to an athlete.

AN ATHLETE PLAYHQ DOESN'T LIST YET

Athlete's name exactly as PlayHQ will show it

PHOTO LIBRARY

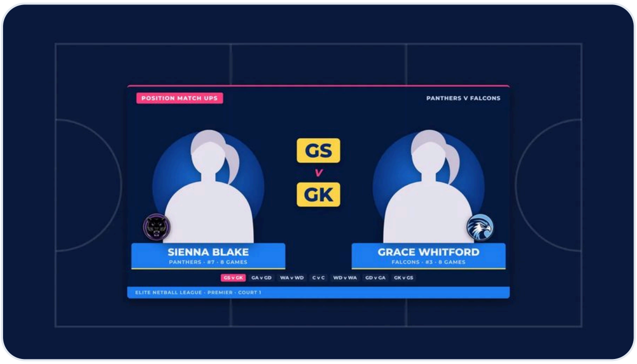
No photos yet.

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

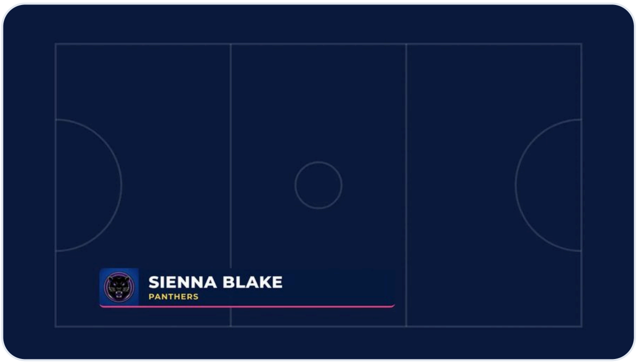
Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.


POSITION MATCH UPS
PANTHERS V FALCONS
GS
V
GK
SIENNA BLAKE
PANTHERS · #7 · 8 GAMES
GRACE WHITFORD
FALCONS · #3 · 8 GAMES
GS v GK
GA v GD
WA v WD
C v C
WD v WA
GD v GA
GK v GS
ELITE NETBALL LEAGUE · PREMIER · COURT 1

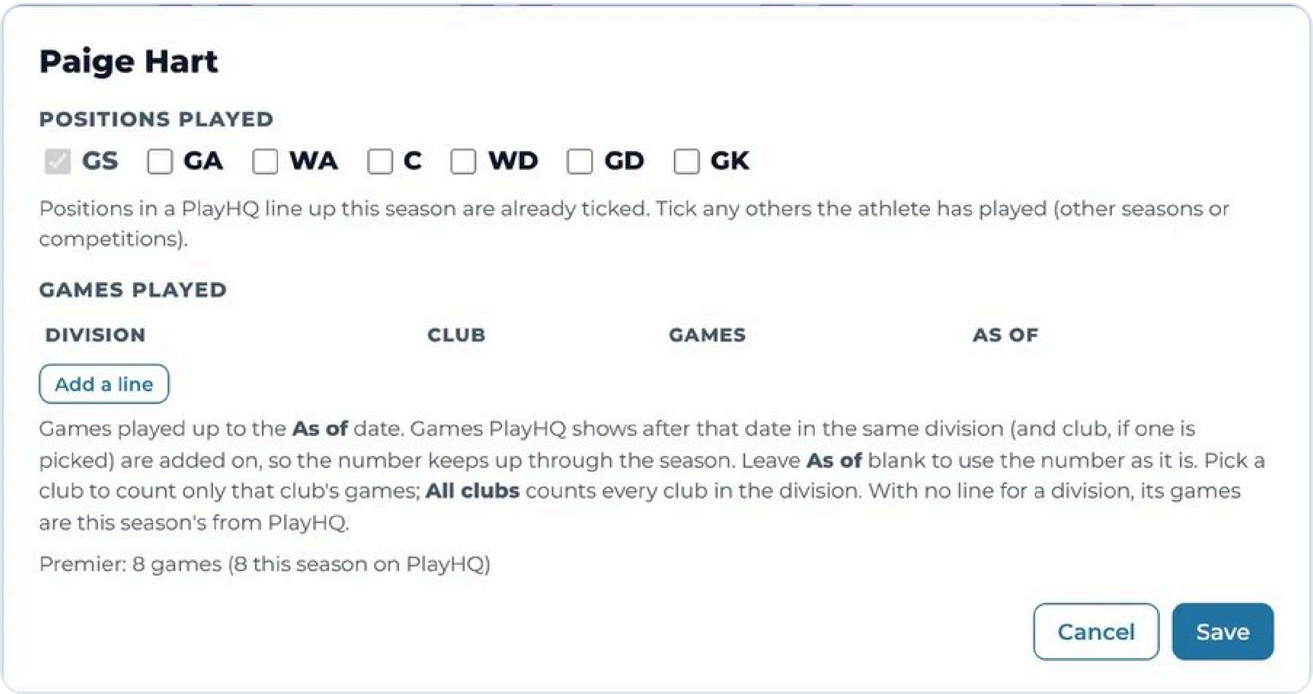
Position match ups use the photos, pairing by pairing.


SIENNA BLAKE
PANTHERS

The athlete lower third, named from the register as you type on the Output page.

Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.



Paige Hart

POSITIONS PLAYED

☑ GS ☐ GA ☐ WA ☐ C ☐ WD ☐ GD ☐ GK

Positions in a PlayHQ line up this season are already ticked. Tick any others the athlete has played (other seasons or competitions).

GAMES PLAYED

DIVISION	CLUB	GAMES	AS OF

Games played up to the **As of** date. Games PlayHQ shows after that date in the same division (and club, if one is picked) are added on, so the number keeps up through the season. Leave **As of** blank to use the number as it is. Pick a club to count only that club's games; **All clubs** counts every club in the division. With no line for a division, its games are this season's from PlayHQ.

Premier: 8 games (8 this season on PlayHQ)

An athlete's positions and games played.

More in the Reference: **Athletes in detail**.

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link. It all lives on the **Devices** tab, under the list of devices allowed to change things (Stream Decks and their link sets, paired phones and programs), which is also where **Devices** in the menu bar icon goes.

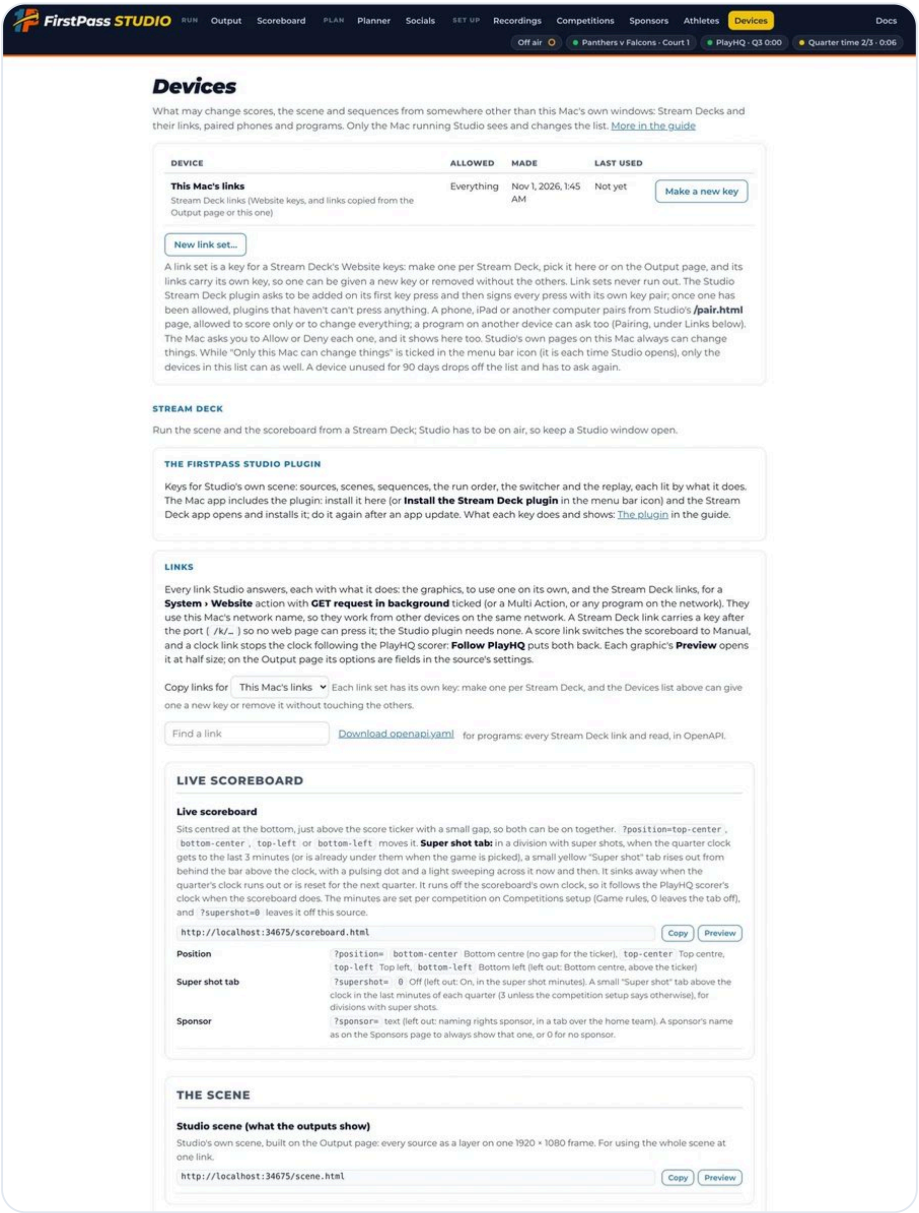
IN THIS CHAPTER

1 The plugin

2 Links for everything else

3 Timed keys and the icon pack

4 The Remote on a phone



Devices

What may change scores, the scene and sequences from somewhere other than this Mac's own windows: Stream Decks and their links, paired phones and programs. Only the Mac running Studio sees and changes the list. More in the guide

DEVICE	ALLOWED	MADE	LAST USED
This Mac's links Stream Deck links (Website keys, and links copied from the Output page or this one)	Everything	Nov 1, 2026, 1:45 AM	Not yet

A link set is a key for a Stream Deck's Website keys: make one per Stream Deck, pick it here or on the Output page, and its links carry its own key, so one can be given a new key or removed without the others. Link sets never run out. The Studio Stream Deck plugin asks to be added on its first key press and then signs every press with its own key pair; once one has been allowed, plugins that haven't can't press anything. A phone, iPad or another computer pairs from Studio's **/pair.html** page, allowed to score only or to change everything; a program on another device can ask too (Pairing, under Links below). The Mac asks you to Allow or Deny each one, and it shows here too. Studio's own pages on this Mac always can change things. While "Only this Mac can change things" is ticked in the menu bar icon (it is each time Studio opens), only the devices in this list can as well. A device unused for 90 days drops off the list and has to ask again.

STREAM DECK

Run the scene and the scoreboard from a Stream Deck; Studio has to be on air, so keep a Studio window open.

THE FIRSTPASS STUDIO PLUGIN

Keys for Studio's own scene: sources, scenes, sequences, the run order, the switcher and the replay, each lit by what it does. The Mac app includes the plugin: install it here (or **Install the Stream Deck plugin** in the menu bar icon) and the Stream Deck app opens and installs it; do it again after an app update. What each key does and shows: The plugin in the guide.

LINKS

Every link Studio answers, each with what it does: the graphics, to use one on its own, and the Stream Deck links, for a **System › Website** action with **GET request in background** ticked (or a Multi Action, or any program on the network). They use this Mac's network name, so they work from other devices on the same network. A Stream Deck link carries a key after the port (`/k/…`) so no web page can press it; the Studio plugin needs none. A score link switches the scoreboard to Manual, and a clock link stops the clock following the PlayHQ scorer: **Follow PlayHQ** puts both back. Each graphic's **Preview** opens it at half size; on the Output page its options are fields in the source's settings.

Copy links for: This Mac's links — Each link set has its own key: make one per Stream Deck, and the Devices list above can give one a new key or remove it without touching the others.

Download openapi.yaml for programs: every Stream Deck link and read, in OpenAPI.

LIVE SCOREBOARD

Live scoreboard

Sits centred at the bottom, just above the score ticker with a small gap, so both can be on together. `?position=top-center`, `bottom-center`, `top-left` or `bottom-left` moves it. **Super shot tab:** in a division with super shots, when the quarter clock gets to the last 3 minutes (or is already under them when the game is picked), a small yellow "Super shot" tab rises out from behind the bar above the clock, with a pulsing dot and a light sweeping across it now and then. It sinks away when the quarter's clock runs out or is reset for the next quarter. It runs off the scoreboard's own clock, so it follows the PlayHQ scorer's clock when the scoreboard does. The minutes are set per competition on Competitions setup (Game rules, 0 leaves the tab off), and `?supershot=0` leaves it off this source.

`http://localhost:34675/scoreboard.html`

Position	`?position=` `bottom-center` Bottom centre (no gap for the ticker), `top-center` Top centre, `top-left` Top left, `bottom-left` Bottom left (left out: Bottom centre, above the ticker)
Super shot tab	`?supershot=` `0` Off (left out: On, in the super shot minutes). A small "Super shot" tab above the clock in the last minutes of each quarter (3 unless the competition setup says otherwise), for divisions with super shots.
Sponsor	`?sponsor=` text (left out: naming rights sponsor, in a tab over the home team). A sponsor's name as on the Sponsors page to always show that one, or 0 for no sponsor.

THE SCENE

Studio scene (what the outputs show)

Studio's own scene, built on the Output page: every source as a layer on one 1920 × 1080 frame. For using the whole scene at one link.

`http://localhost:34675/scene.html`

The Devices tab: the devices allowed to change things at the top, then the plugin, then Links: every link Studio answers, each with what it does, Copy and its settings.

The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Each kind of key has its own colour and picture, so you can see what a key does at a glance. The numbered disc in the top right corner shows which output it acts on, in that output's colour.

KEY	LOOKS	PRESS	SHOWS
Source	Teal, an eye (pink with a speaker for video sound)	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit while it's on, however it was turned on
Scene	Blue, stacked layers	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	Purple, a film strip	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Run order	Yellow, a numbered list	Play next (default), skip what's next, load a saved order, clear the list	What plays next, like **Play next ▶ / Quarter break / +2 more**; lit while something is queued, **empty** when nothing is; a Load key names its order and how many items it holds
Switcher	Red, a camera; no disc	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering
Replay	Grey, a replay arrow; no disc	Play or pause, next or previous moment, nudge, stop the 4K replay or practice game running	What it acts on (**4K replay**, **Practice**, **Demo**); lit while one runs

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). Its disc shows the output's number in its colour (Output 1 orange, 2 pink, 3 blue, 4 green, as on the Output page), so two keys for the same source on different outputs can't be mixed up. A key for a removed output loses its disc and says **no output**.

Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Devices tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Devices tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices can take that one away without touching your other keys. The Devices tab lists every one under **Links**, with what it does, the settings it takes and **Copy**; **Find** narrows the list, a graphic has **Preview** and a status read has **Try**. The useful ones:

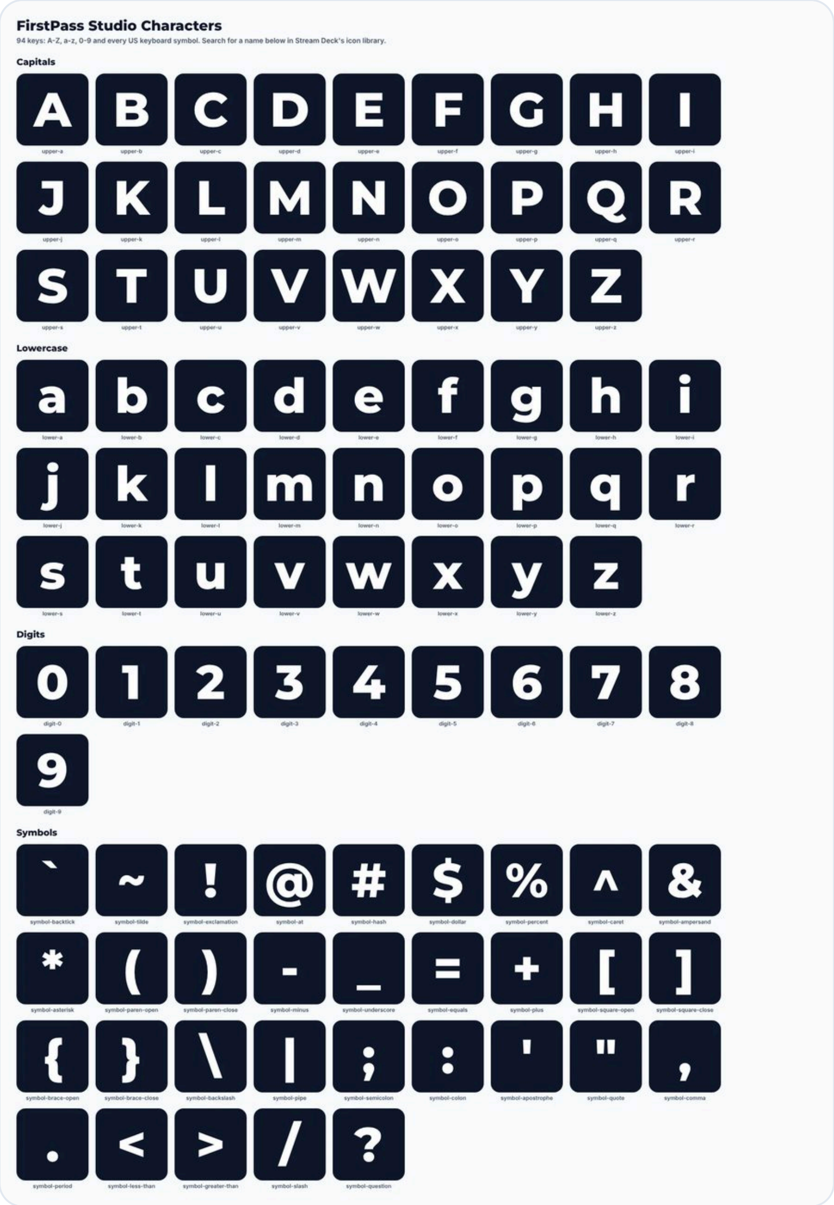
GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Run order	Play what's next, skip it, clear the list, add a sequence or scene, load a saved order.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS

Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.


FirstPass Studio Characters
94 keys: A-Z, a-z, 0-9 and every US keyboard symbol. Search for a name below in Stream Deck's icon library.
Capitals
Lowercase
Digits
Symbols

The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.

The Remote on a phone

No Stream Deck to hand, or away from the desk? The **Remote** puts the status board and a Stream Deck's sequence keys on a phone. On the Mac, press **Open on your phone** at the top of the Socials tab, pick **Remote** and scan the QR code with the phone's camera, on the same Wi-Fi; the first time, choose **Allow everything** on the Mac when it asks, then add the page to the Home Screen to open it like an app (it's the Studio link with `/remote.html` on the end). At the top it says what's on at a glance: on air or off air and which outputs, the game's score, clock and whether PlayHQ is scoring it, the sequence playing and any about to start. Under that, **Play next ▶** and **Skip** for the run order, **Next step**, **Pause** and **Stop** for the sequence playing, and a key for every sequence in its colour, lit while it plays: press it to play the sequence on Output 1, press it again to stop it (which does its Then, as on the Output page). While the scoreboard is on **Manual** (a score press on the Mac or a Stream Deck switches it), a **Scoring by hand** card comes up with each team's score and +1, +2 (where super shots are worth 2) and −1 keys, the clock's Start and Stop and Next quarter; it goes away once the scoreboard follows PlayHQ again. A phone paired to score only can use those keys too.

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Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

IN THIS CHAPTER

1 The 1P menu
2 The data folder
3 Settings in iCloud Drive
4 Backup and restore
5 Updates
6 Links and other devices
7 When something isn't right
8 Data and bandwidth

The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices**; **User guide**; **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up…** and **Restore…**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Appearance** (Follow the Mac, Light, Dark), **Only this Mac can change things**, **What's new…** and Quit.

This guide comes with Studio: **User guide** in the menu bar icon, **Help > FirstPass Studio User Guide** while a Studio window is in front open it in a window of its own, and it is the **Docs** tab (⌘0) in any Studio window, with no internet needed. Phones and other computers on the network can read it too, at the Studio link with `/guide/` on the end.

The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

Every file and folder in it: The data folder in the Reference.

- ## Settings in iCloud Drive

Settings folder > Move to iCloud Drive… moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac…** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac…** undoes the move. Studio goes off air for a moment for each and comes straight back.

In detail: Settings in iCloud Drive in the Reference.

- ## Backup and restore

Back up… saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore…** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

In detail, with the link for scripts: Backup and restore in the Reference.

- ## Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

- ## Links and other devices

Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

Secure phone link. A phone's browser lets a page use its microphone (a phone as a backup commentator headset) or its share sheet only over https, which the `.local` link isn't. Turned on in the **Secure phone link** card on the Devices tab, Studio also answers at a name of its own on the internet, such as `https://k3f9x2.studio.firstpass.pivt.au:8443`, with a free Let's Encrypt certificate it renews by itself, and every **Open on your phone** QR code uses it; a phone pairs once more on the new link. The graphics, the Stream Deck and the http links carry on the same. It needs the internet at the venue, and a few routers refuse a name on the internet that points at their own network: then the https link doesn't open there and the http one still does. It's set up once for PIVT, then each Studio Mac is given the same three things:

1. In AWS, **Route 53 › Hosted zones › Create hosted zone**: domain `studio.firstpass.pivt.au`, Public. Note its **Hosted zone ID** (starts with Z) and its 4 NS values.
2. In GoDaddy's DNS for **pivt.au**, add 4 records: type **NS**, name `studio.firstpass`, one NS value each. It can take up to an hour to work.
3. In **IAM**, a policy `FirstPassStudioDNS` allowing `route53:ChangeResourceRecordSets` and `route53:ListResourceRecordSets` on that hosted zone only, and `route53:GetChange`; a user `firstpass-studio-dns` with it attached and no console access; then **Security credentials › Create access key** (Application running outside AWS). Keep the secret somewhere safe: AWS shows it once.
4. On each Studio Mac, paste the hosted zone ID, the access key ID and the secret into the card and press **Turn on**. It takes about a minute the first time, and the card says each step and, if one fails, why and what to check. The secret goes in the Mac's Keychain.

FirstPass HQ. A club that records FirstPass stats keeps them in its own FirstPass HQ, and Studio signs in to each club, once per Mac, in the **FirstPass HQ** card on the Devices tab. Press **Sign in** beside the club: its Google sign-in page opens in your browser, you sign in with your own account, and the browser says **Signed in**; nothing is typed into Studio. A club admin first gives your account the **Studio** role in HQ: without it Studio keeps nothing and the card says so. The sign-in stays in the Mac's Keychain, never in iCloud Drive or a backup, and is checked again every half hour. **Sign out**, or the club removing or pausing your role, deletes it and every FirstPass stat Studio kept for that club. FirstPass stats are about athletes, many under 18, so they stay private: only this Mac, and phones or browsers paired over the Secure phone link and opened on it, can see them (a device paired at the `.local` link pairs once more on the secure link), whatever **Only this Mac can change things** says. While a club is signed in, Studio reads the FirstPass stats of the game on the scoreboard as it is played, when that club recorded it in HQ, and keeps them up to date on its own. Studio deletes a game's stats 30 days after it, and never writes them to the debug log.

The link name, pairing and what other devices may do: **Links and access** in the Reference.

When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering, or (PlayHQ chip purple, **PlayHQ not live**) the scorebench is off line. Fix the scorebench, or switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game while it's on (or as set, down to one a second; every 30 seconds just before it, every 5 minutes otherwise), the ticker's other courts every 30 seconds, the fixture and ladders about once a minute while a game is on and every 5 minutes when none is, and the rest once when a graphic loads. **Refresh list** on Scoreboard control reads the night's games from PlayHQ there and then. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth. The PlayHQ line above the score on the Output page's status board shows it live: the ping, the bandwidth over the last minute and the data downloaded since Studio went on air.

Page by page: **Data and bandwidth** in the Reference.

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Quick reference

IN THIS CHAPTER

1 Keyboard
2 Stream night in nine lines
3 Where things are set

Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Recordings, Socials, Competitions setup, Sponsors, Athletes, Devices, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

Stream night in nine lines

1. Open Studio from the 1P menu and keep the Output page open all night.
2. Check the competition in use and the court on Competitions setup.
3. On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.
4. Turn on the video output your switcher takes and show one graphic to see it land.
5. Play a few seconds of music and an animation with sound.
6. Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.
7. Go to the Live game scene and let the sequences run the breaks.
8. Pick finals winners as they happen; turn on lower thirds when people appear.

9 Press Stream close, turn the output off, and send the sponsor report.

Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links, and the devices allowed to change things	Devices
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

REFERENCE

Reference

Everything in detail, for when the chapters above aren't enough: what the Mac app needs and keeps, every setting on every page, and what Studio asks of the network.

IN THIS CHAPTER

1 The Mac app
2 Links and access
3 The data folder
4 Settings in iCloud Drive
5 Backup and restore
6 Unsaved changes
7 Data and bandwidth
8 Sponsors in detail
9 Competitions setup in detail
10 Athletes in detail
11 The Output page in detail
12 Scoreboard control in detail
13 Socials in detail
14 Recordings in detail

The Mac app

FirstPass Studio needs an Apple silicon Mac (M1 or later), and fits a 13" MacBook's screen. A plain M1 drives one extra display; Pro and Max chips drive more, which fill and key over two displays needs. A Blackmagic output needs Blackmagic Desktop Video, and NDI needs NDI's free runtime (both under The Output page in detail); the app brings everything else with it, ffmpeg for video conversion included.

The app always starts in the menu bar off air, whether opened at login (tick **Open at login**) or by hand, so nothing talks to PlayHQ until you open Studio. On air it serves the graphics and the Stream Deck links, talks to PlayHQ, and restarts by itself if it stops. **Stop Studio** in the 1P menu closes the Studio windows and goes off air.

The app window is like any Mac app's: no address bar, tabs or bookmarks, the Studio header is the title bar (drag it to move the window), and it shows in the Dock and ⌘Tab while a window is open. Links that open a new tab on Studio (a preview, the control or setup page) open another window; links elsewhere (PlayHQ, GitHub) open in your browser. The menu at the top of the screen has **File > Open in Browser** (⇧⌘O) and **Copy Page Link** (⇧⌘C) for the page in front, **Go** for each Studio page (⌘1 to ⌘9, and ⌘0 for Docs), **View** for reload, back and page zoom (in steps, the level shown in the menu; **Zoom In** ⌘+, **Zoom Out** ⌘−, **Actual Size** ⌘⌥0, or pinch on the trackpad), **Help > FirstPass Studio User Guide** for this guide, **New Window** (⌘N), and **Close Studio Windows** (⌥⌘W) in the FirstPass Studio menu.

Links and access

Studio's links (on the Devices tab, Scoreboard control and in the Mac app's menu, **Copy the Studio link** too) start with this Mac's Bonjour name, such as `http://studio-mac.local:8080`, instead of `localhost`. It is the **Local hostname** in System Settings > General > Sharing, so the same link works on this Mac and from a phone, another computer or a Stream Deck on the same network. The port stays 8080, and `http://localhost:8080` still works on this Mac, so links and Stream Deck buttons you already have carry on. To change it, set `"linkHost"` in `config.json` in the data folder: `"bonjour"` (the default), `"localhost"`, or any name or IP address, then choose **Restart Studio** in the 1P menu.

Only this Mac can change things in the menu bar icon sets `remoteChanges`, and is ticked again every time the app opens: ticked, phones, other computers and Stream Decks on the network that aren't in **Devices** can still open every graphic but can't change scores, the scene, sequences, settings or media, or read private data (the settings and scene backups, the athlete register, the debug log, the sponsor reports, the competition setup); Studio answers those with a 403 that says so. The game stories other Studio Macs copy (`/api/stories`) stay open like the graphics, since copying them only reads. Either way a change must come from Studio's own pages or a program (the Stream Deck plugin, a Website key): a request a browser marks as coming from a web page on another site, or that asks for Studio by a name this computer doesn't answer to, gets a 403 too, and so does a read of private data asked for by such a name. A Stream Deck link must carry this Mac's key (`/k/<key>/deck/…`, as Studio copies them; kept in `devices.json`, never backed up) or be signed by a paired Studio plugin, or come from Studio's own pages, so a web page can't press one. A phone, iPad or another computer pairs from `/pair.html`: the Mac offers **Allow scoring only** (the scores, clock, quarter and game) or **Allow everything**, and the device then works while the box is ticked; Devices can change its level or remove it. (Until a plugin has paired, the plugin's older `x-studio-deck` header still counts, so plugins installed before pairing keep working until a new one is allowed.)

The **Secure phone link** (Devices tab, on this Mac only) adds an https link beside the http one, at `<name>.studio.firstpass.pivt.au` on port 8443 (`"securePort"` in `config.json` changes it); Studio reads the zone's name from the Route 53 hosted zone it's given, so nothing is built in. Each Mac picks its own random name once, so two Macs on one night each have their own. Studio points the name at the Mac's address on the network in Route 53 (again whenever the address changes), proves it owns the name to Let's Encrypt with a TXT record there, and gets a certificate that lasts 90 days, renewed with 30 left. When Studio starts, a certificate it already has is used straight away, so the https link opens within moments; the address is sent to Route 53 only when it has changed. The https link serves the same pages under the same rules (pairing, levels, the lock), and its own pages may change things like the http ones. A step that fails is tried again after 5, 15 and then 60 minutes; a link already working keeps working until its certificate runs out. Its settings, Let's Encrypt account and certificate are in `securelink.json`, the access key's secret in the Keychain.

FirstPass HQ (Devices tab, on this Mac only; `/api/firstpass/clubs`). The clubs are listed in `hqclubs.json` in the app, one Supabase project each (code, name, address and public key), the same list HQ's own app uses; each project allows `http://127.0.0.1:*/**` as a sign-in redirect. A club's test or development project, when listed, ends its name in **(test)** and its code in `TEST`; only the production project holds the club's real stats. **Sign in** uses Google sign-in with PKCE: Studio listens on a random `127.0.0.1` port and path for that one answer, for five minutes, and opens the club's sign-in page in the system browser. It then checks the account holds HQ's `studio` role (`has_role`), and keeps the refresh token in the login Keychain (**FirstPass Studio HQ sign-in**, one item per club) and who signed in when in `hq-signins.json`, both on this Mac only. Every 30 minutes, and as Studio starts, each club's session is refreshed and the role checked; HQ refusing either signs the club out. Signing out deletes the token and the club's folder in `firstpass/`. Everything under `/api/firstpass/` answers only this Mac, and browsers that were paired over the secure link and ask over it (marked in Devices), whatever `remoteChanges` says; their requests are logged without their query, and anything that looks like a token is taken out of the log. Every 5 seconds Studio looks for the scoreboard's game among the signed-in clubs' games (`studio_games`, matched by PlayHQ game id, read once a minute), catches up with `studio_live` (the latest break snapshot and up to 1000 events a page), reads the line-up and who may be named (`studio_lineup`), then follows the game's private Realtime channel; a missed event, the channel coming back, or a minute passing brings another catch-up. `/api/firstpass/status` says which game it follows, in counts only.

On a network you don't control, leave **Only this Mac can change things** ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

The data folder

Everything Studio saves lives in one data folder on this computer, outside the app, so updates never touch it: `~/Library/Application Support/FirstPass Studio` (the setup page shows where, and **Open the settings folder** in the 1P menu opens it).

- **competitions/**: a folder per competition with its setup (`settings.json`), team logos, competition logo, athlete photos and athlete register, its sequences, run orders and the Production Planner's nights (`sequences.json`, `planner.json`), and **active.json**, the one in use.
- **sponsors/**: every sponsor once (`sponsors.json`), a folder of logos each, and **exposure/**, the log the sponsor report is made from.
- **scene.json**, **outputs.json**, **music.json** and **mics.json**: the sources and saved scenes, the video outputs, the music player and playlists, and the mics, shared by every competition; **media/**: the images, videos and songs they use.
- **control.json**: Scoreboard control's choices (the game night picked by hand, automatic game selection and its court, following the PlayHQ clock, how often to check PlayHQ, showing the clock, the ticker speed, finals winners picked by hand).
- **scoreboard-state.json**: the game on the board, the scores and the clock. **followed-games.json** (on this Mac): the night's other games followed in the background, where

each is up to, so a restart carries on. **game-stories/** (on this Mac, not in the settings folder): each streamed or followed game's goals, quarter scores and moments as Studio kept them, for the Socials tab's Game story, Score worm, Moments of the game and Game stats, with copies of the games other Studio Macs kept (copied over the network, or from **iCloud Drive/FirstPass Studio/game-stories**, where each Mac puts its finished games when iCloud Drive is on).

- **config.json** (optional): changes this computer's settings (port, log size, how often to check PlayHQ, `linkHost`, `remoteChanges`) from the app's defaults (see Links and access).
- **devices.json**: the Stream Deck key, link sets and the devices allowed to change things, on this Mac only and never backed up.
- **firstpass/**, **hq-signins.json**: FirstPass stats kept per club and game (deleted 30 days after the game, or at sign-out) and who is signed in to each club's FirstPass HQ, on this Mac only and never backed up.
- **securelink.json**: the Secure phone link's settings, name and certificate, on this Mac only and never backed up.
- **switcher.json**: the ATEM switcher Studio controls (its name on the network, or the address typed) and the stinger's line-up in frames, one for the Mac whichever competition is in use.
- **app-settings.json**: the Mac app's own choices (where the window was).
- **recordings.json**, **demos.json** (saved demo games), **replay.json** (what a 4K replay follows, and each recording's sync) and **Recordings/** (on this Mac, not in iCloud Drive or a backup): whether Studio records each PlayHQ game on air and for how long after full time, and the recordings themselves.
- **logs/**: the debug log. **backups/**: a copy of the folder made before an update changes how settings are saved.

Settings in iCloud Drive

In the 1P menu, **Settings folder > Move to iCloud Drive…** asks for a folder in iCloud Drive (make one called FirstPass Studio, say) and moves the settings into a subfolder named after this Mac, its **Local hostname** in System Settings > General > Sharing: `iCloud Drive/FirstPass Studio/Studio-MacBook-Pro`. Each Mac signed in to the same Apple account gets its own subfolder, so two Studios never save over each other's settings, and a second Mac with the same name is stopped from using the first one's. Studio goes off air for the move and comes straight back. Before anything moves, a copy of the settings is saved in **backups/** in the folder on this Mac, and the originals are removed only once every file has copied across. If the folder already has settings for this Mac (from before), Studio asks whether to use those or replace them.

Some files always stay on this Mac: the debug log, the scores and clock (`scoreboard-state.json`, saved every few seconds during a game), the night's other games Studio follows (`followed-games.json`), PlayHQ caches, the app's window positions and `data-location.json`, which says where the settings are. If iCloud has kept only a placeholder of a file (Optimise Mac Storage), Studio downloads it before going on air; to keep it all on the Mac, right-click the folder in Finder and choose **Keep Downloaded**. If the iCloud folder can't be found (iCloud Drive turned off, signed out), Studio says so and doesn't start, rather than starting with no settings. **Settings folder > Move back to this Mac…** moves them back.

Settings folder > Restore from another Mac… copies another Mac's settings (competitions, scenes, sequences, music, Scoreboard control's choices) over this Mac's: pick the Mac from the list of folders beside this one, or choose its folder. The other Mac keeps its own; this Mac's settings are backed up in **backups/** on this Mac first, and are only replaced once the copy has checked out. If this Mac's settings are on the Mac, they go into iCloud Drive too, in a new folder for this Mac beside the other Mac's, and stay there from then on. Studio goes off air for the copy and comes straight back. **Open the settings folder** opens wherever they are now.

Backup and restore

Back up… in the 1P menu saves every setting in one zip: competitions with their logos and athletes, sponsors, scenes, sequences, music and playlists, the media folder, Scoreboard control's choices and `config.json`, wherever they are kept (on the Mac or in iCloud Drive). The debug log, the scores and clock, PlayHQ caches and stream keys stay out. First it asks for notes (what the backup is for, say) and whether to include the media folder (videos, songs), which can make the zip big. The zip is named `FirstPassStudioBackup-2026-10-24_10-46-02.zip`, the date and time it was made, so each one is new and they sort oldest to newest. Beside it goes a summary with the same name, `FirstPassStudioBackup-2026-10-24_10-46-02.txt`: the notes, then the names of the competitions (and which is in use), sequences, playlists, sources, scenes and sponsors. Inside the zip is one folder of the same name holding the settings as they are in the settings folder, the same `summary.txt`, and `studio-export.json` saying which Mac and Studio version made it. Files iCloud hasn't downloaded yet are downloaded first. Studio stays on air while it backs up, and works off air too. Link for scripts: `/api/settings/export` downloads the same zip (add `?media=0` to leave media out, `&notes=…` for notes).

Restore… in the 1P menu puts a backup back: pick a `FirstPassStudioBackup-….zip`, and Studio shows when and on which Mac it was made, its notes and how many competitions, sequences, playlists, sources, scenes and sponsors it holds, then asks before replacing this Mac's settings with it (on the Mac or in iCloud Drive, wherever they are kept now). This Mac's settings are first copied to **backups/** in the folder on this Mac, and only replaced once the whole backup has unpacked and checked out, so a damaged zip changes nothing. Either way this Mac keeps its own **Only this Mac can change things**, port and link name, and the ffmpeg and PlayHQ addresses in its config.json. A backup made without the media folder keeps this Mac's media. The debug log, scores and clock aren't touched. Studio goes off air while the settings swap and comes straight back. A backup from an older Studio is brought up to date as Studio starts; one from a newer Studio opens with competitions read-only until this Mac's Studio is updated.

Unsaved changes

Changes on the Competitions setup page aren't kept until you press **Save**. With changes not saved yet, leaving the page (a tab, or the Go menu), opening another competition, **Use this one**, **New competition** or **Import a template file** first asks **Save**, **Don't save** or **Keep editing**. **Export as template** asks to save first, since the file is made from the saved settings. Closing the window, reloading or quitting asks whether to leave without saving.

The editors that open over a page (a sponsor, a sequence, a playlist, a mic or a scene's **Switcher** actions on the Output page, an athlete's **Games**) ask the same question: with changes not saved yet, pressing Escape or leaving the page from the Go menu asks **Save**, **Don't save** or **Keep editing**, and closing the window asks whether to leave without saving. Their **Cancel** button still just cancels.

Data and bandwidth

scoreboard.html, **goals.html** and **countdown.html** (and Scoreboard control) follow the scoreboard over the one connection Studio's pages keep open for the scene, so a score press or a PlayHQ reading reaches them the moment it happens, with no asking in between. **ticker.html**, **viewers.html** (every 15 seconds, Studio only) and the automatic **supershot.html** and **streak.html** keep asking for new data; **ticker.html** also asks for its speed every 3 seconds. None of that reaches PlayHQ. **commentatorstats.html** follows the scoreboard live for its clock, moments and run, and with Auto-refresh (on by default there) asks Studio for its stats every 15 seconds and for who is on the mic each minute, neither of which reaches PlayHQ; its team lists load at the start and each quarter break (see below). The super shot and hot streak pages ask Studio every second for the scoreboard and nothing else, so they add no PlayHQ requests. Every other graphic gets its data once when it loads, and again each time it's shown on the Output page (with **Load the page fresh each time it's shown**, the default). The Clash and the winner celebration only retry if Studio isn't running.

Pages only ever talk to Studio on this computer, which uses no internet. Studio is the only thing that contacts PlayHQ, and it caches what it gets. It counts every answer, and the Output page's status board shows the count above the score: **Ping** is how long the last live score request took to answer (or the last request of any kind when the live score isn't being read), timed on a real request rather than a network ping; **Bandwidth** is what PlayHQ sent over the last minute, a second; **Data** is everything PlayHQ has sent since Studio went on air (it starts again each time), counted as the answers' size on the wire. Showing them asks PlayHQ nothing.

- Scoreboard following PlayHQ: one small request every 5 seconds by default (**Check PlayHQ every** on Scoreboard control, 1 to 120 seconds) while the game is on: from 2 minutes before its start until 2 minutes after PlayHQ says it's finished, and whenever PlayHQ says it's under way (started early or running late). From when the game's electronic scoring starts (set on the setup page) until 2 minutes before, one every 30 seconds; nothing is asked before then. After the game, and for a game with no start time or one PlayHQ never finishes (3 hours on), one every 5 minutes. Studio makes this request itself whatever windows are open, so the score, the goal order and the game triggers (quarter breaks, full time) keep up even while the scoreboard is hidden and Scoreboard control is closed; the pages share that reading rather than adding to it.

- The night's other games (**Also follow the night's other games, for their stats** on Scoreboard control): the same small request for each game being scored, at the same rate, from its scoring window until PlayHQ says it's finished; about 2 MB a game. With three other courts going that's about 36 requests a minute. A game whose request fails waits 10 seconds, then twice as long each time, up to 5 minutes. The ticker and the fixture board read their live scores from the same readings.
- Studio scene (scene.html and the Output page): nothing. They keep one connection open to Studio, which only sends when the scene changes; the pages in the scene ask for what they normally do.
- Display output (Mac app): nothing. The output windows show the scene page, so they ask for what the scene's pages normally ask for, once per display (fill and key means the pages run twice). The Blackmagic and NDI outputs run them once more, off screen, and each extra output runs the pages turned on for it once more. NDI's video goes over your local network only, never the internet. In 4K nothing more is asked for, but each NDI source carries about four times the pixels (roughly 250 to 400 Mbit/s at 50 or 60 fps, and fill and key is two sources), so use one source with transparency on a gigabit network, and the Mac does about four times the work per 4K output: the Output page's performance line shows whether it keeps up.
- Mics: nothing over the internet. Each output window mixing mics tells Studio what it hears twice a second, and the Output page asks for it twice a second while its Mics section is open; the sound itself stays inside the Mac. A mic on the switcher is a few small fader messages over the switcher's connection when its level changes (about 20 a second while a dip fades).
- Switcher (an ATEM, if one is set up): nothing over the internet. Studio keeps one connection to the switcher on your local network, a few kilobytes a second of small packets each way; looking for switchers asks the local network once.
- User guide (/guide/): nothing over the internet. It comes with Studio, fonts and pictures included, about 4 MB when it opens and nothing after.
- Stream Deck plugin's keys: nothing. The plugin keeps one connection open to Studio while any of its keys is on screen (and one more while a Switcher key is), and Studio only sends when the scene or the switcher changes.
- Recordings: nothing over the internet. Studio writes what it already has to a file on this Mac every 5 seconds while a game is recording, a megabyte or so a game. A practice game asks PlayHQ nothing at all: the recording answers.
- 4K replay following a HyperDeck: nothing over the internet. While a replay follows it, Studio keeps one connection to the HyperDeck on your local network and asks where it is four times a second, a few hundred bytes each time.
- Countdown: nothing. The clock ticks in the page and the start time comes from the game already picked.

- Live viewers (Competitions setup): only while a video output is on, one small request per stream every 30 seconds to YouTube or Facebook, plus a YouTube search when a channel's live video isn't known yet (at most every 5 minutes). Nothing when no streams are set up.
- Last 10 goals: nothing extra. It shares the scoreboard's 5 second reading, and Studio works out the goal order from it.
- Commentator stats: the stats, shooting % included, share the scoreboard's reading, so auto-refresh adds nothing. Its team lists ask for the game's line up when the page comes on and at each quarter break, about four small requests a game; games played, ladder, form, head to head and up next are read once when the page comes on and for each new game (from the season Studio already keeps); games played reads the division's finished games once (kept in `game-cache.json`, as for the Athletes page).
- Ticker: other courts' scores at most every 30 seconds.
- Automatic game selection: the fixture about once a minute for the current round while a game is on (any game of the night from its scoring opening until 2 minutes after it finishes), every 5 minutes when none is, every 10 minutes for the rest. Ladders the same: once a minute while a game is on, every 5 minutes when none is. Everything is read when Studio starts and when the competition changes, and **Refresh list** on Scoreboard control reads the night's games afresh.
- Socials tab: nothing extra. Its previews and pictures read the season's fixture, results and ladders through the same cache, and the Production Planner, sponsors and logos from Studio. Top shooters and Tournament top shooters read each finished game's line up once and keep it, the same as the Top shooters graphic. Finals bound and the Finals draw (and a finals night's other pictures) read the division's finals the way the Finals graphic does, through the same cache.
- Production Planner: nothing extra. It reads the season's fixture through the same cache as automatic game selection (the current round about once a minute, the rest every 10 minutes), and the commentators graphics ask Studio only. The fixture board's live scores for other courts go through the ticker's reader and cache: nothing extra while the ticker is on, at most one request a minute per live game when it isn't.
- Athletes page: when you open a team, the line up of each of its games this season, a few at a time. Finished games are only ever read once: Studio keeps their line ups and each team's own counts in `game-cache.json` in the data folder, so they're kept through restarts too. A game is kept only once PlayHQ's own game page says it's finished, so a copy read while it was being played is never kept; when a Studio update changes what is kept, each game is read once more.
- Finals: the division's ladder and its season's rounds when the page loads (the rounds are kept for 10 minutes, the current one for a minute).
- Team lists: warmed up ahead, so they show straight away. When a game is picked (by hand or automatically) and when Studio starts, it reads the game's line up and both teams' season squads, and the next game's teams on the same court too. Finished games come from `game-cache.json`, so after the first time that's only a few small requests per game picked. Squads are kept for 30 minutes; the team list itself asks for the game's line up and score when it loads.

Each answer is a few kilobytes, well under 1 MB an hour in total. Two options make a page poll, both off unless you add them: `?refresh=` on the ladder, and `&reload=` on the scoreboard, scene and Scoreboard control.

Animations run at the same speed at any frame rate, 50p or 60p.

Sponsors in detail

A sponsor's QR code is made on this Mac, with tracking tags for the sponsor's website analytics unless they're turned off; **Save QR code as PNG** gives a copy. **Pull up** and **Pull down** take a sponsor from the nearest tier below or above that has one. The vacancy card is never shown without a link or email for its QR code. The Sponsor report covers any run of nights: time on screen and where, ad plays, time with their QR code up, and progress against their targets. What's on an output counts as on air only while that output is on: one that's turned off or set to Web links only isn't, even while another is. **Print or save as PDF** to send it to them.

Sponsors on the graphics: the naming rights sponsor (or what its empty-tier rule gives instead) is a **Presented by** tab over the home team on the scoreboard, a pill under the title banner and a line under the countdown. The game stats, shooting stats and top shooters cards carry the next sponsor in rotation from Major and up in their footer, the ticker leads with the next from Partner and up, and the end of night card (Up next after the last game) thanks the sponsors from Major and up with their logos (`?sponsors=all` every tier, `?sponsors=0` none). On any of them, `?sponsor=0` leaves the sponsor off and `?sponsor=Golds%20Bakery` always shows that one; set it in the source's options on the Output page. With no sponsors they look as they always did.

Competitions setup in detail

When an update improves one of the ready-made setups (AMMNA Nationals, GIG WANL), yours picks up only what you haven't changed yourself. Section by section:

- **Game rules:** quarters or halves and their length, and super shots, for the whole competition or per division. Under the super shots tick box each PlayHQ division can follow the competition, or have super shots or not. A setting can also be saved for a word instead of a whole division name, and then covers every division with that word in its name ("Open" covers "Division 1 Open"; "Men" covers "Men's" but not "Women's"); a division's own setting wins over a word. GIG WANL comes set for super shots in Open only, not 21&Under or Men's, by the words "Open", "21" and "Men". A division without super shots doesn't play the super shot animation for a +2 press or count goals as super shots, and its shooting stats and top shooters leave the Super column off. **Super shots in the last [x] minutes of each quarter** (default 3) is when the scoreboard shows its "Super shot" tab; 0 leaves it off.
- **Finals:** the competition's finals format, and a different one for any division that needs it. Top 2: grand final, or best of three. Top 4: semis then grand final, Page (double chance for 1 and 2), or a placings day (1 v 2 and 3 v 4). Top 6: byes for 1 and 2, or a double

chance for 1 and 2. Top 8: AFL style, straight knockout, or carnival placings (a knockout over three weeks with play-offs, so every team gets a place 1 to 8). Every team: everyone plays on grand final day, 1 v 2 for the title, 3 v 4 for third, 5 v 6 for fifth and so on down the ladder, with nobody knocked out (an odd team out has no game). The ladder marks the finals places by what each earns and the finals graphic follows it.

- **Game night:** the court the scoreboard follows, whether it picks the game automatically, **Electronic scoring starts [x] minutes before the game** (default 10; set 20 or 30 when games are further apart), and how many days ahead counts as this game night. Automatic selection switches to the next game when its electronic scoring starts, and the scoreboard doesn't ask PlayHQ for a game's score before then, even for a game picked by hand: Scoreboard control says when it will start checking. That keeps PlayHQ's "game could not be found or was not electronically scored" error out of the debug log.
- **Colours and font:** nine colours, how see-through the dark panels are, and the font, with a preview. Click a colour's swatch for a colour picker (shade square, hue slider, HEX and RGB boxes, and on Chrome or Edge **Pick from screen**), or type or paste a colour into its box as HEX (#7f4092) or RGB (127, 64, 146). Under **Brand kit colours**, paste any text from a brand kit with HEX or RGB values in it; those colours are saved with the competition and show in every picker, one click to use.
- **Font:** the Font box lists the fonts that come with Studio first (they work on any computer, with no internet), then fonts added to the competition, then every font installed on this Mac (including any you add in Font Book), such as Helvetica Neue, Avenir Next, Futura and DIN Condensed. A Mac font shows only on this Mac: a graphic opened on another computer uses Montserrat instead. **Add a font file** takes TTF, OTF, WOFF or WOFF2 files (one per weight, or one variable font); they're saved with the competition and go in its template file, so it looks the same on any Mac. Add only fonts you're allowed to use, such as open licence fonts from Google Fonts. Any letter a font lacks comes from Montserrat. Every font is sized and centred to match Montserrat, which the graphics were drawn with, so text fills its boxes the same whichever font you pick; narrow fonts stand a little taller so they don't look small. **Text size** (80 to 130%) makes a font other than Montserrat bigger or smaller on every graphic; keep an eye on long team names.
- **Logos:** the competition logo, in a light mode version (for a light picture behind it) and a dark mode version (for a dark one; if you add only one, it is used for both), and the team logos. The logos page and title use the light mode logo unless `?mode=dark` is added; the ladder's corner, on its coloured header, uses the dark mode one unless `?mode=light` is added. Drop team logo files on the page; they match clubs by file name, and the PlayHQ team list shows which logo each team gets, with a box to pick a different one.
- **Clubs:** a team's logo and colour belong to its club, so every team of a club in every division shares them, and changing one row changes them all. The club is the team's name without the division on the end ("Comets 21U", "Comets Men" and "Comets Open" are all Comets; "Stormz Div 2" is Stormz). The **Club** box in the team list links a team to a different club, or to a new one.

- **Club colours:** in the same PlayHQ team list, **Add colour** gives a club its own colour. The scoreboard puts it behind its teams' names and The Clash behind their logo and name, with dark or white text, whichever reads better. Clubs without one (or after ✖) use the competition's colours. The match up page always uses the competition's colours. Pick the colour with the colour picker or paste it as HEX or RGB.

Use this one switches every graphic, Scoreboard control and the Stream Deck to that competition: the game night goes back to automatic and the scoreboard starts on that competition's first game. Saving the competition in use reloads the graphics with the changes. Setups are kept in the **competitions** folder (see The data folder, above).

Athletes in detail

Teams are listed by division, so a club with a team in more than one division (Knights) has a button in each, showing that division's line ups; **All athletes** lists everyone once. A silhouette's gender is the one picked on the page if you've picked one, else PlayHQ's gender for the athlete, else the PlayHQ gender of their division when they play in a men's or women's division (mixed and open divisions leave it neutral). **Silhouette** on each athlete shows where it came from and can override it. **Approve none** takes photo approval away from every athlete listed. Games played can be kept per club as well as per division.

The Output page in detail

- **Sources: Add page** adds any Studio graphic or web address; **Add image** adds a PNG, JPG, GIF, WebP or SVG, and **Add video** an MP4, WebM or MOV (saved in the data folder's **media** folder). Each opens a box to pick it. A new source goes above the selected one (on top if none is selected), hidden, with its settings open under its row so you can set it up straight away. Pages not in the scene yet are listed first, and **Add all unused pages** adds every one of them, hidden. Each row has the source's On / Off (for the output tab picked above the Scene View, once there's more than one), with a speaker to its left on a video to mute it or turn its sound on, an eye to preview it on its own and a pencil for its settings (name, address and page options, size, fade, position, its Stream Deck links, **Move up**, **Move down** and **Remove**); **Done** closes them. The list is top layer first: drag a source up or down to move it between layers.
- **Export sources** saves the scene's sources as a .zip, with the images and videos they use (untick **Export with their images and videos** to leave them out), to keep as a backup or move to another Mac. **Import sources** replaces every source in the scene with the ones in an exported .zip (or its `scene.json`), after asking. Imported sources start hidden and keep their Stream Deck keys. Their images and videos go in the media folder; a file already there with the same name but different contents isn't overwritten, the imported one is kept as ***name (2)*** instead. Exported without media, the files have to be in the other Studio's media folder already; the Output page lists any that are missing.

- **Page options:** a Studio graphic's link options listed on the Devices tab (position, heading, division, round, animate away and so on) show as fields under its **Address**, and are saved in the address.
- **On / Off** fades a source in or out with its own **Fade** time in milliseconds (0 is a cut).
- **Position and scale** place it on the frame; **Full frame** fits it to 1920 × 1080. **Page size** is the size a page is drawn at (1920 × 1080 for full-frame graphics).
- **Load the page fresh each time it's shown** (on by default) reloads a page as it fades in so its animation replays, and unloads it when hidden. Untick it to keep a page running so it shows instantly. **Refresh page** reloads it now.
- **Animations that animate away switch themselves off:** once a graphic has animated away and left the frame clear, its source is switched off in the scene (the Sources list and its Stream Deck key show it off), rather than staying on as an empty layer. That's The Clash and the winner animation with **Animate away after** set, a person lower third with **Animate away after**, positions with **At the end: Animate away**, the logo animation and the break ad with **At the end: Animates away** (their default), and the super shot and hot streak animations when they play once. Pages that stay on screen or wait for something to happen (super shots and hot streaks playing by themselves, the countdown, the scoreboard) stay on. This needs a scene window open (the Output page's Live View, its pop-out or a display or DeckLink output) to see the animation finish.
- **Preview** opens the selected source on its own in a new window, the way the Preview buttons on the Devices tab do; the eye beside a source's On / Off does the same from the list. A preview opens at half size (960 x 540): the page is still laid out at 1080p, just scaled down, on a checkerboard so the see-through parts show, so a few fit on one screen. Resize the window to make it bigger.
- **Videos** start from the beginning each time they're shown (untick to carry on), can loop, and fade out when they finish unless looping. A video that fades out when it finishes is switched off in the scene, so the Sources list and its Stream Deck key show it off and the next On plays it again. **Sound** turns a video's sound on or off, with a volume; it plays through the Mac's audio output, or in the SDI of a Blackmagic output. **Restart video** starts it again.
- **ProRes:** the pages can't play ProRes, so a .mov added to the media folder is converted to a WebM beside it (keeping a ProRes 4444 clip's transparency; the Output page warns when a .mov has none to keep, such as H.264), and sources using it switch over when it's done. The original is kept. The Mac app includes ffmpeg to do this (running from a copy of the code instead, run `vendor/ffmpeg/build.sh` once or `brew install ffmpeg`). MP4 and WebM play as they are.
- **Media folder:** the Output page's **Media folder** (at the bottom, folded until you open it) lists every song, video and image added (under Music, Add image or video, or on the Sponsors page), with its length, size and where it's used: sources, page themes (`?theme=`), sequence music (in any competition), playlists, the super shot song, the music player and sponsors' ads. **Remove** deletes a file for good (with a levelled song's original) after asking, and says where it's used first. A file on air (a source showing it, or the song playing) can't be removed until it's off. Playlists, the super shot song and

sponsors' ads let go of it; sources, page themes and sequence music keep its name, shown as ***(missing)***, so their settings and Stream Deck keys stay until you pick another file. A .mov and the WebM converted from it are separate files.

- **Sequences:** play the scene's sources one after another, for the run-up to a game, a quarter break or full time. **+ New sequence** on the Output page opens a list of steps: each step turns on the sources you pick and stays **for a set time** (seconds), **until it finishes** (its videos end, or pages that switch themselves off when done; it moves on after the video's length plus 3 s, or a minute for a page, if the end never comes), **until the game moves on** (the clock starts, or the next break or full time) or **until I press Next**. As a step ends, sources the next step doesn't use go off with their own fades. A step can also be a **scene** (+ Add scene: a starting or reset point; what the scene turns on stays on under the next steps and after the sequence; it goes straight on or holds like any step) or another **sequence** (+ Add sequence: its steps play there, so a mini sequence is built once and used in many; a sequence can't go inside itself). A sequence can be all sources, all scenes, all sequences or a mix. Tick **Hide** to keep one off the list (a mini sequence used inside others); it still plays inside them, when its start comes and from the Stream Deck, and **Show hidden** brings the hidden ones back to the list. **Starts** is by hand, N minutes before the picked game's scheduled start (only if Studio was running at that moment: opened later, the sequence is left for you to play by hand), at the breaks you tick (quarter, half and three quarter time) or at full time, a few seconds after the siren (from the scoreboard's clock, the PlayHQ scorer's or yours; no extra PlayHQ calls). While it waits to start its card says **Starting** with **Start now** and **Cancel**. **Automatic sequences on** above the cards turns all of that off and on (remembered). **Repeat** plays it once, until stopped, until play resumes or until N minutes before the start (it finishes the step on screen, then ends), and **Then** goes to a scene, turns off what it turned on, or leaves everything as it is; a sequence that plays on more than one output has a Then for each. A sequence only turns off sources it turned on itself. While one plays, its card shows a dot per step, the step's time left and the whole sequence's **Total** (**at least** when part of it waits for Next, the game or a page; **This time round** when it repeats), with **Next** (end the step now), **Pause** and **Stop** (ends it and does its Then). Going to a scene, playing another sequence or hiding everything stops it without its Then; turning the step's source off by hand moves to the next step. One plays at a time, it keeps going with the Output page closed, and nothing is playing after Studio restarts. Sequences belong to the competition in use (saved in its folder as `sequences.json`) and change with **Use this one**; the sources and scenes are shared. Stream Deck: `/deck/sequence/<name>` plays one (`/toggle` stops it on a second press, `/stop` stops it), `/deck/sequence/next`, `pause` and `stop` act on whatever is playing (`/deck/sequence/advance` is Next only when the step on now is **Until I press Next**, and does nothing otherwise), `/deck/sequence/now` and `cancel` on one about to start, and `/deck/sequence/auto/on`, `off` or `toggle` set the switch. The break lengths (Competitions setup, Game rules) let the editor say how many times a break sequence fits in each break.
- **Run order:** tonight's plan, above Scenes and Sequences: a numbered list of sequences and scenes for Output 1 (whichever output tab is picked), in the order they play, under the last two it has played tonight (greyed, with when; earlier ones fold away) and an orange **ON AIR** row for the sequence playing (step, dots, time left) or else the scene on

(Run-up, Walk out, Live, Quarter time, Live, Half time...). **Add...** puts a scene or sequence at the end; rows drag, or move with their arrows, and × takes one out. **Play next** plays the head now (a sequence starts, a scene is gone to) and **Skip** drops it without playing it; stepping through a sequence is the Sequences panel's Next. A sequence that ends looks at the run order first and plays what is queued after it; its own **Then** runs only when nothing is queued, so a sequence built for one night still works alone on another. Whatever is at the head leaves the list as it starts, however it was started, and moves up to the played rows (a skipped or removed row never played, so it doesn't). A game trigger on Output 1 puts its sequence at the head (with its few seconds and Start now / Cancel as usual) rather than starting it blind; Cancel takes it out again, and **Next** plays it when the ball is dead. Playing a scene or sequence by hand leaves the list alone unless it was the head, and nothing in the list starts by itself. The status board's **Run order** line and the bar's **Run order** chip say what Play next will do: ▶ the head, ⏭ what follows it. A sequence the run order starts plays on every output it plays on together when **Play next** is pressed; when the list moves on by itself, another of its outputs joins only if nothing is playing there and it didn't just play the sequence that ended (that one does its own Then). Its row says, for each other output, what follows it there (its own Then). The run order is Output 1's alone: the panel shows it whichever output tab is picked. **Saved...** keeps tonight's order under a name (**Save tonight's order as...**), loads one in place of what is queued, renames (**Rename a saved order...**) or deletes one, or clears the list; the list and the saved orders live with the competition's sequences, so a relaunch picks the night up where it was. A game in the Production Planner can have a saved order assigned (by its id, so a rename keeps it): it loads in place of what is queued as the scoreboard moves to that game, once, and when that drops queued items the panel's note says so until **OK**. Stream Deck: `/deck/runorder/next`, `skip`, `clear`, `add/<sequence or scene>` and `load/<saved order>`; the plugin's **Run order** key presses these, names what plays next in its colour and lights while something is queued. The existing `/deck/sequence/next` still ends the step on screen.

- **Music:** the Output page's **Music** player plays songs from the media folder (MP3, M4A, AAC, WAV or AIFF; use music you're licensed to stream, as Apple Music songs are copy-protected), one song or a **playlist** (shuffled or not, starting again after the last or not). **Add a song** with **Even out its loudness** ticked levels it to the same loudness as the others and keeps the original in the media folder's `originals`. It plays where Output 1's **Sound goes to** says, never in the Scene View. Play fades in over the **fade in** time; Pause and Stop fade out over the **fade out** time (slower by default); **Duck** (lower it to the **Ducking** card's Duck to, keeping it running) goes down over the fade out time and back up over the fade in time; **Mute** keeps it running. While a video with its sound on plays, the music fades out and pauses, then fades back in from where it was, both over the **quick** time (1 second by default; or carries on: untick it on the **Ducking** card; a video without its Sound ticked never touches the music). A **sequence** can bring its own music: a song or playlist starting at one of its steps, from a **start mark** (type it, or **Listen** and press **Mark here**), with a fade in (tick **crossfade from a video before it** and, when the step before plays a video with its sound on, the music starts its fade in time before that video ends and fades in while the video's sound fades out, so they meet as the video ends; it works with either inside another sequence); it fades out so it's gone as the sequence ends (over its whole fade out time, starting in an earlier step when the last steps are shorter than the fade, as long as they have set times; ended early, it fades from then), or cuts as it ends, or **crossfades into the next music** (it plays to the end, then fades out under the different music the next step or sequence brings while that fades in; with no new music straight after, it fades out as usual), or **carries on into the next sequence** when the sequence's Then plays another one (Run-up then Walk out share one song). A sequence with no music of its own plays the music of a sequence inside it while that one plays, from its step, ending as that sequence's music ends. Going to a scene or another sequence fades a sequence's music out over its **Music ends** time (or cuts it, if set to cut; when Music ends gives no time, as when it carries on into the next sequence, over the player's fade out time); going to a sequence with music of its own, the old music fades out under the new. A sequence playing on several outputs that you cut away from on one of them fades its music out there the same way, and it plays on wherever the sequence does. When those outputs fall out of step (a video or sponsor ad that ends at a different moment on each), each fades the music out on its own time as the music ends there, so none cuts it off early. There's one song at a time: start a sequence with its own music while that one plays on elsewhere and the new music takes over, with the old fading out there over its Music ends time. Tick **Ducks the music** on a scene (its pencil) or a sequence for an interview on court: whatever music is playing (the player's or a sequence's) ducks and keeps playing until the next scene, sequence or sequence step without the tick. A **super shot song** (a music setting, with its own volume) cuts in, no fade, when the super shot period starts in a division with super shots and fades out when the period ends. Stream Deck: `/deck/music/toggle`, `play`, `pause`, `resume`, `stop`, `fade`, `next`, `prev`, `/deck/music/mute/toggle`, `/deck/music/duck/toggle`, `/deck/music/load/<song>` and `/deck/music/playlist/<name>` (`?play` starts it too); `?fade=2` on any of them fades over 2 seconds, `?fade=0` cuts.

- **Mics:** live mics plugged into the Mac (wireless receivers for effects mics on court or crowd mics, directly or through an audio interface) go out with the music and video sound. Add each one in the Output page's **Mics** section: a name, the input (**Find inputs** lists the Mac's, in the Studio app) and the channel (both mixed to one, or left or right, so two receivers on one stereo input are two mics). The mics are Output 1's, like the music: heard while Output 1 is on, and never in the Scene View, so a speaker near a mic can't feed back; a commentators' feed, added to Output 1's **Sound goes to**, takes the ones ticked for it. The first time, the Mac asks whether FirstPass Studio may use the microphone. Each mic has a level (100% as it comes in, up to 200%), a mute, a **Low cut** for wind and rumble (on by default) and a meter; **Level** and **Mute all** are over all of them, and a limiter stops a shout clipping the stream. Like the music's duck, the mics **duck** to the **Duck to** level (30% by default) while a video with its sound on plays, while a scene or sequence with **Ducks the music** is on, and while music plays when that's ticked (it starts unticked; each is a tick box on the **Ducking** card, with their Duck to and their times), going down over 1 second and back up over 2; **Duck** ducks them by hand. A mic marked **Never ducks (an external mixer does it)** stays up, since the mixer does the fading, and stays at its level; ticking it shows a reminder that the mixer may need mix minus on its audio out. A device that isn't plugged in says so on its card. A mic can be plugged into the **switcher** instead (an ATEM Mini's Mic 1 or Mic 2): pick **Plugged into: The switcher** and its input. The switcher mixes it, and Studio moves its fader for the same level, mutes and ducks (fading in small steps), changing nothing else on it:

set its On, EQ and dynamics in ATEM Software Control. Moving its fader on the switcher or in ATEM Software Control makes that the mic's level, and Studio takes the faders as it finds them when it connects. Left and Right need the input split into two mono channels in ATEM Software Control. These mics are heard whatever Studio's outputs are doing. Their meters show the sound coming into the switcher's input (before its fader, from the levels the switcher sends, as it does to ATEM Software Control; asking for them changes nothing on it), with a mark where the fader is, and the card says both (**hearing -20.0 dB · fader -6.0 dB**); for an input split in two, the louder half. A mic that hears nothing above -55 dB for 10 seconds while it's listened to says **Silent for 12 s · check its receiver and battery** on its card and **Silent** on the status board, until sound comes in again; a muted mic, and one nobody is listening to (Output 1 off, a switcher that sends no levels), is never silent. A **phone mic** (+ Phone mic, the Commentator page at `/commentator.html` by the Secure phone link) sends the phone's mic straight to the Mac over the network (WebRTC, nothing over the internet); Output 1's window answers it and mixes it like an input on the Mac, one channel, with the browser's call processing off. Studio adds the mic the first time a paired phone (any level) presses Start, named after the phone; deleting it is fine, it comes back the next time that phone starts. It's mixed into Output 1's main sound only, not the other destinations under Sound goes to. The same connection carries **Hear Studio** back to the phone, asked for as music (stereo, full bandwidth, 128 kb/s), and the page tells the phone it plays and records sound rather than making a call. An iPhone may still treat a page using its mic as a call, which can make Hear Studio sound squashed. Hear Studio carries the program as Output 1's window plays it (players and page sounds) and every other mic at its level, but never that phone's own mic, so nobody hears themselves late. The phone plays it straight to its headphones or interface, not through a mixer on the page (iPhone Safari passes nothing through one), so its own volume buttons set the level; Off mutes it. Its card says **Waiting for the phone** before it starts, and **Dropped** (as does the status board) when the connection goes; the phone starts again by itself every few seconds while its page is open. Stream Deck: `/deck/mics/mute/toggle` (`on` , `off`), `/deck/mics/duck/toggle` , `/deck/mics/up` and `/deck/mics/down` (10% a press, `?by=5` for another step), `/deck/mics/level/80` , and one mic by name or number: `/deck/mic/court-mic/mute/toggle` , `/deck/mic/1/up` , `/deck/mic/1/down` .

- **Scene View:** the Output page shows the scene as it is right now (your switcher decides whether it's on air). It runs the scene's pages a second time, so untick **Live View** when the Mac is busy. The Scene View never plays sound: an output's sound plays where its **Sound goes to** says, whether the output is on or off, so nothing plays twice and nothing plays from an Output page that happens to be open. **Pop out** shows the Scene View in its own window to resize, move to another screen or make full screen while testing (silent too), with a floating Sources list to show or hide each source from there; the Output page's own Scene View turns off while it's open and comes back when it closes.

- **Display output (Mac app):** sends the scene full screen to the Mac's extra displays (HDMI into a switcher). **Chroma** puts it over a solid colour on one display (green, blue or any colour) for a switcher that keys on that colour. **Fill and key** uses two extra displays: fill is the scene over black, key is white where the scene shows and black where it's see-through, for the switcher's fill and key inputs. It's only offered with two extra displays connected (a plain M1 drives one; Pro and Max chips drive more). Set those displays to exactly 1920 × 1080 (not HiDPI or scaled) in System Settings, Displays, or the scene looks soft; the Output page warns when they aren't. Turn the switcher's **Pre Multiplied Key** on for clean edges (with the switcher set up, Into the switcher as checks it and can set it). Each output opens full screen in its own window; leave full screen (green button or ^⌘F) to move it to another display or use other apps. While the output is on, its settings are locked: turn it off to change them. **Output on/off**, Esc on an output window, closing it, or `/deck/output/on`, `off` and `toggle` show or blank the displays without changing the setup.

- **Blackmagic output (Mac app):** sends the scene out of a Blackmagic UltraStudio or DeckLink: **fill and key** with the device's external keyer (fill on one SDI out, key on the other) or **fill only**, in the video format you pick (1080p only; pick the one the switcher runs at, usually 1080p50). A format marked ***(device says it can't)*** is one the device reports it can't send (older UltraStudios with HD-SDI rather than 3G-SDI stop at 1080p30); you can still try it, and the Output page says if it fails. The fill is premultiplied: turn the switcher's **Pre Multiplied Key** on for clean edges (clip and gain aren't used then), and leave **Straight** off: straight fill into a Pre Multiplied Key leaves a light fringe round logos and text. Tick Straight only for a switcher that can't do a premultiplied key (every ATEM can); ticking it asks if you're sure. **Sound goes to** the SDI to start: 48 kHz stereo, in time with the picture (with fill and key, on the fill), so on the switcher turn on the sound of the input the fill comes into (On, or AFV to follow the picture); an SDI to HDMI converter passes it on. The Mac's sound output can be picked instead, or added as a second destination. It needs Blackmagic Desktop Video installed (the Output page says if it isn't, with where to get it; quit and reopen Studio after installing) and the device free, so quit any other app using it first. Output on/off and the `/deck/output` links work the same.
- **NDI® output (Mac app):** sends the scene over the network as NDI sources, in the video format you pick (1080p at 23.98 to 120 fps, the same rates as a Blackmagic device, or 720p at 25 to 60 fps): **fill and key** as two sources (***Source name Fill*** and ***Source name Key***) for a switcher's fill and key inputs, **chroma** as one source over a solid colour (the same colour setting as the chroma display), or **with transparency** as one source with alpha for receivers that key on NDI's own transparency. They show up on the network as this Mac's name with the **Source name** in brackets (the output's name, for example ***Output 2 Fill***, and it follows the output when it's renamed; tick **Use my own name** to type another; a source that changes name has to be picked again on the receiver). Each source only sends while something is receiving it, and the Output page says how many are. It needs NDI's runtime for Mac installed (free, the libNDI installer from ndi.link/NDIRedistV6Apple; Studio also uses the copy inside NDI Tools' apps if it finds one, as long as that copy is signed by NDI's makers, Vizrt or NewTek: a library runs inside Studio, so an unsigned file by that name is left alone and the log says so); the Output page says if it isn't, with where to get it, and finds it as soon as it's installed. Use a wired network: each 1080p source is roughly 100 to 150 Mbit/s at 50 or 60 fps and about twice that at 100 or 120 (fill and key is two of them), more than Wi-Fi carries steadily, so over Wi-Fi the receiver loses frames and moving graphics such as the ticker stutter. Pick the frame rate your switcher or stream runs at (1080p30 for a 30 fps stream): a higher one costs more of the network, and the receiver drops the extra

frames anyway. Once a minute while something is receiving, the app log says how many frames a second went out and how many new pictures the scene drew; both at the full rate with a choppy picture means the network lost them. Sound plays through the Mac's audio output, not over NDI. Output on/off and the `/deck/output` links work the same. NDI® is a registered trademark of Vizrt NDI AB.

- **Picture, HD or 4K (Mac app):** the **HD** and **4K** buttons at the top of Video output set the picture for every output at once. In 4K each NDI and Blackmagic output sends 2160p at the frame rate picked for it (2160p50 for 1080p50) and keeps everything else: its scenes, sources, NDI source names, Stream Deck keys and links, and its switcher key. Press HD and it's exactly as before. The graphics keep their 1920 × 1080 layout and are drawn at twice the density, so Studio's text, shapes and logos come out sharp; pictures and videos are only as sharp as their files (a 1080p video or a small logo looks softer). A Blackmagic device without that 2160p format (an HD UltraStudio), or a 720p NDI format, stays as it is and its card says so; displays are left alone, as a 4K screen set to look like 1920 × 1080 already draws at 4K. In 4K a Blackmagic output's video format list also has the device's own 4K formats (2160p50 and so on, as the device lists them) to pick directly; HD hides them again and sends the 1080p format at the same rate. Use 4K only with 4K gear: an HD switcher can't take 2160p. Any graphic's own link also draws sharp at 4K: in a browser source or window set to 3840 × 2160 it draws itself at twice the size, with nothing added to the link.
- **More than one output (Mac app): Add an output** sends different graphics to up to four outputs at once, for example the downstream key on the Blackmagic and an upstream key on a camera over NDI, so those graphics switch with the camera. Each output is set up like the first (a chroma display, fill and key displays, the Blackmagic device or NDI) and has a name, a number and a colour: 1 orange, 2 pink, 3 blue, 4 green. Two outputs can't share a display, the Blackmagic device (one output) or an NDI source name: the earlier output keeps it and the Output page says so. Several outputs can each send NDI, each under its own source name (its own output name unless one is typed), and two chroma outputs can go to two displays over HDMI, each with its own key colour. **Output 1 is the primary**: the plain `/deck/layer` links act on it. **Every output plays its own video sound; the music and the mics are Output 1's**: each output plays the videos and animations on it, and Output 1 alone plays the music (the player's, a sequence's and the super shot song; a sequence played on another output alone plays without its music) and the Mac's mics. **Sound goes to** (in each output's settings) lists where its sound plays. The first destination is where the output itself carries it: the SDI of a Blackmagic output, or the Mac's sound output; on Output 1 every Mac mic goes with it. **Add a destination** sends the sound somewhere else as well: the Mac's sound output on a Blackmagic output, or any of the Mac's sound outputs such as the commentators' mixer (**Find sound outputs** lists them, in the Studio app). The status board's **Sound out** shows every destination of every output with a meter for each side, every half second: the SDI's from the sound as it goes into the Blackmagic device (with how often it ran dry, when the device had nothing to play, over the last minute), the Mac's sound outputs' from what each window plays to its device, taken from a copy of the sound so the meter never changes what plays. The SDI always takes PCM 16-bit, 48 kHz, stereo (1.5 Mbps); a Mac sound output shows the sample rate and channels macOS gives it. An added destination carries the output's videos, music and

page sounds at the levels and fades they go out at, and only the Mac's mics ticked for it (none to start with); switcher mics are mixed on the switcher, and pages from other websites aren't in it. Added destinations and their mics can be changed while the output is on; the first waits for the output to be off, since it changes how the output is sent. An older setup's **Sound** and **Also send the sound to** become the list. It starts when the output comes on and stops when it goes off, says **Sending to** the device or what's wrong (such as the device not plugged in), and plays nothing while the device is missing rather than play on the Mac's speakers. On the mixer, not in Studio: keep what comes in from the Mac out of the mixer's audio out back to the Mac (its mix minus, or the USB return left out of the USB send), or the music comes back through the commentators' mics and goes out twice, a little late. To check, play music and watch their mics' meters stay still while nobody talks. **Scenes and sequences are shared by every output**, and each output plays its own: a click on a scene or sequence plays it on the output picked over the Scene View. Each output has its own current scene and plays one sequence at a time, so the same scene or sequence can be on two outputs at once; coloured dots on scene cards, and on sequence cards each output's tick in its colour with how it is there (Playing, Paused, Starting or Stopped (it ended there, played out or cut short); a hollow dot and "off" while that output isn't sending, as it still plays there for when it's turned on), show where. **Hide all** (under Sources) clears one output only and its label names it: the one the list shows when **Show** is set to one output, else the picked output. A sequence's **Plays on** ticks (on its card, in each output's colour, and in its editor) pick the outputs it plays on together: started by itself, or played on any of them (a click with that output's tab picked, or its Stream Deck key), it starts on all of them at once, and each output has its own **Then** for when it ends there (Output 1 back to the live scene, a camera's keyer output off, say). Played on an output it isn't ticked for, it plays there alone, and `/deck/output/<output>/sequence/<name>/only` plays it on that output alone even when it's ticked for more. Stop on a card, `/deck/…/sequence/stop` and Next act on one output; `/toggle` and `/deck/…/sequence/<name>/stop` stop it on every output it plays on, each doing its own Then. A sequence plays on its outputs whether or not they're sending, so turning one on mid-sequence shows it where it's up to; an output it wasn't started on doesn't join in until it's played again. An output it plays on that is removed is left out (stopping what played there); with all of them removed, the sequence doesn't start by itself (rather than going to air on Output 1) until another is ticked, and the log says so. A Sponsor ads step off Output 1 plays its ads in that output's own Sponsor ad sources (named after it, and removed with it). Only Output 1 and outputs with **Counts for sponsors** ticked (on the output's card, off to start) go in the sponsor report, as Studio can't tell whether a camera's keyer or a commentary monitor is on the stream; time on a ticked output is listed as "Ladder (Wide camera)", and a sponsor on two outputs at once counts once. A Sponsor ads step on an output that doesn't count shows whose turn it is without taking it, so the ads on air keep their order. `/deck/scene/<name>` and `/deck/sequence/<name>` play on Output 1; `/deck/output/<output>/scene/<name>`, `next`, `prev` and `again`, and `/deck/output/<output>/sequence/<name>` (with `/play`, `/toggle`, `/stop` or `/only`), `next`, `advance`, `pause`, `stop`, `now` and `cancel` play on that output. Sources are built once and shared; each output has its own on and off for every source, as coloured numbered buttons in the Sources list (a new output starts with everything off). **Show** above the list narrows it to one output, the tabs over the Scene View show each output, and **Pop out** opens the one picked. Stream Deck: `/deck/output/<output>/on`, `off` and `toggle` for one output (its name as a link, or its number), `/deck/output/all/…` for every one, and `/deck/output/<output>/layer/<source>/on`, `off`, `toggle` and `/deck/output/<output>/layers/off` for one output's sources (the Source settings list them). Each output with its own graphics runs its pages once more, so watch the Mac's load with three or four: while an output is on, its card shows its CPU (a percentage of one processor core), memory and the frame rate its pages draw at against what they should (in amber when it falls under 90%), with Studio's total and the graphics processor under the status, and the log has a ***Performance*** line once a minute.

- **Chroma key safe:** a chroma key cuts out one colour, so anything half see-through (see-through panels, soft shadows and glows, fades, soft edges) mixes with that colour and keys with a fringe or holes. With **Chroma** (a display, or NDI's chroma signal) picked, **Chroma key safe** is ticked unless you untick it, and the Output page warns that the sources are switched to it: every graphic and animation goes out with solid panels, no soft shadows, glows, light rays or flashes (hard shadows stay, made solid), and nothing half see-through, so fades become cuts (sources cut instead of fading) and soft edges are hard. Options that only fade, like the countdown's fade and the ad's glow, are skipped, and the source's options say so. Titles, logos, light and dark modes and every other setting work as usual. It never changes a saved setting: pick another output, or untick it, and everything is back as it was. The Scene View shows the scene chroma key safe while it's on. The Output page also lists any competition or club colour close to the key colour (a keyer would cut holes in it); logos can't be checked, so look at those. For a graphic used on its own at its link and keyed on a colour, add `?chromaKey=true` to its link.
- **Switcher (a Blackmagic ATEM):** the **Switcher** card under Video output controls an ATEM over the network: the Mac and the switcher on the same network, ideally wired. **Set up**, then **Look for switchers**, finds every ATEM on the network the way ATEM Software Control does (the first time, macOS asks to let Studio use the local network: say Allow); pick one and Studio connects and remembers it by name, so it finds it again if the venue's router gives it another address. **Or type its address** (as set in ATEM Setup; an ATEM ships as 192.168.10.240) works too. The card shows the switcher's inputs by the names set in ATEM Software Control, red on program and green on preview: click one to put it on preview, then **Cut** or **Auto** (the switcher's own transition) to take it. **Fade to black**, and each upstream and downstream key on and off (red while on air; a downstream key's **Auto** takes it at its own rate), are there too. Studio reads what the switcher has and offers only that:
    - **ATEM Mini and Mini Pro:** 4 inputs, one upstream key, one downstream key. Fill and key into the downstream key takes two inputs and leaves two cameras; a chroma key needs the upstream key (one input, three cameras left).
    - **ATEM Mini Extreme:** 8 inputs, four upstream keys, two downstream keys.

  A link or button for something the switcher doesn't have (a second downstream key on a Mini Pro, say) is refused with what it does have.
  Into the switcher as: each output's setup (under Video output) says which of the switcher's keys its graphics come into and which inputs its fill and key are plugged into

(a chroma output needs just one). The usual key is downstream key 1 for Output 1 and upstream keys 2 to 4 for Outputs 2 to 4; a chroma output goes on an upstream key, as only those key on a colour. Pick another key, or ***Not keyed by the switcher***. Studio then checks that key's settings in the switcher, read only, and lists what doesn't suit the output: its fill and key sources, its type (Chroma for a chroma output, Luma for fill and key) and **Pre Multiplied Key** (on), and asks for a Blackmagic device's **Straight** to be off. It also says when the output can't be keyed as it is: a chroma output on a downstream key, NDI with transparency or fill only (an ATEM needs a key signal or a colour), two outputs on the same key, or a key the switcher hasn't got. **Set up this key for me** changes just those settings on the switcher (and unticks Straight, which restarts the Blackmagic output for a moment), after asking; it leaves the key's clip, gain, mask, position and whether it's on air alone, and the log says what it changed. For a chroma key, pick the key colour in ATEM Software Control with its sample picker. Nothing is sent to the switcher until it's pressed.

Cameras per output: an output on an upstream key can say which cameras its graphics **go with** (ticked under Into the switcher as; the outputs' own fill and key inputs aren't offered). Studio's cuts, autos and stingers, from the Switcher card, a Stream Deck key or a sequence or scene, then take that key on with those cameras and off with the others in the same transition, so the camera and its graphics change on the same frame (Studio picks the keys in the switcher's next transition and puts the BKGD and KEY buttons back as they were afterwards). A straight-to-program press brings the key straight after. With **Keys follow the camera** on (the Switcher card, on unless turned off), a cut made on the switcher itself or in ATEM Software Control brings the keys a frame or two later. It only acts on a change of camera, so a key turned on or off by hand stays as it is, and it sends nothing on connecting. An output with no cameras ticked, on a downstream key (over every camera), or whose key check shows a problem is left alone. The output's card says ***On air with Bench*** or ***Waiting for Bench***, and the Switcher card underlines each key in its output's colour. The switcher's own buttons, ATEM Software Control and other controllers keep working alongside Studio, and the card follows what they do. Studio sends nothing to the switcher by itself: not when it connects, and not when it reconnects. If the switcher stops answering (a pulled cable, a power cut), or is off when Studio connects, the card turns red within about five seconds and Studio keeps trying; anything pressed meanwhile is refused, not saved up for later, and the log says so. Studio connects when it goes on air and lets go when it goes off air. Stream Deck: `/deck/switcher/cut`, `auto` and `ftb`; `/deck/switcher/preview/<input>` and `program/<input>` (its number, or its name such as `Camera%202`); `/deck/switcher/usk/<n>/on`, `off` or `toggle`; `/deck/switcher/dsk/<n>/on`, `off`, `toggle` or `auto`; and `/deck/switcher/macro/<n>` (a macro saved on the switcher, numbered from 1 as in ATEM Software Control). The Stream Deck plugin's **Switcher** key presses any of these and shows the tally: a camera's key lights red while it's on program and green on preview, a key or Fade to black red while on air, and every key goes dark with a yellow dot while the switcher isn't answering.

Stinger: with the Stinger page as a source in the scene, the card's **Stinger** button (and `/deck/switcher/stinger`, `/deck/switcher/stinger/<input>`) plays it on the output on a downstream key (Output 1, unless another output that's switched on is Into the switcher as a downstream key, as only a downstream key covers the whole picture) and cuts to the preview input underneath it. The stinger tells Studio when it starts and when it covers the frame, from its own settings, and the cut lands in the middle of the cover (0.72 s in with the usual half second hold), so the cut comes about a second and a half after the press, once the page has loaded. That output's picture reaches the switcher a few frames late (more over NDI than HDMI): set its **line-up** in frames (each output keeps its own, so moving the stinger to another output keeps both) (−25 to 25, at the switcher's own frame rate) to move the cut, and press **Test the line-up** before you're live: it cuts just 2 frames after the bands cover the frame, so if the old picture flashes before them, add frames (run it two or three times and go by what you see each time: one run can be a frame out either way). Only that output's own video output starts the timing, not a preview or another output, so a stinger needs that output on. **Music under it** is the stinger source's own Music option (the same choice as in its settings, saved with it): ***Carries on as it is***, ***Ducks for the next scene*** (down by the time the bands cover the frame and kept down, as a scene that ducks the music, until a scene, sequence or step that doesn't) or ***Ducks while it plays*** (down the same way, back up over the music's **quick** time as it leaves). It goes by the output the stinger plays on, so it works on a night Output 1 is off; the stinger shown by hand on another output leaves the music alone. With no stinger in the scene, a stinger is an auto; if the page doesn't start within 3 s, it's a plain cut, and the log says which. A second stinger while one plays is left out (a press, a Stream Deck key, or a sequence or scene's own Stinger to), so a double tap never makes two cuts. **Stinger to** a camera that's already on program plays the stinger with no cut (for a change of graphics under it). In a sequence or scene, actions at the same moment as a stinger wait for its cut, so a key change lands under the bands too.

Switcher actions in sequences and scenes: once a switcher is set up, a sequence's sources or scene step has a 🎥 button, and each scene has 🎥 **Switcher** in its settings (the pencil on its tile). Each adds up to eight actions: **Cut to**, **Auto to** or **Stinger to** an input (or what's on preview), **Put on preview**, an **Upstream key** or **Downstream key** on, off or toggled (a downstream key can go **at its rate**), **Fade to black** or back, and **Run macro**, each as the step starts or the scene is picked, or up to 60 s in (in tenths of a second). Actions at the same moment go in order, so a preview then a cut works. A step's actions still to come are called off when the step ends or the sequence stops; a scene's when another scene is picked. A scene step does the scene's actions too. They happen only when the scene or sequence plays on Output 1 (its picture is the one going into the switcher); on another output it plays without touching the switcher. The editor offers the inputs and keys the connected switcher has; one it hasn't got (upstream key 2 on a Mini Pro, say) is shown in red and skipped, with a line in the log. If the switcher isn't answering when an action's moment comes, it's skipped and logged, never done late. Scenes and sequences with switcher actions show 🎥 by their name.

The scene only changes when you change it, so it adds no PlayHQ requests beyond those of the pages in it.

Scoreboard control in detail

Every graphic uses the game night, game and court set on the Scoreboard control. That page is laid out for the operator: the game on the board in one bar with **Follow**

PlayHQ or **Manual** and the next game the board will switch to, then the two scores and the clock as the biggest things on the page, with their keys printed on the buttons (**A**/**Z** and **K**/**M**, **Shift** for a super shot, **Space** for the clock); the graphics, team lists, finals and the rest are tabs beneath, with a preview of the board beside them, and the game's own set-up (night, court, automatic selection, how often PlayHQ is read) is the **Game** section at the bottom, which **Change game…** opens and which folds to its title. A setting added to a page's own link (like `?division=2` or `?round=6`) overrides it for that page only.

- **Game night:** the games list has every division's games on one date, whatever round each division is up to, so one division on Round 10 and another on Finals Round 1 the same night are listed together, each game with its own round. **Automatic** picks the first game night in the next 5 days, counting today, so a Tuesday night is picked from the Friday before (the number of days is on the competition setup page). Otherwise it's the last night played. Each night in the list names its rounds. Pick a night to use it, then set it back to **Automatic**. Pages that show one round (like the fixture) use each division's round on that night. On a day with several rounds, it moves on to each round as its first game comes up.
- **Always a game on:** when Studio starts, or the scoreboard is cleared, it puts a game on straight away with **Follow PlayHQ** on: the game for now on the court, otherwise this game night's first game on that court, otherwise its first game. The board is only left empty if PlayHQ can't be reached or the competition isn't found (check the PlayHQ link and competition on the competition setup page).
- **Game: Pick the game automatically for court** is ticked at the start of every game night, on the night's court (e.g. `Indoor 01` or just `1`). It shows the first game of the day on that court, then switches to each next game when its electronic scoring starts (10 minutes before by default, set on the setup page), with the quarter back to Q1 and a full clock. A final (a round with "Final" in its name) stays on until PlayHQ marks it finished, in case of extra time. With a game night picked by hand, it follows that night's games on the court by their time: the game due now, otherwise the next one to come, otherwise (a night already played) its last game.
- **Production Planner:** the Production Planner plans the court Studio streams each game night and who is on the mic for each game. When it has a court for tonight (the game night picked by hand, otherwise today), that court is set here when the night arrives, which also ticks automatic selection, and again if the plan's court changes during the night. A night with no plan (the night before's plan counts, so mostly before the Planner is first used) uses the competition's **Fallback court to follow**. Picking another game night, or another competition, is a new night. The **Commentators** graphic (its lower thirds with their headshots, if asked) and **Commentator stats** show whoever is planned for the game on the scoreboard, and a night that hasn't been planned takes the plan of the night before it. Beside the court it says where it came from: ***from the Production Planner***, ***fallback from Competitions setup*** or ***changed by hand for tonight***. A court changed or a game picked by hand stays for the rest of that night, restarts included, and the Planner flags it.
- **The night's other games: Also follow the night's other games, for their stats** (on to start with, and kept) follows every other game of the game night in the background

while Studio is on air, from when its scoring opens until PlayHQ says it's finished: the goal order, quarters, clock and moments, as for the scoreboard's game, kept as each game's story for the Socials tab, marked followed. Nothing plays on air for them. The note under it lists each one: its court, the quarter and score, ***from*** its start time before it starts, ***final*** at the end, or ***PlayHQ slow*** while a failed read waits (10 seconds, doubling to 5 minutes). Picking one for the scoreboard carries on from its goals, quarter and clock (no Q1 reset); the game the scoreboard leaves before it ends goes on being followed from where it was. Turning it off lets them go, keeping what they have. Kept through a restart (followed-games.json, on this Mac).

- **Overrides:** picking a game from the list turns automatic selection off until you tick it again, change the court, or the next game night starts. Pressing a score button switches scores to manual, and any clock button switches the clock to manual.
- **Follow PlayHQ:** the score and quarter come from PlayHQ's live scoring every 5 seconds, and **Follow the PlayHQ scorer's clock** runs the clock from the scorer's clock too. Switching scores between **Follow PlayHQ** and **Manual** (on Scoreboard control, the Stream Deck, by pressing a score button or by picking a game) turns the clock option on or off to match, and Scoreboard control says so. Changing the clock after that still wins: untick the box or press a clock button to run the clock by hand while scores follow PlayHQ.
- **Pages that use the picked game's division:** ladder and finals (without `?division=`), ladder bar, top shooters, title, match up, The Clash, winner celebration, team lists, shooting stats and full time. The fixture, results, ticker and up next use the round.
- **Stream Deck:** `/deck/auto/on`, `/deck/auto/off`, `/deck/auto/toggle`, `/deck/court/2`, `/deck/clock/playhq/on` and `/deck/ticker/faster` / `slower`. The Devices tab has the rest under Links, with Copy buttons, and the plugin. A score key switches scoring to Manual (Follow PlayHQ stops for the score and the clock), and a clock key stops the clock following the PlayHQ scorer.

Socials in detail

The Socials tab makes pictures of a game night for Instagram and other social media, from the fixture, results and ladders Studio already has. Pick the **game night** (by date, so every division playing that night is in it), then **Upcoming** (before the night) or **Results** (after it; a night with results opens on Results) and the sizes: **Square** 1080 x 1080, **Portrait** 1080 x 1350 and **Story** 1080 x 1920. Upcoming pictures:

- **Game night fixture:** every game of the night on every court, by time, with the logos, division and court.
- **Match up:** one per game, the two logos big with the date, time, court and venue.
- **Head to head:** one per game whose teams have met this season: the wins each way and each meeting's score, round and date.
- **Team sheet:** one per game: both squads this season (everyone in their line ups), each player's usual position and games played in the division, counting games typed in on the Athletes tab.

- **Form guide:** one per game, each team's last five results this season, oldest first, with the scores (a win in the main colour), and their meetings this season ("First meeting this season" until they've met).
- **Round preview:** one per division, its games with each team's ladder place and points as it stands, the game of the round (the two teams with the best places added together; Top of the table when it's 1st v 2nd) outlined, and "A win can lift Titans into the top 4" for a team outside the finals places within a win of the points in the last of them.
- **Byes:** the teams of each division playing that night that aren't, from the season's fixture. Not offered when every team plays.
- **Milestones coming up:** athletes in either team this season whose next game would be a 50th, 100th or another 50 (line ups aren't known before the night). Games played are the Athletes tab's: this season's line ups plus any typed in under Games.
- **Finals race:** one per division with a finals format in its last three rounds (and more teams than finals places): each team on the ladder as in the finals (no more teams than the finals places can still reach its points), out (it can't reach the points in the last finals place even winning every game left) or still a chance, with its points and games left. A draw on points counts as a chance, since percentage decides it.
- **Countdown:** in the last four rounds before the finals, the rounds to go (tonight's included, "Last round before finals" on the last); on a finals night, the days or weeks to the grand final.
- **Season draw:** on a division's first night, one per team, every game of its season with the date, time, court and opponent, and the result once played.
- **On the stream:** the court Studio streams that night, each game's division and who is calling it, from the Production Planner (plan the night first).
- **Commentary team:** a picture for each game someone calls, with that game's commentators (up to three) side by side and its time and teams once under the headshots, so the social team can post them one by one or group them; each commentator has their headshot, to share before the livestream. Add each commentator's headshot on the Production Planner (click the circle beside their name); their initials show until then, and a guest gets a Special guest pill.
- **Polaroid:** a tilted white print for a match photo. Drop a photo on it, drag the preview to frame it and set the zoom; with no photo its window is see-through, so your social team can put a photo under it in Canva or a phone app.

Results pictures (games show once PlayHQ has their final score):

- **Results:** every finished game's final score, division by division, the winner in the highlight colour.
- **Round in review:** the night's biggest win, its closest game (a draw is the closest) and every upset, each with its score and what made it one ("4th beat 1st"). An upset is a win by a team that stood at least half the ladder lower before the night, or 4 places lower, whichever is fewer: 2 places in a ladder of 4, 3 in 6, 4 in 10. A game that is two of these shows once, saying both. Upsets need the ladder from before the night, so they show only on the division's latest night with results. Needs two or more results.

- **Milestones:** athletes in the finished games' line ups who played their 50th, 100th or another 50, or a first game. A first game counts only when their team had played before without them and nothing is typed in for them under Games, since PlayHQ only knows this season: type in a returning player's earlier games so they don't show as new.
- **Full time:** one per game, the two logos and the score big, with who won and by how much.
- **Ladder:** one per division that played, as PlayHQ has it now, saying what it's after ("After Round 5": that night's round when it is that division's latest night with results, else its latest round with results, or the date where games have no round), with the finals line. It can be made before a night's results are in. On that latest night, with **Highlight movers** ticked (the same tick as Tournament top shooters), each team shows the places it climbed (green ▲) or fell (red ▼) since before the night. Studio works out where each team stood by taking that night's results back off PlayHQ's ladder, with a win worth what the ladder shows (a draw half that); bye points that night aren't known, so a team with a bye keeps its points.
- **Top shooters:** one per division, its top six shooters that night, from each finished game's line ups (goals, plus super goals and points where there are super shots).
- **Tournament top shooters** (**Season top shooters** when the fixture runs 15 days or more, or as picked under Game night in Competitions setup): one per division, its top eight shooters over every finished game so far, labelled after this night's latest round, or after an earlier round that night (**As at**, for a carnival day with several rounds). With **Highlight movers** ticked (kept for the competition), each shows the places they climbed (green ▲) or fell (red ▼) since the round before, even on the same day (the day before for games with no round), and NEW for anyone new to the top eight. It reads the same kept line ups as Top shooters.
- **Sharpshooter:** one per division, its most accurate shooters that night (goals and super goals over attempts), counting only those with 10 attempts or more.
- **Super shot leaders:** one per division that plays super shots, its top three for super shots that night and over the season so far.
- **Team super shots:** one per division that plays super shots: every team that has played, with its super shots that night (a dash if it didn't play) and over the season so far, most first. It uses each team's own count in PlayHQ, so a super shot PlayHQ gave the team without naming the shooter counts too; a game PlayHQ couldn't give is noted, not counted as none.
- **On fire:** one per division, up to five shooters who played that night whose goals a game (super goals counted as goals) over their last three games rose most on their games before, with those three games' goals.
- **Night in numbers:** the night's finished games added up: games, points (goals where there are no super shots), super shots, games won by 3 or fewer (points where super shots are played), the biggest win, the highest score, the night's shooting % (goals and super shots over attempts, as a whole percent), the upsets (by Round in review's rule) and any draws. A night without an upset shows the average winning margin instead, to a whole goal, leaving draws out. Shooting % and super shots come from each team's own count in PlayHQ, so one PlayHQ gave the team without naming the shooter still

counts; a game whose line ups PlayHQ couldn't give is noted under the number, not counted as none. Three more come from the goal order, only when every finished game that night has a whole story (streamed or followed in the background, with no gap, and its score PlayHQ's final), so no game is left out: the lead changes, the most goals in a row (with the team) and the runs of 5 goals; Night in numbers on the Socials tab says how many games have one so far and which don't.

- **Club round up:** one per club with teams in more than one finished game that night (teams belong to clubs as set on Competitions setup), each game's result division by division.
- **On a roll:** the teams unbeaten all season (3 games or more) or on 3 wins or more in a row, after that night's results, longest run first. It isn't there when no team is.
- **Team season card:** pick a team: played, won and lost, its ladder place and percentage, its biggest win and its top shooter so far.
- **Season wrap:** on a division's last night: its champions once the grand final has a winner, the season's top shooter, sharpest shooter (10 attempts a game on average or more), most super shots (where it has them) and biggest win.
- **Player of the match:** one per game: pick a name from either team's line up (PlayHQ's, read once the game finishes). Their photo shows when they approved its use (else their team's logo), with their goals, shots and accuracy that game if they shot. Type a **Medal name** (optional) for a special round and it becomes the title.
- **Team of the round:** one per division that played: pick an athlete for each position, GS to GK, from that night's line ups in the division (those who played it there, and anyone without a position). Positions left empty don't show.
- **Athlete spotlight** (Upcoming and Results): pick an athlete with an approved photo: their teams, the positions they play, their games played per division (as typed in on the Athletes tab, plus this season) and their goals this season, up to that night.
- **Sponsor spotlight** (Upcoming and Results): pick any of the competition's sponsors: their logo big, their tier and tagline, and after a streamed night, how many minutes they were on screen (from the sponsors' report).
- **Thanks for watching:** a thank you with the next game night and every naming, elite and major sponsor. It has no live viewer numbers yet.
- **Game story:** one per game Studio streamed or followed in the background: the score quarter by quarter (a quarter Studio didn't see all of shows –), then quarters won, the best quarter, the longest run of goals, the biggest lead and the lead changes, as on the stream's Game stats and Score worm. Studio keeps them on this Mac (game-stories/) as it follows each game, so they're there after the scoreboard moves on, and copies the finished ones another Studio Mac kept (over the same network every few minutes, or through iCloud Drive/FirstPass Studio/game-stories), so a second court's games are there too; if the goals Studio saw don't add up to PlayHQ's final score, it says Studio didn't see every goal.
- **Score worm:** with Game story: the margin over the game, the home team above the line and the away team below, in their colours.

- **Moments of the game:** with Game story: the lead changes, comebacks, runs and super shots Studio saw, oldest first, each with its quarter, the time left (when the clock ran) and the score then; the 8 that matter most when there are more.
- **Game stats:** one per finished game: team against team from PlayHQ's team counts (super shots and goals where super shots are played, attempts, accuracy; a goal PlayHQ gave the team without naming the shooter counts too), the better figure highlighted and a drawn figure left plain; when Studio streamed or followed the game, also quarters won, the longest run, the 3 and 5 goal runs (a run of 5 counts as a 3 goal run too) and the biggest lead. There are no misses: having the most isn't a win.
- **Shooting duel:** one per finished game: each team's shooters, most points first, with goals from attempts and accuracy, under the team's total.
- **Runs of the night:** every game Studio streamed or followed that night, its longest run of goals first, with each team's runs of 3.
- **Super shot time:** for the streamed or followed super shot games: the points each team scored in each quarter's super shot period. The goals need clock times, so the clock has to have run (following PlayHQ or by hand).
- **Comeback of the night:** of the streamed or followed games, the winner who came back from furthest behind (the most the losing team led by), with the game's Score worm.
- **Watch it back:** a QR code for the replay's web link (pasted on the card; your first stream on Live viewers to start with), with the night's streamed games beside it when the night is planned.
- **Polaroid:** one per game, with the score under the photo ("Blizzardz win 31–26") and the opponent, division, round and date under that.
- **Finals bound:** after a division's last home and away round (when it has a finals format in Competitions setup), the teams in its finals in ladder order, each with what their place earned: Double chance, Week off, Elimination final.
- **Finals draw:** with Finals bound, and again on each finals night (under Upcoming too, to post before the games): the division's finals week by week, who plays who ("Winner minor semi final" until a team is known), with the scores and winners filled in as the games finish, from PlayHQ or entered by hand on Scoreboard control. The grand final is outlined.
- **Champions** (or **Premiers**, as picked under Finals in Competitions setup): once the grand final (or a best of three) has a winner, the winners' logo lit in their colour, their name, and the grand final's score.

On a finals night, the Match up, Full time and Results pictures name each final (Major semi final, Grand final) instead of the round.

Each picture has a **Sponsor strip** tick (your naming, elite and major sponsors along the bottom), remembered for next week, and Under **Options**, **FirstPass Studio mark** (bottom right) and **PIVT logo** (bottom left) turn those on or off for every picture. **Template** picks the look of every picture: **Default** (the stream's own), Dynamic, Junior, Retro, Professional, Thunder, Gradient, High contrast, Neon, Poster, Splash, Game Day, Hero or Final (these two stand a picked athlete in front of the title; cut-out PNGs look best), the netball sets Court, Bib and Goal, or the sporting sets Bottle, Cones, Sneakers, Stadium and Locker room, each with its own shapes and lettering. A star beside a template makes it a favourite, and **Favourites only** lists just those (and the one in use), remembered for the competition. Any template works with any colour set. Every picture's title stays on one line beside the competition logo, shrinking a little at a size where it would otherwise wrap. A short ladder or a couple of games grow to fill the picture, and a Story keeps its words and logos out of the top and bottom, where Instagram puts its own bar and reply box. A long team name shrinks to stay on one line in its row. Athletes' photos come from the Athletes tab, and only for athletes ticked **Approved for photo use**: Top shooters shows each shooter's photo, and on a Match up, Full time or Polaroid card you can pick an athlete by name, narrowed by division and team. **Colours** picks a colour set for every picture: **Competition colours** (the stream's own look), or a set made from the competition's main, highlight and accent colours (Accent first, Highlight ground, Accent ground, Night, Day, Two colours), or your own with **New colour set**. A set whose titles would be hard to read says ***low contrast***. The choice and your own sets are kept per competition. **Select all** and **Select none** tick or untick every picture at once. **Save pictures** saves the ticked ones as PNGs in **Pictures/FirstPass Studio/<competition> <date> Upcoming** (or **Results**) and opens the folder; with **As a zip** ticked it also saves them as one .zip beside the folder, and Share sends that zip; with **As a carousel** ticked they're saved in the carousel's order (one size, up to 20, each marked 1/6, 2/6…) in **… Upcoming carousel** (or **Results carousel**); **Share…** saves them too and opens the Mac's Share menu (AirDrop to your phone, Messages, Mail). Saving needs the Mac app. Each picture is also a page of its own, `/social.html?kind=fixtures&size=square` (kinds `fixtures`, `matchup`, `onstream`, `commentators`, `polaroid`, `results`, `fulltime`, `ladder` with `&division=`, `topshooters` with `&division=`, `tournament` with `&division=` (and `&after=HH:MM`, `&movers=0|1`); `&date=`, `&game=`, `&sponsor=0|1`, `&mark=0|1`, `&pivt=0|1`, `&colours=`, `&template=`, `&home=`, `&away=`, `&athlete=`, `&title=`, `&page=2/6` for its place in a carousel). Photos dropped on a polaroid stay on this Mac (the newest 40), never in a backup or iCloud.

Captions: the Caption box under each picture writes its post text with AI, in a style (Hype, Professional, Emotional, Neutral, Fun), optionally short, from that picture's own facts (the same data its page reads, for the division, game, team or athlete it's about). The rules every AI gets: only the facts given, never invent quotes or numbers, never mention or highlight a negative stat, Australian English, no hashtags (Studio adds yours). The text can be changed or typed by hand; **Undo** brings back the text before the last one written. Saving writes the ticked pictures' captions to `Captions.txt` beside them and into the zip; on a phone they come back with the pictures. The **Captions** card (on the Mac only) picks the AI: Claude (Anthropic), ChatGPT (OpenAI, not stored by OpenAI) or Gemini (Google), each with its own key, checked when saved and kept in the Mac's Keychain (never in a backup, iCloud or a page; only the last four characters show). Defaults: Claude Haiku 5.5, GPT-5.4 mini, Gemini 3.5 Flash-Lite; **Model** lists the models the key can use. Prices are US dollars per million tokens from LiteLLM's public price list, checked weekly; this month's spend adds up each caption's cost (a model not in the list shows its tokens and an amber warning). One caption at a time, at most 30 in 10 minutes. Without the internet the box says so, and the caption can still be typed.

Recordings in detail

For each PlayHQ game on the scoreboard while on air, Studio keeps a recording of what PlayHQ said and what went to air, timed: PlayHQ's answers (live score, clock, quarter, line ups, fixtures, ladders), each scoreboard press and Stream Deck press, and every change to the scene, the board and the answers the graphics read. There's no video or sound in it, only what the graphics need, a megabyte or so a game. Each recording follows a PlayHQ game: it starts when the game is due on the scoreboard (as many minutes before its scheduled start as **automatic game selection** on the competition setup page says, 10 by default) and stops a set time after PlayHQ says it's finished (**Keep recording for … minutes after full time**, 10 by default, 0 to 60), when the scoreboard moves on to another game, or when Studio goes off air. Being on air with no PlayHQ game due records nothing, and a game PlayHQ already says is finished isn't recorded. Each recording starts with the last of everything Studio already knew (the fixtures and ladders PlayHQ gave, the scene, the board), so it stands on its own. The Recordings tab lists each game's recording (named after when it started and the teams), its length and size, with **Download** (the .jsonl.gz file, to keep or send in as test data) and **Remove**, and says when the next one starts. **Record each PlayHQ game on air** is on unless you turn it off; turning it off stops the recording being made. The Output page's status board has the same switch as **Record** and **Stop**, so a recording can be stopped or started without leaving it. A recording older than 60 days is removed when Studio starts, and so is an old one with no game in it. Recordings stay on this Mac (in **Recordings/** of the data folder), never in a backup or iCloud, and hold nothing secret: no keys, devices, cookies or private answers. A recording also replays a game's graphics at 4K (below).

The night's other games: each game Studio follows in the background (Scoreboard control's **Also follow the night's other games**) gets a recording of its own, from Studio's first reading of it to the same minutes after PlayHQ says it's finished, or until it's no longer followed (picked for the scoreboard, following turned off) or Studio goes off air. It's named after when it started and the teams with **(followed)**, and listed as **followed, not streamed**. It holds what PlayHQ said (the fixtures, ladders and line ups Studio read that night, and that game's live scoring), a few hundred KB, and nothing that went to air, so it has Practise and Demo but no Replay. Played with Practise, anything a graphic asks that the recording doesn't have is read from PlayHQ (see Graphics on another court's game). The same switch turns them on and off.

A practice game's controls: Jump to Start (as many minutes before the scheduled start as **A practice game starts in** says), each quarter (the moment PlayHQ's scorer started its clock, else when PlayHQ first said it was on) and Full time; **Pause**/**Play**; **Nudge** a second either way, to line it up with a video; **Stop**. The Stream Deck links are `/deck/practice/play`, `pause`, `toggle`, `next`, `prev`, `stop` and `nudge/<ms>` (+ or −).

4K replay: (step by step: Put 4K graphics on a recorded game). **Replay** beside a game puts that game's graphics back on every output exactly as they went to air: the scene, the scoreboard and what each graphic read, from the recording. Turn on **4K** on the Output page and the outputs send it at 2160p, sharp, to key over the 4K camera recording of the game in a 4K switcher or recorder; Studio doesn't play the video. It starts paused at the game. To sync, pause the video on a moment you can see (the first centre pass), pick the same moment in the list of what went on air (***Scoreboard on***, ***Q1***...) and **Jump there**, then press **Play** on both together; **Nudge** moves the graphics a frame (40 ms) or a second either way. While it runs the live scene and scoreboard are kept as they were and can't be changed (the outputs can), nothing counts for the sponsors and the replay isn't recorded, and a red strip on every Studio page says so, with Stop. Stop puts the live graphics back.

Over the recording in another program: to lay the graphics over the 4K recording in a program that takes a web page as a source instead of a switcher, add that output's **Output canvas** link (Devices tab, Links, The scene; one link per output) as one source set to 3840 × 2160. It shows exactly what the Output page's picture of that output shows, live: scenes, sequences, the run order and the replay, drawn sharp at 4K. It's silent (the recording has its own sound); `&sound=1` on the link plays the output's music and video sound too. Nothing has to be sending: 4K and the output can stay off.

Following a HyperDeck: with the 4K video on a HyperDeck, set **Follow** to ***A HyperDeck's timecode*** and type its IP address (on its setup page). The replay then goes where the HyperDeck's timecode says, as a time of day on the game's night: play, stop and scrub the HyperDeck and the graphics go with it, so they never drift, and Studio's own Play is the HyperDeck's. Studio only reads its timecode over the network (its control protocol, port 9993) and never plays, stops or records anything on it. If the HyperDeck's clock and this Mac's differed on the night, sync once: stop the HyperDeck on a moment you can see, pick the same moment in the list and press **Jump there**. **Nudge** then moves the graphics against the video. The sync is kept for each recording. The recording needs time-of-day timecode (the HyperDeck's timecode set to Time of day, or timecode from the switcher).

Following the SDI input: any player can drive the replay through the UltraStudio instead. Send its SDI into the UltraStudio's SDI input (a HyperDeck's monitor or SDI output is enough; the 4K picture can go to the switcher as usual) and set **Follow** to ***Timecode into the UltraStudio (SDI)***. Studio reads the timecode that comes with each frame (only the timecode, not the picture) and follows it the same way: playing, stopped and scrubbing, with **Jump there** and **Nudge** to sync. HDMI carries no timecode, so the HDMI input can't be used. If the UltraStudio can't read its input while it sends the graphics, Studio says so beside Follow. Only in the Mac app.

Stream Deck: the plugin's **Replay** key plays or pauses, jumps to the next or previous moment, nudges or stops the 4K replay or practice game, whichever is running, and lights while one runs; or the links `/deck/replay/toggle`, `next`, `prev`, `nudge/-40`, `stop` and `/deck/practice/toggle`, `next`, `prev`, `stop` (the Devices tab lists them all under Links).

Practice games: (how to use one: Redo a night's production). Studio asks PlayHQ nothing while one runs; the recording answers instead, with every date and time in it moved by the gap to now, so the game is due as many minutes from now as **A practice game starts in** says (5 to begin with, 0 to 60, remembered on this Mac). Only PlayHQ's answers are played back: the scoreboard and Stream Deck presses in the recording aren't. **Run game-timed sequences again** (on to begin with, remembered on this Mac) is read as the practice starts: that start and each **Jump to** (and the Replay key's next and previous) forget which of the game's moments have fired, so its pre-game, break and full time sequences queue again even for a game practised before or played live tonight. A pre-game moment more than a minute before the start or jump is skipped; unticked, moments that have already fired stay fired. **Pause** stops the clock where it is (the gap to now grows).

Demo games: Demo beside a recorded game plays it as one of the competition in use's games, for showing Studio to a league in their own colours. Pick a **division** and one of its finished **games**: Studio takes that game's two teams (with their logos and colours) and athletes, and the division's other teams, and everything else (the score, clock, quarters, moments and timing) comes from the recording. Athletes are matched by position, so the recorded GS's goals are their GS's; anyone with no match is left unnamed and their goals go to the team. If PlayHQ has no line up in the recording for one team (it happens), the team on that side plays with its own line up and its goals go to the team; the **Athletes** table says so. A game with no line up (or a short one) takes the team's season squad, by the position each played most. While the demo runs, its **Athletes** table shows who each recorded athlete is shown as, and any of them can be changed. **Save demo** keeps it by name (teams, athletes, Swap) under **Saved demos**, to start again in one click; saved demos stay on this Mac. **Swap home and away** puts the teams the other way round, so either can be the winner. The competition's setup comes from PlayHQ, so a demo needs the internet. Studio never plays video: play the recorded game wherever you like and lay Studio's output over it. A red strip says a demo game is running, as for practice.